

**CROSS-DOMAIN CLUSTERING AND LOCALIZATION
OF MULTIPLE PARTIAL DISCHARGE SOURCES VIA
DEEP LEARNING**

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**DEPARTMENT OF ELECTRICAL ENGINEERING,
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KUALA LUMPUR**

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LOCALIZATION OF MULTIPLE PARTIAL
DISCHARGE SOURCES VIA DEEP LEARNING**

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**DEPARTMENT OF ELECTRICAL ENGINEERING,
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ABSTRACT

Partial Discharge (PD) poses a significant threat to high-voltage insulation systems, often leading to catastrophic breakdowns if undetected. While Time Difference of Arrival (TDOA) methods are effective for localizing single PD sources, real-world substation environments frequently present multiple, simultaneous PD sources that complicate detection and localization. This project proposes a comprehensive methodology to address the simultaneous localization of multiple PD sources by integrating a Digital Twin simulation environment with advanced deep learning clustering architectures. A digital twin was developed to simulate electromagnetic (EM) waves from PD sources, allowing for the generation of robust synthetic datasets. The study evaluates and compares multiple deep learning pipelines, contrasting Convolutional Neural Networks (CNNs) operating on Short-Time Fourier Transform (STFT) spectrograms against Vision Transformers (ViT) processing 2-channel complex bispectra. To bridge the domain gap between simulated models and real-world physical measurements, Supervised Contrastive Learning (SupCon) and Domain Adversarial Neural Network (DANN) pre-training frameworks were implemented. Results indicate that bypassing non-linear dimensionality reduction techniques (such as t-SNE or UMAP) yields the highest clustering stability. The upgraded ViT architecture with DANN achieved an 82.98% classification accuracy on unseen, real-world measured PD waveforms, successfully isolating two primary defect clusters with a near-perfect silhouette score of 0.997. Future work is needed to overcome limitations in the current static distance thresholds used for pulse instance grouping.

Keywords: Partial Discharge, Vision Transformer, Deep Embedded Clustering, Domain Adversarial Neural Network, Complex Bispectra.

ABSTRAK

Nyahcas Separa (PD) menimbulkan ancaman besar kepada sistem penebat voltan tinggi, yang sering membawa kepada kerosakan katastropik jika tidak dikesan. Walaupun kaedah Perbezaan Masa Ketibaan (TDOA) berkesan untuk menyetempatkan punca PD tunggal, persekitaran pencawang sebenar sering mempunyai pelbagai punca PD serentak yang merumitkan pengesanan dan penyetempatannya. Projek ini mencadangkan satu metodologi komprehensif untuk menangani penyetempatan serentak pelbagai punca PD dengan mengintegrasikan persekitaran simulasi Kembar Digital bersama seni bina pengelompokan pembelajaran mendalam termaju. Satu kembar digital telah dibangunkan untuk mensimulasikan gelombang elektromagnet (EM) daripada punca PD, membolehkan penghasilan set data sintetik yang mantap. Kajian ini menilai dan membandingkan pelbagai saluran pembelajaran mendalam, membezakan Rangkaian Neural Konvolusional (CNN) yang beroperasi pada spektrogram Transformasi Fourier Masa Singkat (STFT) berbanding Vision Transformers (ViT) yang memproses dwispektrum kompleks 2-saluran. Untuk merapatkan jurang domain antara model simulasi dan ukuran fizikal dunia sebenar, kerangka pratihan Pembelajaran Kontrastif Diselia (SupCon) dan Rangkaian Neural Adversarial Domain (DANN) telah dilaksanakan. Keputusan menunjukkan bahawa melangkau teknik pengurangan kematraan tidak linear (seperti t-SNE atau UMAP) menghasilkan kestabilan pengelompokan tertinggi. Seni bina ViT yang dinaik taraf dengan DANN mencapai ketepatan klasifikasi 82.98% pada bentuk gelombang PD ukuran sebenar yang belum pernah dilihat, dan berjaya mengasingkan dua kelompok kecacatan utama dengan skor siluet yang hampir sempurna iaitu 0.997. Langkah-langkah perlu diambil untuk mengharungi batasan pada ambang jarak statik yang digunakan untuk pengumpulan acara reruang.

Kata kunci: Nyahcas Separa, Pengubah Visi (Vision Transformer), Pengelompokan Terbenam Mendalam (Deep Embedded Clustering), Rangkaian Neural Adversarial Domain, Dwispektrum Kompleks.

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LIST OF SYMBOLS AND ABBREVIATIONS

AC	:	Alternating Current
ADMM	:	Alternating Direction Multiplier Method
AIS	:	Air-Insulated Substation
ANN	:	Artificial Neural Network
AOA	:	Angle Of Arrival
AOK	:	Adaptive Optimal-Kernel
Avg.	:	Average
AWGN	:	Additive White Gaussian Noise
BiLSTM	:	Bidirectional Long Short-Term Memory
BPNN	:	Back Propagation Neural Network
BSS	:	Blind Source Separation
CBLAG	:	CNN-BiLSTM-Attention-GAN
CC	:	Connecting Cable
CD	:	Coupling Device
CGAN	:	Conditional Generative Adversarial Network
CNN	:	Convolutional Neural Network
CSA	:	Crow Search Algorithm
CWRU	:	Case Western Reserve University
CWT	:	Continuous Wavelet Transform
DBSCAN	:	Density-Based Spatial Clustering of Applications with Noise
DC	:	Direct Current
DEC	:	Deep Embedded Clustering
DEDO	:	Double Exponentially Damping Oscillation
DEP	:	Double Exponential Pulse
DFT	:	Discrete Fourier Transform
DL	:	Deep Learning
DOA	:	Direction of Arrival
DOP	:	Damped Oscillatory Pulse
DRN	:	Dilated Residual Network
DUT	:	Device Under Test
DWT	:	Discrete Wavelet Transform
EM	:	Electromagnetic
Est.	:	Estimated
FCNN	:	Fully Connected Neural Network
FDTD	:	Finite-Difference Time Domain
FFT	:	Fast Fourier Transform
FIT	:	Finite Integration Technique
FM	:	Frequency Modulation
FN	:	False Negatives
FP	:	False Positives
F-P	:	Fabry-Perot
GAN	:	Generative Adversarial Network
GELU	:	Gaussian Error Linear Unit
GIL	:	Gas-Insulated Line
GIS	:	Gas Insulated Switchgear
GMM	:	Gaussian Mixture Model
GRL	:	Gradient Reversal Layer
GRU	:	Gated Recurrent Unit

GW-GAN	:	Gradient-Penalized Wasserstein GAN
HDBSCAN	:	Hierarchical Density-Based Spatial Clustering of Applications with Noise
HFCT	:	High-Frequency Current Transformer
HHT	:	Hilbert-Huang Transform
HV	:	High Voltage
HVDC	:	High Voltage Direct Current
KDE	:	Kernel Density Estimate
KL	:	Kullback-Leibler
kNN	:	k-Nearest Neighbors
LSTM	:	Long Short-Term Memory
MI	:	Measuring Instrument
ML	:	Machine Learning
MLE	:	Maximum Likelihood Estimate
MLP	:	Multi-Layer Perceptron
MMFCC	:	Modified Mel-frequency Cepstrum Coefficients
NLoS	:	Non-Line-Of-Sight
NT-Xent	:	Normalized Temperature-scaled Cross Entropy
PCA	:	Principal Component Analysis
PD	:	Partial Discharge
PMA	:	Planar Monopole Antenna
PRPD	:	Phase-Resolved Partial Discharge
PSO	:	Particle Swarm Optimization
RF	:	Radio Frequency
RMS	:	Root Mean Square
RMSE	:	Root Mean Square Error
RNN	:	Recurrent Neural Network
RSS	:	Received Signal Strength
SAE	:	Stacked Autoencoder
SEDO	:	Single Exponentially Damping Oscillation
SELMS	:	Sign-Error Least Mean Square
SimCLR	:	Simple Framework for Contrastive Learning of Visual Representations
SMG	:	Sinusoidally Modulated Gaussian
SNR	:	Signal-To-Noise Ratio
STFT	:	Short-Time Fourier Transform
SupCon	:	Supervised Contrastive Learning
SVM	:	Support Vector Machine
SWT	:	Scatter Wavelet Transform
TDOA	:	Time Difference of Arrival
T-F	:	Time-Frequency
TN	:	True Negatives
TOA	:	Time of Arrival
TP	:	True Positives
t-SNE	:	t-Distributed Stochastic Neighbor Embedding
UHF	:	Ultra-High Frequency
UMAP	:	Uniform Manifold Approximation and Projection
ViT	:	Vision Transformer
VMD	:	Variational Mode Decomposition
WD	:	Wavelet Decomposition

CHAPTER 1: INTRODUCTION

1.1 Background

Partial Discharge (PD) is a localized electrical discharge that partially bridges the insulation between conductors, due to imperfections in the insulation. Under high voltage stress, the electric field in the defect exceeds a critical threshold, causing a discharge. Over time, this damage degrades the insulation, causing void expansion and chemical breakdown, ultimately leading to complete insulation breakdown (Sharma & Bhanddakkar, 2015). The detection, and localization of PD have been under research since 1898 (Stone et al., 2022). Traditional methods of PD detection use voltage measuring devices across an impedance to detect a current spike in the device. This process is expensive and limited to devices electrically connected to the measuring instrument. Traditionally, according to IEC 60270, PD would be measured by supplying AC voltage, U , to a coupling capacitor, C_k in parallel with the test object, C_a to supply charge for the current spike. Using an impedance, z_{mi} to convert the current spike into output voltage, it is measured via a suitable coupling device (CD), connected via a connecting cable (CC) to a measuring instrument (MI).

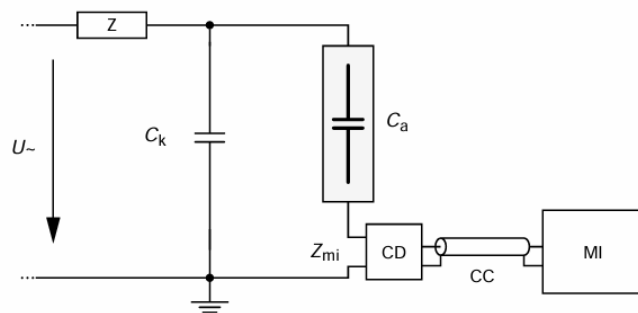


Figure 1.1: Traditional method of measuring PD occurrence (IEC60270:2025).

A high-frequency current transformer (HFCT) can also be used to obtain a voltage signal of the current spike. A newer technology is to use Ultra-High Frequency (UHF) sensors to detect electromagnetic (EM) waves emitted from the discharges. This method

exhibits strong anti-interference ability, high sensitivity, and allows for online detection (Yuwei et al., 2023).

Utilizing the time difference of arrival (TDOA) of the signals at different receivers, the PD location can be determined (Chan et al., 2025). Multiple optimization methods to localize PD from the TDOA have been developed.

The classic root finding algorithm to this problem is the Newton-Raphson method. It is a classic iterative numerical technique used to solve the non-linear system of equations derived from Time Difference of Arrival (TDOA) measurements. For a system with N sensors, the relationship between the PD source location (x, y, z) and the i -th sensor location (x_i, y_i, z_i) relative to a reference sensor (sensor 1) is given by:

$$c * \Delta t_{i,1} = R_i - R_1 \quad (1)$$

$$R_i = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} \quad (2)$$

Where c is the speed of light, $\Delta t_{i,1}$ is the TDOA between sensor i and sensor 1, and R_i is the distance from the source to sensor i . To solve for (x, y, z) , the algorithm minimizes the residual error vector $\mathbf{F}(\mathbf{X})$ iteratively:

$$\mathbf{X}_{k+1} = \mathbf{X}_k - \mathbf{J}^{-1}(\mathbf{X}_k)\mathbf{F}(\mathbf{X}_k) \quad (3)$$

Where $\mathbf{X}_k$ is the estimated coordinate vector at iteration k , and $\mathbf{J}$ is the Jacobian matrix of partial derivatives. While precise, this method requires a good initial guess to converge and is sensitive to local minima.

To overcome the sensitivity of Newton-Raphson, Particle Swarm Optimization (PSO) is often employed. PSO is a stochastic optimization algorithm inspired by social behaviour, such as bird flocking. In this algorithm, a "swarm" of particles moves through the search space. Each particle i has a position X_i (representing potential PD coordinates)

and a velocity V_i . The movement of each particle is updated based on its own best-known position (P_{best}) and the entire swarm's best-known position (G_{best}):

$$V_i^{k+1} = wV_i^k + c_1r_1(P_{best,i} - X_i^k) + c_2r_2(G_{best} - X_i^k) \quad (4)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1} \quad (5)$$

Where w is the inertia weight, c_1 and c_2 are acceleration coefficients, and r_1, r_2 are random values between 0 and 1. This global search capability makes PSO robust against the non-linear errors found in real-world environments.

For detecting multiple sources, the localization process extends beyond simple equation solving into a multi-stage workflow:

- i. **Signal Acquisition:** UHF sensors capture the continuous electromagnetic spectrum.
- ii. **Pulse Isolation:** An energy-based or amplitude-based trigger isolates transient pulses from the background noise floor.
- iii. **Feature Extraction:** Each isolated pulse is transformed into a feature vector (e.g., time-equivalent bandwidth, cumulative energy, or principal components) to reduce dimensionality while retaining signal characteristics.
- iv. **Clustering:** Unsupervised algorithms (e.g., DBSCAN, K-Means) group these feature vectors. Signals in the same cluster are assumed to originate from the same PD source.
- v. **Localization:** TDOA values are calculated specifically for these clusters, allowing for the independent localization of each unique PD source.

1.2 Problem Statement

In practice, there will always be multiple PD sources discharging repeatedly, overlapping, attenuating, refracting, reflecting and taking multiple paths as they travel through obstacles in a substation to reach the receiver. Signals often overlap, and higher amplitude signals can mask lower amplitude ones. In addition, PD often occurs within high voltage equipment, such as gas insulated switchgear (GIS) or transformers. Copper windings in transformers block the direct line-of-sight of UHF signals to the receiver, causing signals to take longer routes and distort the TDOA. Complex structures like GIS have different attenuation characteristics for each component within it, and are easily affected by signals from neighbouring HV devices (Yizhi Fang, 2023).

Separating multiple PD sources can be done if 4 PD antennae are used to collect the signals. By performing localization using time-difference-of-arrival (TDOA), different signals origins will yield different PD locations, effectively distinguishing the signal sets. However, the problem arises when there are less than four antennae, or two PD discharging very close to each other (He et al., 2020). In these scenarios, separation and clustering using the shape and features of the waveform are needed.

Beyond the issues of signal overlap and attenuation, the clustering of PD sources in real-world environments faces significant hurdles regarding feature robustness. In complex substations, the phenomenon of multipath propagation—where signals reflect off walls, transformers, and metallic casings—introduces "ghost" signals and distorts the pulse shape. This means that two pulses originating from the exact same PD source may look drastically different if they arrive at the sensor via different reflection paths. Consequently, standard feature extraction methods that rely heavily on pulse shape (such as Equivalent Time-Frequency bandwidth) may incorrectly classify these as different sources. Furthermore, the stochastic nature of Partial Discharge presents a challenge for

unsupervised learning. PD pulses from a single defect naturally vary in amplitude and rise time due to the physics of the discharge process. If a clustering algorithm is too sensitive, it may fragment a single source into multiple clusters. Conversely, if it is too lenient, it may merge distinct sources—such as a void discharge and a nearby corona discharge—into a single cluster, rendering the subsequent localization inaccurate. This necessitates the development of features that are invariant to propagation path distortions but sensitive to the fundamental physics of the discharge type.

Furthermore, obtaining real world PD data for development purposes poses many challenges. First, real world data would contain interference from locally transmitted radio, as well as signal attenuation (Upton et al., 2021). This would require signal denoising for proper development to be done. In addition, accurately emulating PD would require a lot of resources to fabricate the PD emulator and obtain HV test equipment. Even so, emulators may not represent a wide range of possible PD types that can occur. Obtaining access to a substation to perform testing also proves to be difficult, requiring permission from the relevant authorities. Not to mention, for the signals collected from actual defects in substation equipment, there would be no ground truth to know the exact PD type, for every collected signal. Therefore, a method to accurately simulate UHF signals obtained from PD sources in different environments needs to be developed, allowing for clustering algorithms to be tested preliminarily before deploying them to real world scenarios. In fact, this field has been at the forefront of PD research for many years (Loubani et al., 2019).

Another current issue is the fact that a vast number of research efforts in PD clustering and classification use only a limited number of PD types. The proposed clustering algorithms are only tested on the specific PD type and environment in which the signals were collected. A truly robust algorithm/model would be one that is trained on signals

from one domain but exhibits good performance when tested on data from a different domain. For example, a clustering model trained using only simulated data, but accurately clusters real-world, measured data.

1.3 Objectives

- i. **Digital Twin Development:** To develop a digital twin environment of EM wave propagation with multiple partial discharges (PD) using 3D EM wave solvers, incorporating white Gaussian noise to mimic real-world conditions.
- ii. **Experimental Verification:** To obtain real-world measured data of PD waveforms and verify the simulation's accuracy by ensuring a model trained on simulated datasets can achieve >90% accuracy on measured datasets. This is termed “**cross-domain validation**”
- iii. **Algorithmic Innovation:** To develop and implement a novel clustering method (e.g., a hybrid approach) that demonstrates a quantifiable improvement in computational efficiency and accuracy from existing methods in literature.
- iv. **Localization Evaluation:** To perform localization on the obtained model clusters, to determine the location of the PD. Predicted PD location must be less than 1 m from the actual coordinates.

CHAPTER 2: LITERATURE REVIEW

2.1 PD Localization

There are several methods to localize a PD source from its emitted UHF waveforms. Time difference of arrival (TDOA) utilizes the difference in time between arriving signals at different receivers to estimate the distance of the PD from each of them. The exact moment of signal arrival is usually determined from the first peak, cumulative energy or cross-correlation plot. A set of equations that represent the relationship between TDOA and PD location is solved using optimization algorithms, such as Newton-Raphson, particle swarm optimization (PSO) and ant colony optimization (Chan et al., 2025). PSO is a classic optimization method that emulates a group of particles moving with different velocities and trying to find the global minimum of an objective function. The direction of arrival (DOA) method uses phased antenna arrays to determine which direction the PD is coming from, while the received signal strength (RSS) method uses the principle that the strength of received signal corresponds to the distance travelled by signal (Chen, 2019) (Upton et al., 2021).

In recent years, the use of advanced localization techniques for UHF based PD signals has been increasingly common. The study by Cao et al. (2023) introduces a hybrid localization method that combines time-difference-of-arrival (TDOA) and angle-of-arrival (AOA) measurements, optimized using a crow search algorithm (CSA) enhanced by particle swarm optimization (PSO). The approach addresses nonlinear equation-solving challenges in non-line-of-sight (NLoS) environments by modifying the maximum likelihood estimation fitness function and inserting initial solutions into the population to improve convergence and accuracy. Simulation experiments demonstrated that the PSO-CSA algorithm consistently outperformed other methods and showed resilience to increasing measurement errors and larger cell radii, maintaining smoother error curves and higher stability than competing methods. The PSO-CSA achieved a 3D localization

error as low as 16.64 m, compared to 130.5 m for Taylor's method, 122.9 m for Chan's algorithm, 123.9 m for hybrid TDOA/AOA, 101.8 m for PSO, 29.23 m for chaotic PSO, and 19.95 m for chaotic CSA.

Next, another study (Dhara et al., 2023) proposes a hybrid localization method that combines TDOA with relative RSS to improve accuracy in detecting PD sources within air-insulated substations (AIS). The two components are combined and interpreted via weighted least-squares optimization to obtain PD location. It was shown that the proposed method produces a 7.17% localization error, compared to pure TDOA and relative RSS methods with 11.49% and 14.4% respectively when tested under laboratory conditions. In field testing, the results were 7.92%, 11.4% and 22.36% respectively.

Another study (Lee et al., 2024) introduces a compressive spherical frequency-difference beamforming technique for localizing partial electrical discharges (PDs) in high-voltage systems, using ultrasonic emissions captured by acoustic sensor arrays. A two-step normalization process was implemented to eliminate bias from nonplanar waves, preserving only phase and relative amplitude information, while a distributed convex optimization algorithm based on the alternating direction multiplier method (ADMM) solved the localization problem efficiently. Through simulation and experimental validation, the near-field localization of PD sources was validated, with the method achieving robust performance even under limited sensor conditions.

2.2 Multiple PD Separation/Clustering

Table 2.1: Comparison of existing clustering methods in literature.

<i>Reference</i>	<i>Core Methodology</i>	<i>Primary Result</i>
<i>Hao et al. (2011)</i>	Wavelet decomposition (WD) + PCA + DBSCAN	Successfully clustered PD sources
<i>Chan et al. (2015), Wang et al., Nimmo et al., Zhu et al.</i>	Waveform envelopes & time-frequency sparsity maps	Separated non-overlapped signals
<i>Tang et al. (2009)</i>	Blind source separation (BSS)	Disentangled mixed overlapping signals
<i>Yang & Judd (2003)</i>	Wavelet analysis on UHF EM waveforms	Separated modelled and actual PD signals
<i>Pinpart & Judd (2010)</i>	Envelope comparison on UHF EM waveforms	Clustered same-source PDs across different locations
<i>Almehdhar & Prochazka (2025)</i>	Time-frequency analysis (SWT/HHT) + Deep Learning	100% experimental separation accuracy
<i>Carvalho et al. (2024)</i>	ML (K-means, GMM, SVM) on statistical features	90%–93% clustering accuracy
<i>Fu et al. (2024)</i>	t-SNE on wavelet time–frequency spectra	100% accuracy, but highly computationally expensive
<i>Z. Chen et al. (2025)</i>	Adaptive filtering + t-SNE + DBSCAN	Separated 4 discharge types in realistic conditions
<i>Chang et al. (2022)</i>	CNNs on pulse sequence data	Improved accuracy via spatial-temporal features
<i>Jiang et al. (2023)</i>	Stepwise transfer learning with CNNs	High accuracy despite limited training samples
<i>Zhang et al. (2022)</i>	Fabry-Perot array + AOK + ResNet-18	98% accuracy across 5 PD types
<i>Rauscher et al. (2024)</i>	2D CNN + continuous wavelet transform	99.75% accuracy for PD vs. noise classification
<i>L. Chen et al. (2025)</i>	Adaptive VMD + Hybrid CNN-LSTM	96.2% accuracy with near real-time processing
<i>Ding et al. (2025)</i>	Reinforcement learning-guided CNN compression	85%–96% parameter reduction with <1% accuracy loss

<i>Reference</i>	<i>Core Methodology</i>	<i>Primary Result</i>
<i>Fu et al. (2025)</i>	CNN + BiLSTM + Attention + GW-GAN augmentation	98.8% overall recognition accuracy
<i>Han et al. (2021)</i>	Spectrogram + MMFCC features + GRU	96.7%–100% recognition accuracy
<i>Hong et al. (2024)</i>	Compared SVM, CNN, and RNN on GIS datasets	87%–100% accuracy; RNN was sensitive to conditions
<i>Das et al. (2022)</i>	VGGNet16 deep feature extraction + BiLSTM	97%–99% accuracy with reduced computational load
<i>Jie et al. (2023)</i>	Dilated Residual Network (DRN) + SVM fusion	89.64%–98.33% classification accuracy
<i>Jing et al. (2022)</i>	Stacked Autoencoder (SAE) + Transfer learning	87.13% accuracy utilizing mostly unlabelled data

When multiple PD sources are repeatedly discharging and emitting signals, it becomes difficult to differentiate each received waveform. Multiple methods to separate and cluster signals obtained from the voltage output waveform obtained from IEC 60270/HFCT measurements have been developed. Typically, features are extracted from the signals in the time, frequency or time-frequency domain, and a clustering algorithm is then applied. The signals arriving at a sensor can either be overlapped or non-overlapped. The non-overlapped case is simpler. One example of a developed method for that case is wavelet decomposition (WD) + principal component analysis (PCA) + density-based spatial clustering of applications with noise (DBSCAN). WD passes the signal through a series of complementary high pass and low pass filters, to decompose it into detail and approximate coefficients. The signal can then be represented by scaled and shifted versions of the Mother Wavelet, a basis function defined by the user. PCA reduces the dimensionality of the sources to 3 and DBSCAN uses density to form clusters (Hao et al., 2011). They managed to successfully cluster different PD sources when testing using the IEC 60270 method. Another method is to use simple geometrical structures to

form envelopes of the waveform and create a time-frequency sparsity map (J. C. Chan et al., 2015). Similar methods can be found in (Wang et al., 2015), (Nimmo et al., 2017) and (Zhu et al., 2017). The overlapping case is more complex and requires blind source separation (BSS), a process to disentangle mixed signals (Tang et al., 2009). Research has shown that wavelet analysis and envelope comparison can also be used to separate sources from UHF EM waveforms (Yang & Judd, 2003) (Pinpart & Judd, 2010). Envelope analysis by Pinpart & Judd (2010) managed to achieve margins between 1.7:1 and 5:1, successfully clustering sources from the same source at different locations, using multiple sensors. However, no further localization was done. The wavelet analysis by Yang & Judd (2003) revealed that it was able to separate signals from modelled PD sources with a similarity value of $S < 0.12$ and $S > 0.41$, and from actual PD in a transformer with $S < 0.06$ and $S > 0.73$.

Machine learning has also been explored and applied to the field of PD clustering and separation. Applications of time-frequency analysis methods, like scatter wavelet transform (SWT) and Hilbert-Huang transform (HHT) to extract PD features, and deep learning (DL) models to cluster them are detailed by Almehdhar & Prochazka (2025). They achieved 100% experimental accuracy in PD separation. Carvalho et al. (2024) experimented with machine learning (ML) models (K-means, gaussian mixture model, mean-shift, support vector machine, SVM) to cluster UHF PD signals according to features such as crest factor, skewness and kurtosis, achieving 90% to 93% accuracy for the different methods.

Among some of the more recent and advanced methods of PD clustering, Fu et al. (2024) proposes a novel approach to overcome the poor applicability of traditional PD features across sensors with different amplitude–frequency characteristics by directly extracting features from wavelet time–frequency (T–F) spectra using the t-distributed

stochastic neighbor embedding (t-SNE) algorithm. By aligning waveforms and applying wavelet transforms, the dimensionality was reduced with t-SNE and the perplexity parameter was optimized using a root mean square error (RMSE)-based method. Verification across sensors with different bandwidths was done. Compared to PCA, isometric mapping, and traditional feature extraction (which used the best 2 out of 38 common PD features), the t-SNE method achieved superior separation accuracy, with clustering accuracy of 100%, although at 10 times the computational cost of the next best method.

Z. Chen et al. (2025) proposes an integrated framework for interference suppression and signal separation of multi-source partial discharge (PD) signals in HVDC gas-insulated lines (GILs), addressing the challenge of mixed PD sources and overlapping noise. Narrowband noise was suppressed in the frequency domain using a moving median threshold with Akima cubic interpolation, while white noise was reduced via a sign-error least mean square (SELMS) adaptive filter, which achieved an SNR improvement to 8.14 dB with RMSE = 0.0509. From 240 signals, eight features across time, frequency, and time–frequency domains were extracted, standardized, reduced via t-SNE, and clustered using DBSCAN, successfully separating all four discharge types without prior knowledge of source number. Under near-realistic experimental conditions, the algorithm correctly clustered the PD signals when ≥ 160 signals were used.

Beyond traditional statistical analysis, recent advancements have increasingly leveraged Deep Learning (DL) for the separation and clustering of PD signals. While manual feature engineering relies on calculating specific metrics like skewness and kurtosis, DL models, particularly Convolutional Neural Networks (CNNs), are designed to process grid-like topology data such as time-series waveforms or Phase-Resolved Partial Discharge (PRPD) patterns. By utilizing learnable filters that scan input data,

CNNs can detect local patterns like sharp rise times or oscillatory decays that are characteristic of specific PD types. For instance, Chang et al. (2022) demonstrated that applying CNNs to pulse sequence data significantly improved pattern recognition accuracy in underground cable joints by effectively capturing spatial-temporal features that traditional methods often miss. Similarly, Jiang et al. (2023) utilized a stepwise transfer learning approach with CNNs to enhance PD pattern recognition in generator stators, achieving high accuracy even with limited training samples. This shift towards automated feature extraction represents a significant evolution in PD diagnostics, reducing the reliance on expert domain knowledge for signal processing.

Another study (Zhang et al., 2022) presents a partial discharge (PD) pattern recognition method for oil-immersed transformers using a multifrequency fiber-optic Fabry–Perot (F–P) ultrasonic sensing array, adaptive optimal-kernel (AOK) time–frequency representation, and a deep learning framework (ResNet-18). Three F–P probes with resonant frequencies at 41, 75, and 112 kHz were deployed to capture PD signals across different bands, improving sensitivity and interference immunity compared to conventional piezoelectric sensors. The AOK method was used to generate high-resolution time–frequency matrices (300×375) that highlighted attenuation effects, especially for high-frequency components blocked by solid insulation. A database of 1,100 samples (220 per PD type) was used to train the modified ResNet-18, achieving 98% recognition accuracy across five PD types, outperforming piezoelectric sensor-based detection (92.9%). Confidence analysis showed probabilities >0.99 for most known PD types, though signals attenuated by pressboards and windings had lower diagnostic confidence, while unknown PD or noise signals produced dispersed or low confidence predictions.

The study by Rauscher et al. (2024) introduces a systematic framework for improving PD versus noise classification in end-of-line stator testing under surge voltage. Six networks: fully connected neural networks (FCNNs), 1D/2D convoluted neural networks (CNNs), long-short term memory (LSTM), bidirectional LSTM (BiLSTM), and gated recurrent unit (GRU), were combined with different time-frequency transforms and were optimized via hyperparameter tuning. Results showed that a 2D CNN with continuous wavelet transform achieved the highest accuracy of 99.75% on unseen test objects, with detection rates above 95% even for PD signals near the noise floor. Receiver operating characteristic analysis confirmed a true positive rate >99.5% and false positive rate <0.14%, outperforming conventional threshold-based detection. The concepts of PD versus noise separation detailed here can be applied to PD-PD separation.

L. Chen et al. (2025) proposes a method to classify PD acoustic emissions. First, Adaptive Variational Mode Decomposition (VMD) that uses sample entropy optimization, processes the signal by automatically selecting the number of modes and penalty factors. Next, a hybrid deep learning model (CNN + LSTM) is trained. It consists of a 1D-CNN branch that extracts spatial features (pulse shapes, high-frequency discharge signatures), and a Bidirectional LSTM branch that captures sequential dependencies in discharge activities. Together, they learn both transient discharge morphologies and their evolutionary patterns. The method achieved 96.2% classification accuracy for multi-source defects ($\approx 38\%$ better than wavelet methods), 5.8 mm mean localization error ($\approx 53\%$ better than traditional TDOA) and 4.2° angular accuracy even under high noise. The processing time was ~ 0.8 s, making it practical for near real-time monitoring.

Another study by Ding et al. (2025) compresses CNN models for transformer partial discharge (PD) diagnosis using reinforcement learning-guided filter pruning via geometric median, achieving 85–96% parameter reduction with accuracy losses under

1%. Deployment tests on Raspberry Pi 4B and Jetson Nano confirmed real-time feasibility, with power consumption reduced by $\sim 40\%$ and memory usage halved. While the paper does not address PD source separation directly, its results show that compressed CNNs can still classify UHF PD signals reliably.

Fu et al. (2025) propose a transformer partial discharge (PD) recognition framework that integrates convolutional neural networks (CNN), bidirectional long short-term memory (BiLSTM), and an attention mechanism, enhanced with a gradient-penalized Wasserstein GAN (GW-GAN) for data augmentation. Using wavelet time–frequency spectrograms of PD signals, CNN extracts local features, BiLSTM captures bidirectional temporal dependencies, and the attention mechanism adaptively weights features to improve classification. To address data imbalance, GW-GAN generated realistic synthetic samples, boosting generalization. Results showed that the CNN-BiLSTM-attention-GAN (CBLAG) model achieved a recognition accuracy of 97.36%, outperforming CNN+CGAN by 1.8% and traditional SVM by 5.8%. The CBLAG model reached the highest overall accuracy (98.8%) among six tested methods, while baseline models like BPNN lagged at 64.8%.

A study by Han et al. (2021) attempts to differentiate four different PD models (column-plate, particle, point, and air-gap) using time-frequency characteristics. First, a spectrogram of the UHF signal is generated, followed by feature extraction using modified Mel-frequency cepstrum coefficients (MMFCC). Next, the extracted features are analyzed using recurrent neural networks (RNNs). The paper compares the performance of standard RNNs, LSTMs, GRUs, SVMs, and kNNs. The results demonstrate that while feeding raw time signals directly into RNNs causes non-convergence and low accuracy, the proposed MMFCC-GRU method achieves a recognition accuracy between 96.7% and 100%.

Research by Hong et al. (2024) presents a comparative analysis of machine learning and deep learning approaches for recognizing four types of PD (corona, surface, particle, and floating) using a UHF sensor in gas-insulated switchgear (GIS). The authors evaluated three distinct pipelines: statistical features paired with an SVM classifier, high-dimensional phase-resolved partial discharge (PRPD) matrix data processed by a CNN, and time-series data analyzed by an RNN. Tested across 400 datasets with optimized hyperparameters, the best models yielded accuracies ranging from 87% to 100%. The study also noted that while altering gas types or testing periods did not significantly impact results, changing the sensor type, pressure, or testing location substantially degraded the RNN's performance.

Addressing the computational expenses of deep learning and the struggles of conventional time-of-arrival (TOA) methods with multiple PD events, Das et al. (2022) proposes a deep transfer learning framework for localization and identification. Using acoustic sensors in transformer oil, the method first extracts time-frequency features via the Smoothed Pseudo Wigner-Ville distribution. Next, a pre-trained VGGNet16 model extracts deep features, which are then fused using average feature fusion and fed into a BiLSTM network to predict the location of single and multiple PD sources. Evaluated on 47 PD events, the framework achieved 97% to 99% across accuracy, sensitivity, and specificity metrics, slightly outperforming standard STFT and CWT methods while significantly reducing the training computational load.

To overcome the limitations of standard signal decomposition techniques and single-model architectures, Jie et al. (2023) introduces a hybrid fusion model for recognizing four types of GIS defects. The researchers constructed an augmented dataset of 4,000 images by adjusting the brightness and chroma of 240 experimental PRPD plots. A Dilated Residual Network (DRN) using a softmax classifier and cross-entropy loss was

developed as the primary architecture. Concurrently, statistical features were extracted from the signals using wavelet transforms and classified via an SVM. The study found that combining these approaches into a DRN-SVM fusion model outperformed the standalone DRN, delivering classification accuracies between 89.64% and 98.33%.

Finally, a study by Jing et al. (2022) addresses the research gap of insufficient labeled data by utilizing a Stacked Autoencoder (SAE) combined with transfer learning. The proposed architecture features an SAE trained in an unsupervised manner to extract features, with a top-connected soft-max regression classifier. Network parameters are then fine-tuned using a small subset of labeled data via stochastic gradient descent and backpropagation. By transferring the learned parameters from a 1,000-sample original dataset to a 400-sample target dataset (50% labelled to further train the model) with similar but distinct fault conditions (free metal particle, metal tip, floating electrode, and insulation void defects), the model successfully leveraged unlabelled data to achieve an 87.13% recognition accuracy.

2.3 Digital Twin Creation

Most attempts at creating a digital twin of PD occurrences represent partial discharge using a gaussian pulse of varying physical lengths, amplitudes and pulse widths, in software such as XFDTD, Ansoft HFSS, or CST Microwave Studio. The software can numerically solve the E and H fields surrounding the PD source using the principles of finite-difference time domain (FDTD) or finite integration technique (FIT). Other attempts to model PD use equations such as single exponentially damping oscillation (SEDO), double exponentially damping oscillation (DEDO), double exponential pulse (DEP), and damped oscillatory pulse (DOP). A comparison of previous works to simulate PD is given in Table 2.2 and Table 2.3.

Table 2.2: Comparison of PD simulation techniques using software.

Reference	PD simulation technique	Purpose	What the simulation tries to represent	Findings (related to simulation)
(He et al., 2020)	Gaussian current impulse in Ansoft HFSS, with added white noise.	Separating multisource PD to localize.	UHF PD signals	Good agreement between simulation and measurement results
(Loubani et al., 2019)	Gaussian current impulse in XFDTD	Analyzing propagation characteristics of PD.		-
(Qin et al., 2020)		Mixing characteristics of PD signals in GIS		Verified PD mixing characteristics
(Tian & Kawada, 2014)		Estimating DOA of PD		-
(Zheng et al., 2023)	Dielectric Barrier Discharge model + FDTD	EM wave transmission in GIS.		-
(Umemoto & Tenbohlen, 2018)	CST to model EM wave propagation of antenna emitted signal.	Comparing simulated and measured PD signal propagation in transformer		Similarly shaped simulated and measured signals.
(Toshihiro et al., 2008)	Gaussian current impulse in FDTD	Comparing simulated and measured propagation of EM waves in GIS		-
(Ohtsuka et al., 2006)	Geometrically shaped current impulse in FDTD	Estimating PD charge quantity from signals detected in GIS		-
(Jaber et al., 2016)	Gaussian current impulse in CST	Compare PD emulator pulse with simulated.		Good agreement between simulated and measured signals
(Zanjani et al., 2012)		Analyze EM wave propagation in distribution transformer.		

Table 2.3: Comparison of PD simulation techniques using equations.

Reference	PD simulation technique	Purpose	What the simulation tries to represent	Findings (related to simulation)
(Lin et al., 2022)	DEDO & SEDO + Gaussian white noise	Denoising	Acoustic PD emissions	-
(Ma et al., 2002)	DEP, DOP	Exploration of wavelet analysis on PD detection	Current in the detection circuit caused by PD	-
(Sahnoune et al., 2024)	DEP, DOP	Denoising	Current in the detection circuit caused by PD	-
(Kumar et al., 2023)	Mixture of SEDO + DEP	Denoising	Acoustic PD emissions	-
(Friebe & Jenau, 2024)	Mixture of SEDO + DEP	Denoising	UHF PD signals	Simulated waveform made to closely resemble measured signals

Zheng et al. (2023) and Ohtsuka et al. (2006) were the only ones that didn't use Gaussian current pulses to model PD. Most of the simulations were done with the purpose of validating the accuracy of single simulated PD waveforms compared to experimentally measured waveforms. Only (Yun et al., 2020) and (He et al., 2020) investigated the simulation of multiple PD sources for the purpose of applying separation/clustering algorithms to them. Research regarding PD simulation using equations like SEDO, DEDO, DEP and DOP are not specifically done for PD clustering either. In fact, some of them do not even focus on modelling UHF emissions, but rather acoustic waves or electric current pulses. A clear gap exists in this area, because more research is needed to confirm the eligibility of testing separation algorithms on simulated PD waveforms. Verification needs to be done on whether a simulation would be capable of producing different types of waveforms that accurately resemble experimentally collected samples, with extractable features that are unique to each different simulated PD.

2.4 Research Gap

Despite the extensive literature on PD detection, significant gaps remain in the development of high-quality Digital Twin environments for multiple-source scenarios. While previous studies have successfully simulated single PD sources using Gaussian pulses, there is a lack of comprehensive research that models the simultaneous propagation of multiple distinct PD types (e.g., void, surface, etc) within a 3D space, for the specific purpose of clustering. For clustering, the PD pulse shape is important, as simulating the wrong pulse shape would lead to the model extracting wrong features from them. Current research that simulates PD for other purposes, i.e. TDOA estimation, propagation characteristics, denoising, do not place importance on how accurate the shape of the pulse is, because the shape is redundant for those applications.

Furthermore, current research efforts typically evaluate their models on isolated, single-domain datasets. A predefined set of PD defects is either simulated or tested in a controlled physical environment, and the dataset is simply split into standard training, validation, and testing subsets. Consequently, the constructed models learn to classify and cluster PD by overfitting to the specific environmental artifacts of that single laboratory or digital twin. For this field to advance, models must demonstrate cross-domain adaptability. A model trained on a specific, fully labeled dataset with a constrained number of PD types must be capable of deploying and clustering real-world signals seamlessly—especially given the absence of empirical ground truth available in the field for labeling and fine-tuning.

CHAPTER 3: METHODOLOGY

3.1 General Flow of Methodology

3.1.1 Model Training Pipeline

The methodology of this research distinguishes between the training and testing pipelines. For the **Model Training Pipeline**, the process begins with **Data Acquisition** from four distinct sources: 3D electromagnetic solver (HFSS & MATLAB), equation-based waveform mixing, the CWRU Bearing Dataset (to prevent overfitting), and a subset of real-world experimental measurements. During the **Preprocessing** phase, a specialized waveform isolation algorithm is applied *exclusively* to the measured dataset to extract transient PD pulses from the ambient noise floor, while the other three datasets bypass this step. In the **Feature Extraction** phase, all acquired 1D time-domain signals are transformed into robust 2D representations utilizing either Short-Time Fourier Transform (STFT) or Signal Bispectra techniques. Finally, the pipeline concludes with the **Model Training** phase, where these extracted features are used to train and optimize advanced deep learning architectures, specifically Convolutional Neural Networks (CNNs) and Vision Transformers (ViT).

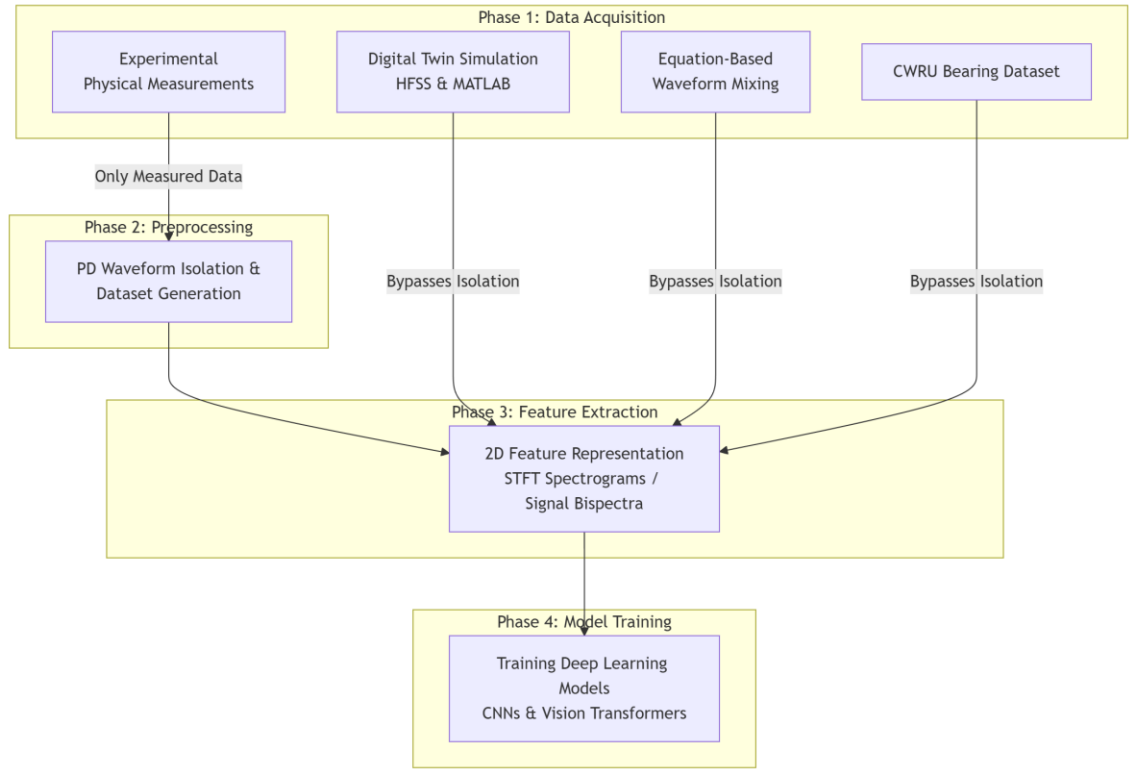


Figure 3.1: Model training pipeline.

3.1.2 Model Testing & Localization Pipeline

For the **Model Testing (Inference) and Localization**, the methodology evaluates the trained networks on entirely unseen data. The process begins with **Data Acquisition** restricted exclusively to the experimental physical measurements of the incision (PD2) and delamination (PD3) defects. In the **Preprocessing** phase, the raw signals undergo the waveform isolation algorithm to extract the transient PD pulses from the ambient noise. These pulses are then converted into 2D representations during **Feature Extraction**. Next, the trained deep learning models (CNNs or ViTs) are used for **Inference** to extract 128-dimensional feature embeddings without further weight updates. These embeddings undergo optional **Dimensionality Reduction** (t-SNE or UMAP) before being grouped by the HDBSCAN algorithm during the **Clustering** phase. Finally, the pipeline proceeds to **Localization & Evaluation**, where pulses from the same predicted instance are matched across channels, their Time Difference of Arrival (TDOA) is computed via cross-correlation, and Particle Swarm Optimization (PSO) calculates the physical coordinates for evaluation.

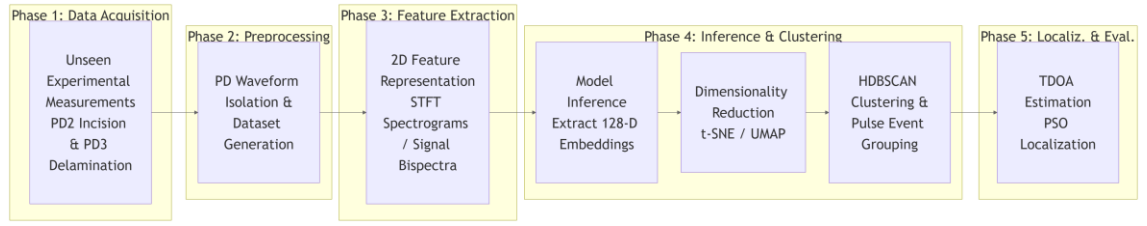


Figure 3.2: Model testing and localization pipeline.

3.2 Digital Twin Simulation Method

A digital twin environment would be created to generate data for clustering analysis. Two different methods of digitally modelling a PD occurrence will be explored. The first is to use 3D EM software to model the antennae and PD sources, and then obtain their transfer characteristics. The signals are then obtained by injecting Gaussian current pulses into the system by convoluting the transfer characteristics with current pulses in MATLAB. The second is to utilize a linear combination of equations with normally distributed parameters to model the received UHF signals.

3.2.1 3D EM Modelling + Gaussian Pulse Injection

3.2.1.1 Ansoft HFSS Simulation

The first method utilizes Ansoft HFSS 13.0 (He et al., 2020) which utilizes the Finite Element Method to model the transfer characteristics between PD and antennae. The partial discharge sources are modelled as dipole antennae with a lumped port. This is because, in the real world, the receivers who are far away from the PD will ‘see’ the radiation pattern that resembles a dipole’s. Many scholars also use this model to predict the radiation characteristics of PD (Yan et al., 2019). The sensors are planar monopole antennae (PMA). The simulation was designed to be a digital twin of the real-world experimental setup in Section 3.4.

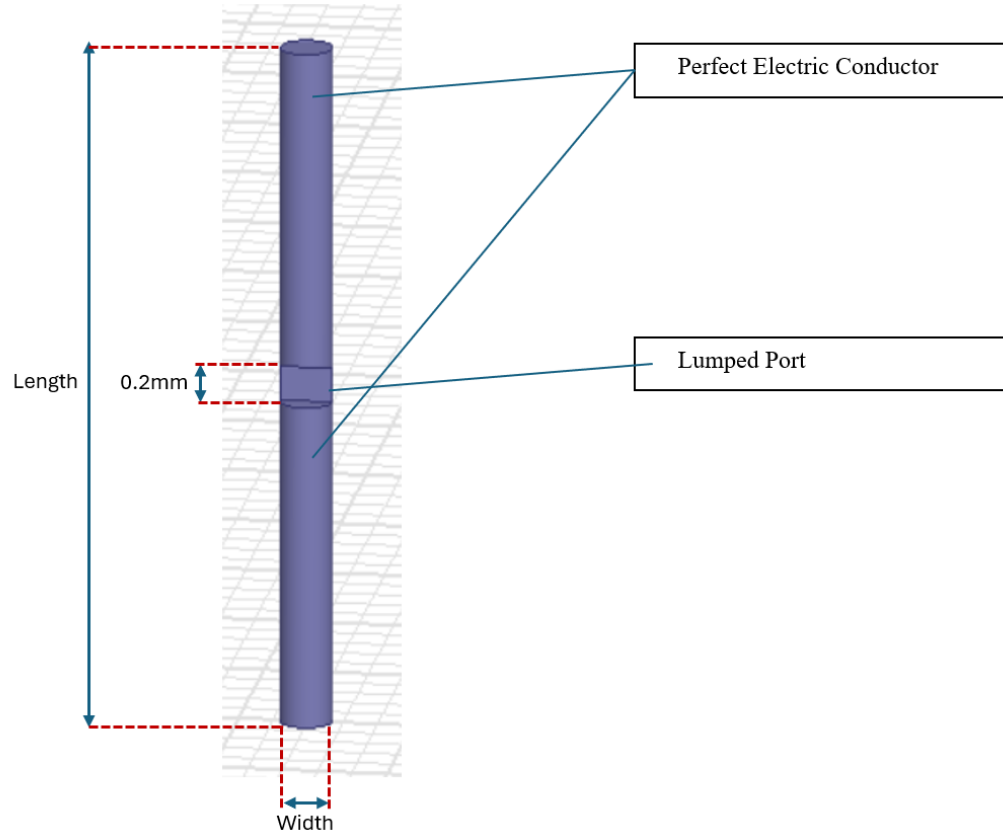


Figure 3.3: Shape and materials assigned to the PD model (dipole antenna).

The dipole antenna's length and width are modelled after the estimated size of the arc. For example, if the size of the void in an insulator is 4mm, it can be said that an electric arc of length 4mm will form during a PD event. Thus, the length of the dipole antenna is 4mm. For the width, according to Qi et al. (2018), the width of a streamer discharge is 0.3mm. In the simulation, 2 PD sources are modelled after real world PD emulators: void defect (4mm length, 0.4mm width) and incision defect (6mm length, 0.4mm width). The emulators are developed by prior research effort (Madhuranga et al., 2025) and will be used to obtain real-world measurements for analysis too. 0.4mm is used as the width, as the solver faced problems computing 0.3mm.

The antennae are modelled based on the work by Kayser Azam et al. (2024), which would also be used in the experiment to obtain real-world measurements.

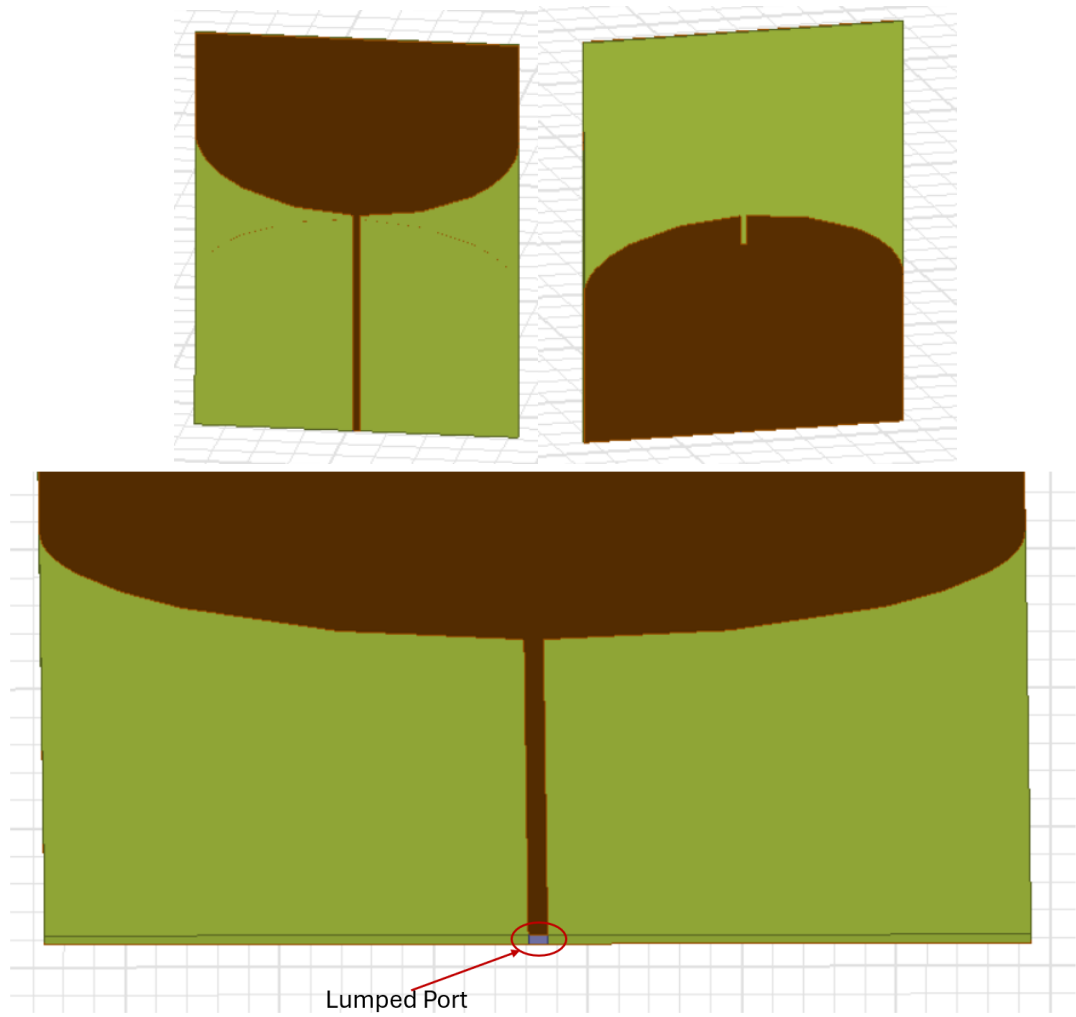


Figure 3.4: Antenna model: front (top left), back (top right) and lumped port position (bottom).

Table 3.1: Location of PD and antennae in the digital twin.

PD/Antenna	Coordinates (m)
PD1 (Void defect)	3.7, 5, 0.1
PD2 (Incision defect)	3.7, 7, 0.1
Antenna 1 (S1)	1, 3, 1.4
Antenna 2 (S2)	1, 5, 1.8
Antenna 3 (S3)	1, 7, 1.4
Antenna 4 (S4)	1, 9, 1.8

The solution type was set to ‘Driven Terminal’, which simulates the transfer characteristics of voltage and current. The lumped ports on the antennae were given 50-ohm resistance. The faces of the air box surrounding the components was assigned radiation boundaries on the top and sides, and a perfect E boundary on the bottom. Some other simulation parameters are given in Table 3.2.

Table 3.2: Simulation parameters for Ansoft HFSS 13.0.

Parameter	Value
Adaptive solution: maximum number of passes	50
Adaptive solution: maximum delta S	0.03
Solution frequency	1GHz
Lambda refinement	Enabled (Default lambda target)
Order of basis functions	Mixed order

Default values were chosen for the other simulation parameters and settings. The z-parameters are obtained from the simulation, and MATLAB was used to inject current waveforms into the inputs and obtain the output voltage waveforms from the PMAs.

The current injected follows Gaussian waveforms, as shown below.

$$I(t) = I_{max}e^{-t^2/2\sigma^2} \quad (6)$$

Where I_{max} is maximum current amplitude, σ is the pulse width (Loubani et al., 2019).

3.2.1.2 MATLAB Current Pulse Injection

After simulation in Ansoft HFSS, touchstone files containing the Z-parameters of the model are exported. MATLAB is then used to inject gaussian impulse currents of varying time widths and amplitudes.

Step 1: Noise Calculation: Before current injection, the required noise amplitude is first calculated. The script performs a quick mathematical "dry run." It pushes a theoretical, average-sized Partial Discharge (PD) pulse through the Z-parameters and measures its peak voltage. Based on that peak, it calculates an absolute noise floor at the requested SNR. In this project, the SNR value was 20dB.

Step 2: Current Injection: Depending on the defect type (PD1 or PD2), the current pulse is randomly assigned a specific amplitude and time-width.

Table 3.3: Parameters of Current Impulse for PD1 and 2.

PD	PD1 (Void Defect)	PD2 (Incision Defect)
Amplitude (A)	$A_{PD1} \sim N(0.9, 0.2)$	$A_{PD2} \sim N(1.4, 0.2)$
Time Width (ns)	$\tau_{PD1} \sim N(0.9, 0.1)$	$\tau_{PD2} \sim N(1.1, 0.1)$

The current pulse is then injected into the system (represented by Z-parameters) to obtain the output waveform, as detected by the modelled UHF antennae (sensor). Output waveforms for all combinations of PD type and antennae number are obtained. The script then looks at the 3D coordinates of the 2 defects and the 4 sensors. It calculates the TOA to each sensor based on the coordinates, discharge time and speed of light.

Step 3: Sensor Reading Construction: The script then constructs the final output signals (the sensor ‘readings’). For each channel (4 channels, representing 4 sensors/antennae), it arranges the output signals from Step 2 at different locations in the time domain, based on the calculated TOA and PD discharge time (it assumes that the two PD discharge multiple times randomly). At the end, 4 long signal outputs (one for each antenna), consisting of many PD pulses scattered in the time domain, are obtained. This represents the real-world scenario where the four sensors continuously detect PD signals at random intervals of time, outputting four continuous streams of data. Finally, the script adds Gaussian noise ($\text{SNR}=20\text{dB}$) and FM radio interference (at $f = 100\text{MHz}$, $\text{amplitude} = 0.05 \times \max(\text{PD signal value})$).

Step 4: Labeling: Labels are generated for deep learning model training. Each PD discharge instance is given a unique label number. The bounding boxes of the PD signals, TOA, and PD type are also labelled.

Step 5: Exporting the Signals: The simulated “S1 – S4” output aren’t exported as one continuous stream, but rather in batches of $5\mu\text{s}$ (‘Scenes’). This is to allow them to be shuffled in random order during model training. A group of Scenes are then packaged into one HDF5 file (Shard), and the dataset consists of multiple Shards.

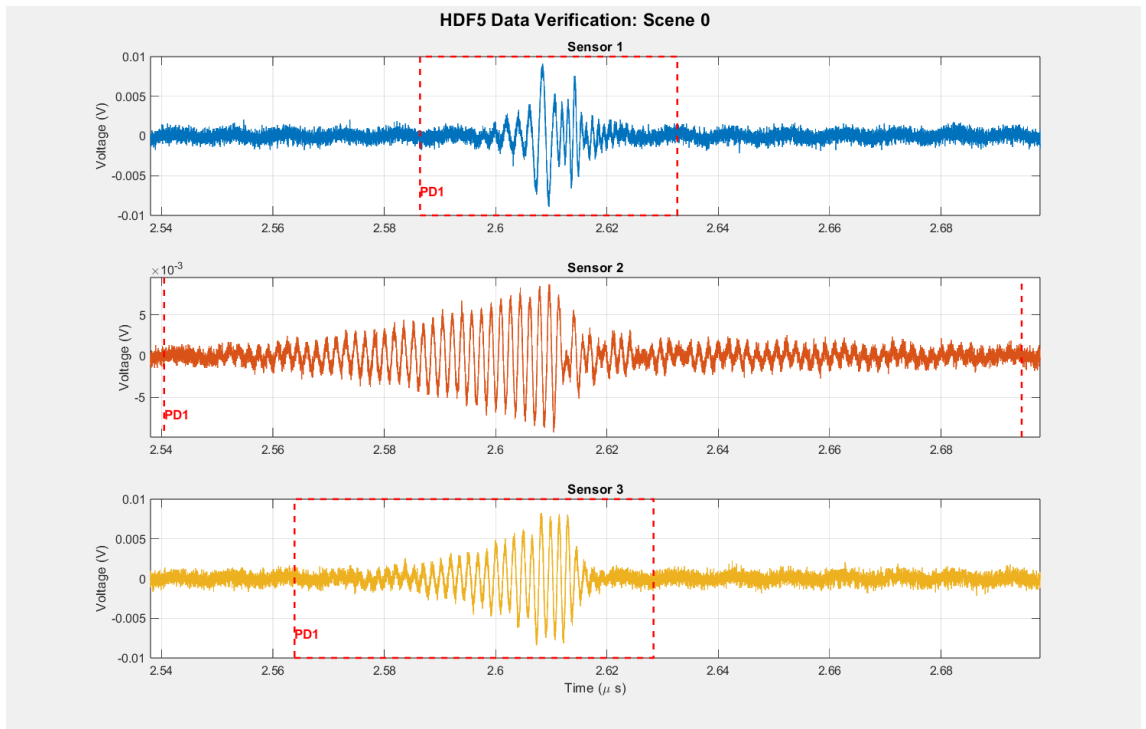


Figure 3.5: Simulated waveforms using HFSS and MATLAB.

3.2.2 Waveform Mixing

The second method is to simulate PD emitted EM waves as a combination of multiple basic waveforms. Different types of PD contain superpositions of different base waveforms, of various time delays and amplitudes. The waves are assumed to emit from a specified point in space and attenuate based on how far they travel to each receiver. Gaussian noise is added to the received signal to model real-world conditions. Although this method was originally meant to model transient current pulses in an IEC60270 setup rather than UHF signals, it was used to introduce variations in the dataset to prevent the model from overfitting to the EM modeler + current pulse injected waveforms.

Several base waveforms used in the modelling are given below:

1. Single Exponentially Damping Oscillation (SEDO)

$$V_0 e^{-\frac{(t-t_d)}{\tau}} \sin(2\pi f_c(t - t_d) + \theta) \quad (7)$$

Where V_0 is peak amplitude, t_d is time delay, τ is decay time constant, f_c is center frequency of oscillation, θ phase offset. The values of the signal for $t - t_d < 0$ are forced to zero, and this is done for all subsequent signals. This is a modified version of the equation used in (Lin et al., 2022).

2. Double Exponentially Damping Oscillation (DEDO)

$$V_0(e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)}) \sin(2\pi f_c(t - t_d) + \theta) \quad (8)$$

Where α and β control the rise and fall times, ($\alpha < \beta$) (Lin et al., 2022). This model better represents what a PD EM wave would look like, containing a voltage rise and fall, with oscillations.

3. Double Exponential Pulse (DEP)

$$V_0(e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)}) \quad (9)$$

This is simply a DEDO without the oscillations, as introduced and used in (Ma et al., 2002).

4. Sinusoidally Modulated Gaussian (SMG)

$$A e^{-\frac{((t-t_0-t_d)^2)}{2(\tau^2)}} \sin(2\pi f_c(t - t_0 - t_d)) \quad (10)$$

A is the amplitude, t_0 is the pulse center. This signal has an equal rise and fall time, and is symmetrical about t_0 (Miljanovic et al., 2021).

Six different PD sources were simulated with Python, with the parameters as shown in Table 3.4. The sampling rate is set to be 5GSs^{-1} . The values are in the format of $N(\mu, \sigma)$, where N is a normal distribution with mean μ and standard deviation σ .

Table 3.4: Parameters used for PD Simulation.

PD1	DEDO	$V_0 \sim N(2, 0.8)$ $\alpha \sim N(1e7, 0.4e7)$ $\beta \sim N(1e8, 0.4e8)$ $f_c \sim N(1e7, 0.4e7)$ $\theta = 0$ $t_d = 0$	+	SMG	$A \sim N(0.7, 0.28)$ $t_0 = 1.5e - 7$ $\tau \sim N\left(\begin{smallmatrix} 0.4e - 7, \\ 0.16e - 7 \end{smallmatrix}\right)$ $f_c \sim N(0.5e7, 0.2e7)$ $t_d = 2e - 7$			
PD2	SEDO	$V_0 \sim N(3, 1.2)$ $\tau \sim N\left(\begin{smallmatrix} 0.1e - 6, \\ 0.4e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 2.3e7, \\ 0.92e7 \end{smallmatrix}\right)$ $\theta = 0$ $t_d = 0$	+	DEDO	$V_0 \sim N(0.9, 0.36)$ $\alpha \sim N(1e7, 0.4e7)$ $\beta \sim N(1e9, 0.4e9)$ $f_c \sim N\left(\begin{smallmatrix} 1.7e7, \\ 0.68e7 \end{smallmatrix}\right)$ $\theta = 0$ $t_d = 1.8e - 7$			
PD3	SEDO	$V_0 \sim N(0.5, 0.2)$ $\tau = \infty$ $f_c = 0$ $\theta = \pi/2$ $t_d = 0$	-	SEDO	$V_0 \sim N(0.5, 0.2)$ $\tau \sim N\left(\begin{smallmatrix} 0.1e - 6, \\ 0.4e - 7 \end{smallmatrix}\right)$ $f_c = 0$ $\theta = \pi/2$ $t_d = 0$	×	SEDO	$V_0 \sim N(9, 3.6)$ $\tau \sim N\left(\begin{smallmatrix} 0.15e - 6, \\ 0.6e - 7 \end{smallmatrix}\right)$ $f_c = 1.68e7$ $\theta = 0$ $t_d = 0$
PD4	DEP	$V_0 \sim N(3, 1.2)$ $\alpha \sim N(5e6, 2e5)$ $\beta \sim N(1.5e7, 6e6)$ $t_d = 1.88e - 7$	×	SEDO	$V_0 \sim N(1.5, 0.6)$ $\tau \sim N\left(\begin{smallmatrix} 0.18e - 6, \\ 0.72e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 0.9e7, \\ 0.36e6 \end{smallmatrix}\right)$ $\theta = 0$ $t_d = 1.88e - 7$	+	SEDO	$V_0 \sim N(3.8, 1.52)$ $\tau \sim N\left(\begin{smallmatrix} 0.12e - 6, \\ 0.48e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 1.7e7, \\ 0.68e7 \end{smallmatrix}\right)$ $\theta = 0$ $t_d = 0$
PD5	SMG	$A \sim N(1.2, 0.48)$ $t_0 = 1e - 7$ $\tau \sim N\left(\begin{smallmatrix} 0.2e - 7, \\ 0.08e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 2.5e7, \\ 1e6 \end{smallmatrix}\right)$ $t_d = 0$	+	SMG	$A \sim N(0.2, 0.08)$ $t_0 = 2.5e - 7$ $\tau \sim N\left(\begin{smallmatrix} 0.2e - 7, \\ 0.08e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 2.5e7, \\ 1e6 \end{smallmatrix}\right)$ $t_d = 0$			
PD6	DEDO	$V_0 \sim N(2.5, 1)$ $\alpha \sim N\left(\begin{smallmatrix} 1.5e7, \\ 0.6e7 \end{smallmatrix}\right)$ $\beta \sim N\left(\begin{smallmatrix} 1.5e8, \\ 0.6e8 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 1.5e7, \\ 0.6e7 \end{smallmatrix}\right)$ $\theta = 0$ $t_d = 0.5e - 7$	+	SMG	$A \sim N(0.3, 0.12)$ $t_0 = 3.5e - 7$ $\tau \sim N\left(\begin{smallmatrix} 0.3e - 7, \\ 0.12e - 7 \end{smallmatrix}\right)$ $f_c \sim N\left(\begin{smallmatrix} 3e7, \\ 1.2e6 \end{smallmatrix}\right)$ $t_d = 0$			

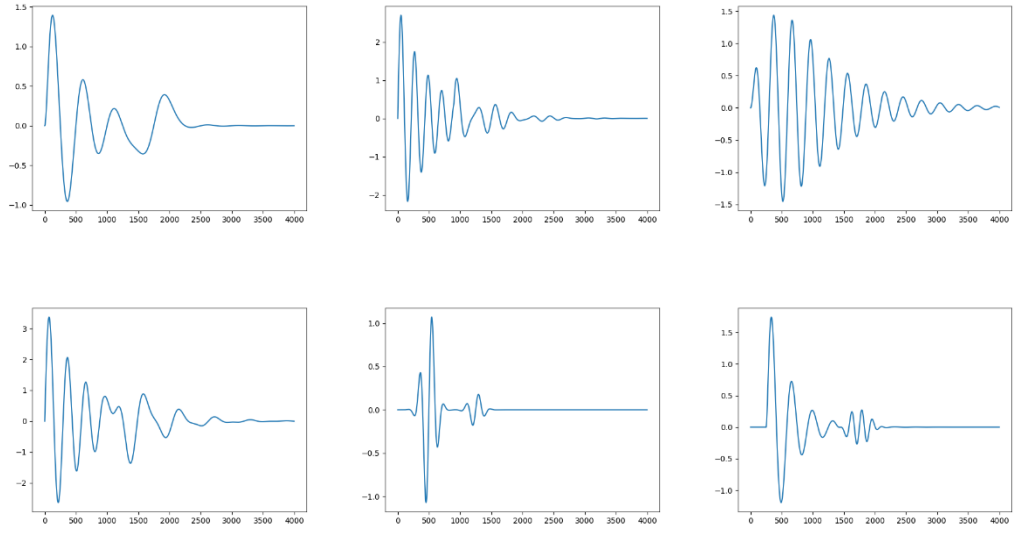


Figure 3.6: The generated waveforms, in order from 1 to 6.

Each PD source is given a different geometrical coordinate, as shown in Table 3.5.

Table 3.5: PD source locations.

PD Source	Coordinate (m)
PD1	1.2, 2.3, 3.1
PD2	2.9, 6.3, 5.4
PD3	4.5, 7.8, 6.2
PD4	6.1, 1.7, 3.5
PD5	7.8, 1.2, 4.9
PD6	9.2, 3.5, 2.8

The simulated sensor coordinates are [1,9,1.8], [1,7,1.4], [1,5,1.8], [1,3,1.4] for sensor 4 to 1 in that order. They are arranged at different x, y positions, and different heights, z, which improve precision and accuracy as compared to layouts with all the sensors at the same height (Chan et al., 2026). Each PD is set to discharge 8 times, uniformly distributed in a time span of 0.4ms. Each PD waveform would then be added to the simulated sensor output signal, with the appropriate time difference of arrival (TDOA) for each sensor. The amplitude of the signal detected at each sensor was scaled by the factor $e^{-0.2l}$, where l is the distance between PD source and the sensor to account for attenuation.

Additive white gaussian noise (AWGN) was added at $SNR = 20 \text{ dB}$ per channel, with the noise power in watts as the variance of the noise signal being added. The final simulated sensor outputs consist of a series of PD waveforms at different times, from different sources and with different amplitudes. The term ‘sensor’ is synonymous with ‘channel’ in this report, because the sensor output is detected by the channels of an oscilloscope. The output is then split into the same Scene and Shard structure of the measured dataset.

3.3 Case Western Reserve University Bearing Dataset

The Case Western Reserve University (CWRU) Bearing Dataset is a standard benchmark for testing fault diagnosis models (Smith & Randall, 2015). It consists of vibration signals from a test motor with artificially damaged bearings, recorded under varying motor loads to capture a wide range of operating conditions. A small portion of this dataset (consisting of only Inner Race fault of 0.007 inches and 0.014 inches) was also used during training to prevent the model from overfitting (Yuwei et al., 2023). By exposing the model training pipeline to these diverse, out-of-domain signals early on, the network was forced to learn general feature extraction. This prevented the model from simply memorizing the specific patterns of the simulated partial discharge data before being tested on real-world measurements.

3.4 Experimental Measurement

An experiment was conducted to obtain a set of real-world measured PD waveforms, to test the capabilities of the clustering methods. It was conducted in an indoor environment, in a high voltage cage. Three different types of physical PD models were used. The first was a void defect in epoxy resin insulation (PD1), mimicking internal gas pockets with a 1.5mm energized copper wire and 0.5mm grounded copper strip. The second was a similar model, but with an incision in the insulation, instead of a void (PD2). The third was like the second, but the defect was in the form of a delamination in the insulation layers (PD3). The models were immersed in transformer oil to suppress surface discharge. Next, the electrodes on the model were energized to AC voltages of 8kV for the void defects, 14kV for the incision, and 10kV for delamination. The complete design details of the models are specified in a prior work by Madhuranga et al. (2025).

Four UHF sensors were used to collect PD EM wave emissions. The sensors were planar monopole antennae (PMA) developed by Kayser Azam et al. (2024). Each sensor was connected to a four-channel digital oscilloscope via RG58 coaxial cables, operating at 5GS/s. According to the Nyquist-Shannon sampling criterion, this allows continuous-time signals of up to 2.5GHz to be reconstructed from the sampled data.

In addition, the PRPD plots of each defect were taken using the MPD 600 measurement/analysis system to confirm the presence of the correct PD.

The experimental setup is given in Figure 3.7. The sensor coordinates are chosen to improve accuracy and coverage area of PD localization, based on preliminary testing done by Chan et al. (2026).

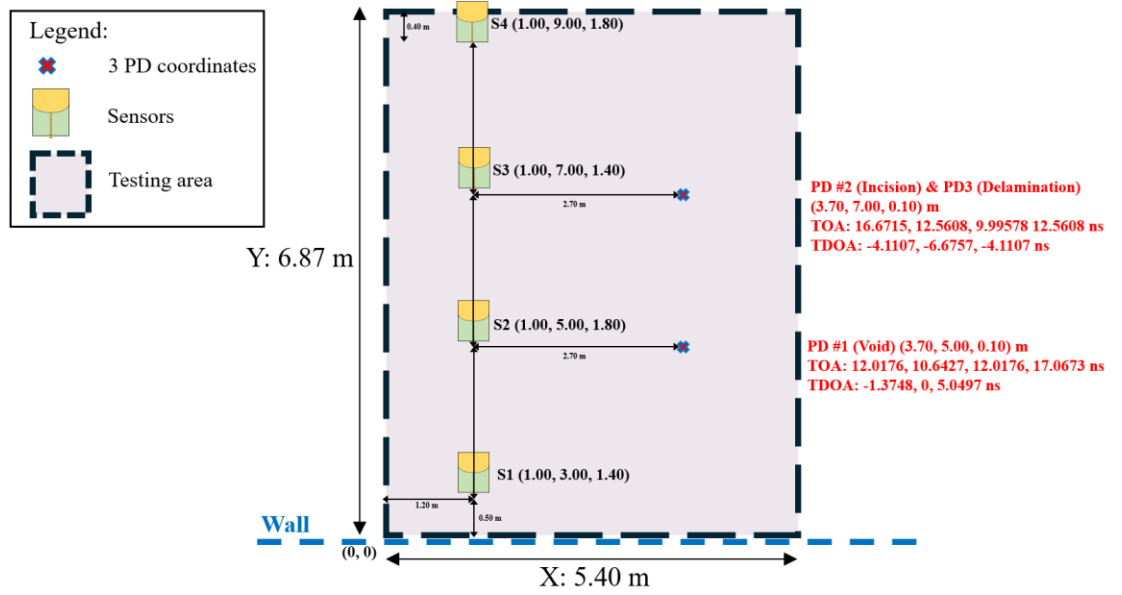


Figure 3.7: Schematic diagram of experimental layout.

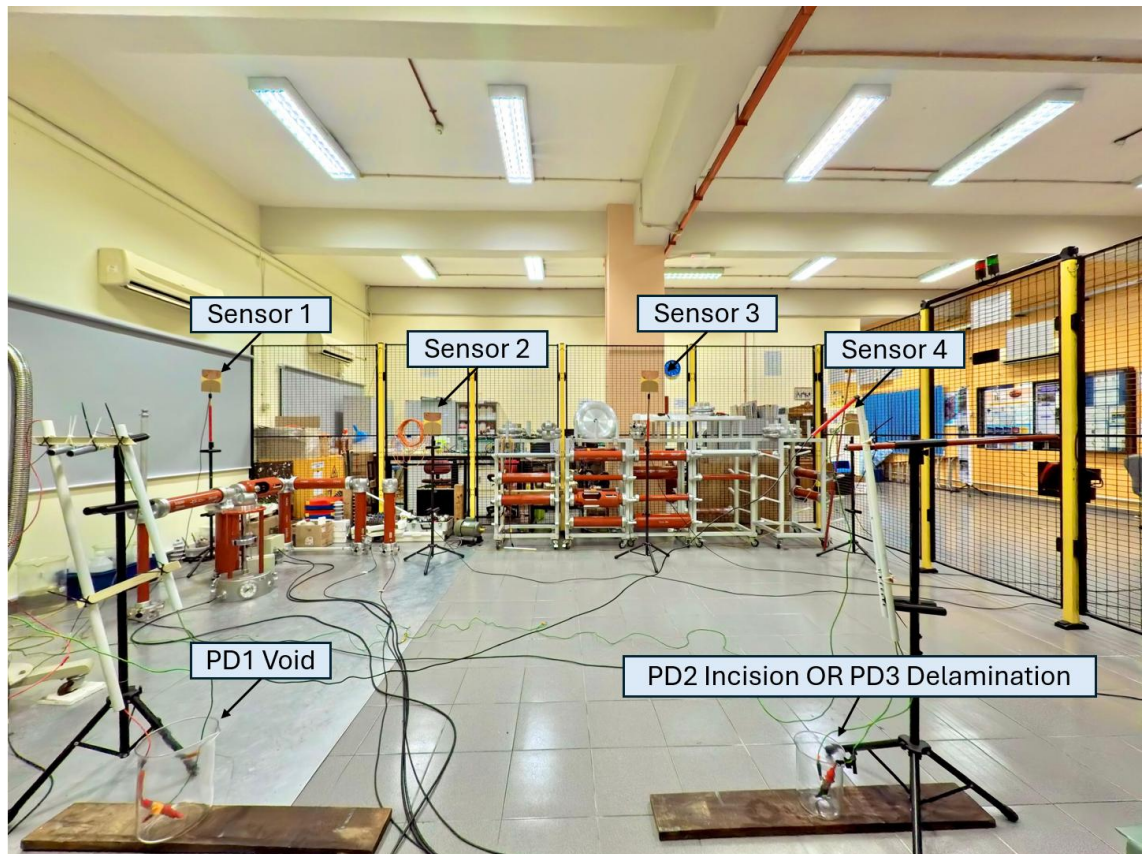


Figure 3.8: Actual experimental layout.

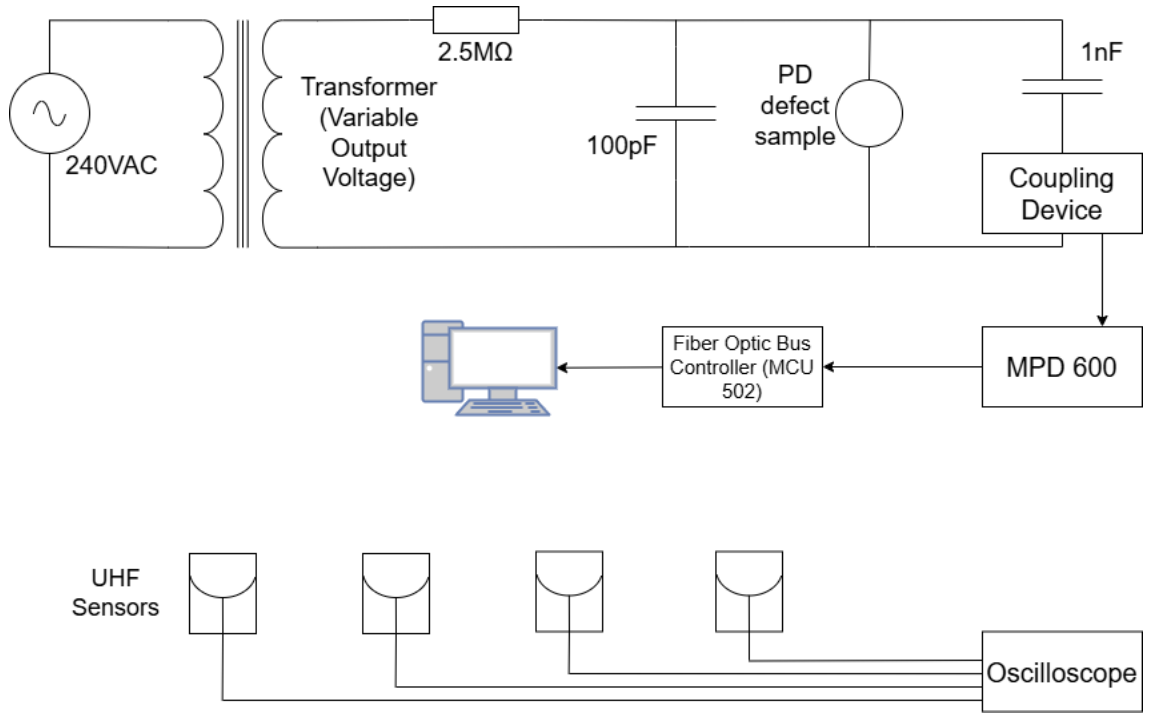


Figure 3.9: Circuit Diagram of Experiment.

When the PD model is energized, a threshold is set on the oscilloscope to capture the instant in time where the reading exceeds it (which is when a PD waveform is observed). The readings of all four channels are then saved as one set. Only 3 sets of signals of PD1 were recorded, and PD1 was used as part of the model training regime. For PD2, 304 signals sets were recorded, and for PD3, 531. These two PDs were used to test the models' performance.

Some of the obtained PD waveforms are shown in Figure 3.10 and Figure 3.11.

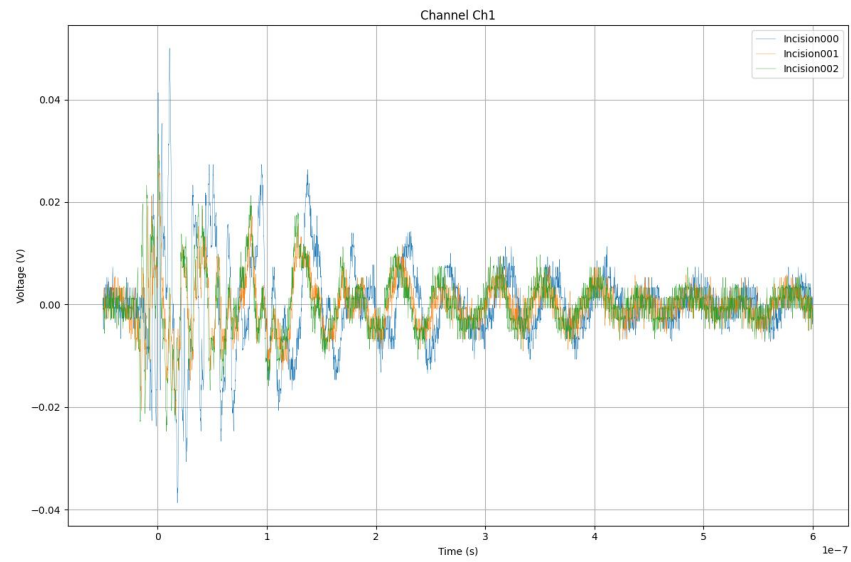


Figure 3.10: Measured PD waveforms of the incision defect on sensor 1 (Ch1).

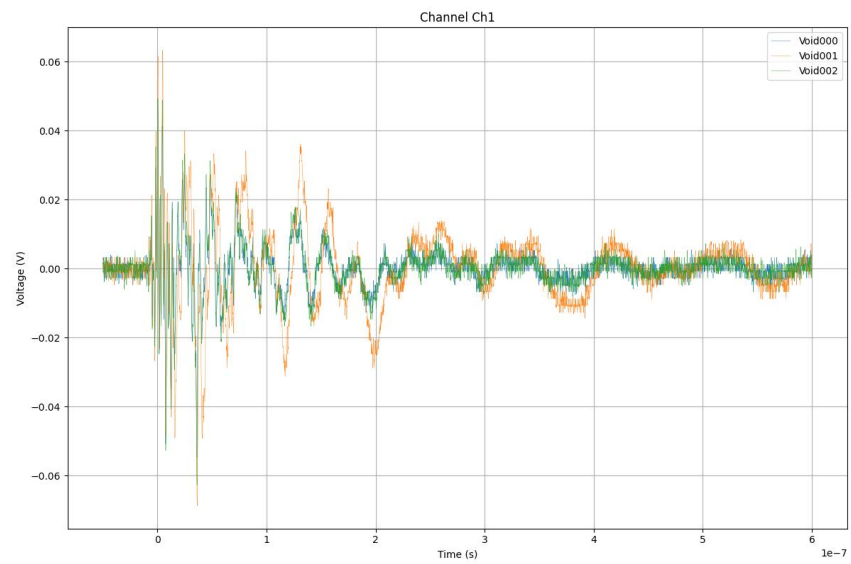


Figure 3.11: Measured PD waveforms of the void defect on sensor 1 (Ch1).

3.5 Measured PD Waveform Isolation & Dataset Generation

For clustering and classification to be performed on actual measurements of Partial Discharge (PD) occurrences, an algorithm was developed to robustly isolate the transient PD pulses from the real-world ambient noise floor.

The raw sensor outputs are first centered by removing the DC offset. To suppress high-frequency background noise while preserving the sharp, transient features typical of PD events, the signal undergoes Discrete Wavelet Transform (DWT) denoising. A Symlet 4 (sym4) mother wavelet is applied up to a decomposition level of 4, utilizing universal soft-thresholding based on the noise variance of the finest detail coefficients.

To isolate the non-noise portions, the algorithm evaluates the signal's energy profile using a rolling Root Mean Square (RMS) over a 30-sample sliding window. A dynamic threshold is established based on the statistical distribution of this baseline energy:

$$Threshold = Baseline_{RMS} + (0.7 \times \sigma_{RMS}) \quad (11)$$

where the *Baseline* is the median RMS value and σ_{RMS} is the standard deviation. Time indices where the rolling RMS exceeds this threshold are flagged as active events.

To ensure the entire pulse geometry is captured without fragmentation, active regions separated by fewer than 100 samples are merged into a single continuous event. A guard margin of 50 samples is then appended to the beginning and end of the region. If a single waveform contains multiple distinct events, the algorithm isolates the region exhibiting the highest mean squared power.

A Hanning window is applied to the very edges of the extracted segment. This smooths the transition to zero, preventing spectral leakage during subsequent frequency-domain processing.

The isolated pulses that were extracted from the oscilloscope-obtained inputs are then randomly scattered across four channels, forming long, continuous streams of data, emulating what would be detected in the real world by four UHF sensors, and multiple PDs discharging simultaneously. The four channel inputs are also broken down and saved in the same Scene and Shard structure of the other datasets.

3.6 Training & Testing Dataset Metrics

Table 3.6: Dataset Metrics.

<i>Dataset Category</i>	<i>Shards</i>	<i>Total Scenes</i>	<i>Total Signals (Pulses)</i>	<i>Train:Val Split</i>
<i>Training</i>				
<i>CWRU</i>	16	80	320	13:3
<i>Waveform Mixing (Equation Generated)</i>	20	1,000	8,052	8:2
<i>HFSS + MATLAB</i>	20	1,000	5,936	8:2
<i>Measured (PD1 Void)</i>	5	25	204	6:4
<i>Testing</i>				
<i>Measured (PD2 Incision & PD3 Delamination)</i>	20	2,000	15,924	-

As can be seen, models are trained on datasets from various sources, and tested on a completely unseen, measured dataset. This is to achieve **cross-domain validation**.

3.7 Feature Extraction Methods

Two methods to extract features from the simulated/experimental PD UHF waveform have been identified: Bispectra (He et al., 2020), and Short-Time Fourier Transform (STFT).

3.7.1 STFT

To construct a more robust feature space, the 1D waveforms were transformed into 2-dimensional (2D) time-frequency representations utilizing the Short-Time Fourier Transform (STFT). The STFT operates by dividing a non-stationary time signal into shorter, localized segments of equal length using a sliding window function. A discrete Fourier transform (DFT) is then computed on each windowed segment to extract the frequency content present at that specific localized instance in time. The discrete formulation of the STFT is expressed as:

$$X(m, \omega) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n} \quad (12)$$

where $x[n]$ is the discrete time-domain PD signal, $w[n]$ represents the sliding window function (e.g., a Hann window), m denotes the time index, and ω is the angular frequency.

To eliminate phase dependencies, the absolute magnitude of the complex STFT output, $|X(m, \omega)|$, is computed to generate a 2D spectrogram.

This spectrogram acts as a dense energy heatmap, visually mapping the intensity of frequency components across the duration of the physical Partial Discharge (PD) event. Even if the real-world measurement environment heavily distorts the time-domain amplitude or shape of a pulse, the underlying high-frequency energy "fingerprint" characterizing a specific PD fault type remains largely invariant in the frequency domain.

3.7.2 Signal Bispectra

The bispectrum of a signal is defined as the two-dimensional Fourier transform of the third-order cumulants of a signal.

Third order cumulants are defined as the expected value of three time-shifted points of a signal $x[n]$,

$$C_{3x}(\tau, \rho) = E[x[n]x[n + \tau_1]x[n + \tau_2]] \quad (13)$$

And consequently, the bispectrum:

$$B(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} C_{3x}(\tau, \rho) e^{-j(\omega_1 \tau_1 + \omega_2 \tau_2)} d\tau_1 d\tau_2 \quad (14)$$

To estimate the discretized bispectra, the signal is first split into K sections with M observed samples in each section. These are denoted as $x^{(i)}(1), x^{(i)}(2), \dots, x^{(i)}(M)$, where $i = 1, 2, \dots, K$. Then, the Fourier transform coefficients of each section are obtained.

$$X^{(i)}(\lambda) = \frac{1}{M} \sum_{n=1}^M x^{(i)}[n] e^{-j\left(\frac{2\pi(n-1)\lambda}{M}\right)} \quad (15)$$

And $\lambda = 0, 1, \dots, M/2$. The bispectra of each section of the signal is

$$\begin{aligned} & \tilde{b}^{(i)}(\lambda_1, \lambda_2) \\ &= \frac{1}{\Delta^2} \sum_{i_1=-L}^L \sum_{i_2=-L}^L X^{(i)}(\lambda_1 + i_1) X^{(i)}(\lambda_2 + i_2) X^{(i)*}(-\lambda_1 - \lambda_2 - i_1 - i_2) \end{aligned} \quad (16)$$

With $\Delta = f_s/N_0$, and $M = (2L + 1)/N_0$. Finally, the bispectra estimation of the whole signal is:

$$\tilde{B}(\lambda_1, \lambda_2) = \frac{1}{K} \sum_{i=1}^K \tilde{b}^{(i)}(\lambda_1, \lambda_2) \quad (17)$$

This method is presented by He et al. (2020), because the bispectra are insensitive to Gaussian noise, leaving only the non-Gaussian PD signal to be visible.

3.8 Clustering/Separation Methods

Several different clustering/separation methods have been identified during the literature review, such as the K-means algorithm, Gaussian mixture model, density based spatial clustering of applications with noise (DBSCAN), and deep learning (DL). However, only one version will be used, which is the Hierarchical Density Based Spatial Clustering of Applications with Noise.

3.8.1 Hierarchical Density Based Spatial Clustering of Applications with Noise (HDBSCAN)

HDBSCAN improves upon classic DBSCAN by finding clusters of varying densities. Like DBSCAN, it does not require the pre-determination of the number of clusters. Moreover, it eliminates the need for a strict global cluster radius threshold. Instead, it relies primarily on a single intuitive user-defined parameter, with optional secondary and ternary parameter:

- **min_cluster_size**: The primary parameter representing the minimum number of data points required to form a valid, independent cluster.
- **min_samples** (*Optional*): The number of neighboring points used to calculate a point's core distance. This controls how conservative the clustering is regarding noise (larger values force more points into noise).
- **Epsilon (ϵ)** (*Optional*): The distance between clusters such that the algorithm starts merging them together. If set to 0, HDBSCAN maintains its original density based split that was obtained without using any cluster radius threshold.

Instead of classifying points using a static radius, HDBSCAN transforms the space based on density using two key distance metrics:

i. **Core distance:** The distance from a point to its k -th nearest neighbor (where k is `min_samples`). Dense areas have small core distances, while sparse areas have large ones.

ii. **Mutual reachability distance:** A distance metric between two points that is equal to the maximum of either point's core distance or the actual Euclidean distance between them. This effectively pushes sparse noise points further away from dense clusters.

Finally, the HDBSCAN algorithm operates as follows:

i. **Transform the space:** Calculate the core distance for all points and compute the mutual reachability distance between all pairs of points.

ii. **Build the Minimum Spanning Tree (MST):** Construct a graph connecting all points using the mutual reachability distances as edge weights, and extract the minimum spanning tree.

iii. **Construct a cluster hierarchy:** Sort the edges of the MST by distance (from shortest to longest) to build a single-linkage dendrogram (a tree of connected components).

iv. **Condense the tree:** Traverse the hierarchy. When a cluster splits, if a resulting branch has fewer points than `min_cluster_size`, it is considered "points falling out of the cluster" rather than a true split. If both branches have at least `min_cluster_size` points, it is a true cluster split.

v. **Extract stable clusters:** Calculate the "stability" (persistence) of each condensed cluster over varying density thresholds. The algorithm selects the clusters with the highest stability as the final output.

vi. **Merge close clusters:** If requested ($\epsilon \neq 0$), HDBSCAN will merge close clusters based on the selected ϵ value.

After processing, any points that fall out of a cluster before it becomes a stable, extracted cluster are classified as noise points.

3.9 TDOA Estimation Method

TDOA is estimated using the cross-correlation technique. The script finds the time-domain shift in which maximum cross-correlation is acquired between two signals. This time-shift is taken as the TDOA.

3.10 Localization Technique

The particle swarm optimization (PSO) technique was used to produce results of localization, for comparison purposes.

3.11 Deep Learning (DL) Models

This section is dedicated to explain the specifics of DL model training and testing. It includes the specific input features fed into the model, and how the model outputs are used for training (to minimize model loss) and testing (HDBSCAN clustering & localization). Four model architectures (experiments) are developed to perform the task of deeper feature extraction, before clustering using HDBSCAN. They take in either STFT or bispectra generated from the raw signals and convert them into 128-dimensional vectors representing signal features. HDBSCAN is then performed on those embeddings, with the option of first reducing them to two or three dimensions before clustering.

3.11.1 Architecture of the Semi-Supervised Deep Embedded Clustering Model (exp01)

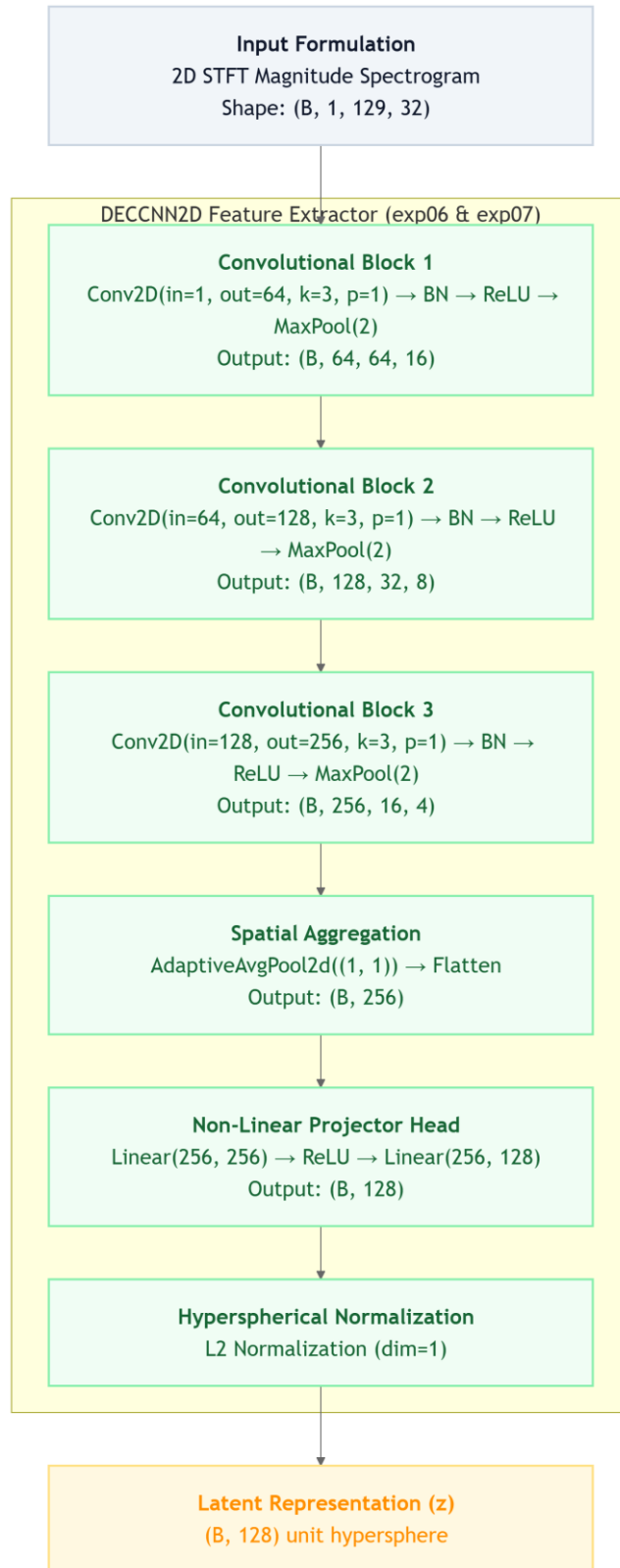


Figure 3.12: Architecture of exp01.

The methodology employed in this experiment utilizes a Two-Phase Semi-Supervised Spherical Deep Embedded Clustering (DEC) framework to categorize partial discharge (PD) signals. The network takes 2D Short-Time Fourier Transform (STFT) magnitude spectrograms as input. **To compute the STFT, the periods of time from the datasets that contained PD signals were extracted and fed into the algorithm to produce the plots.** This means that no noise signals were extracted to produce STFT representation, only valid PD signals. The architecture relies on a lightweight 2D Convolutional Neural Network (CNN) backbone which serves as a feature encoder, followed by an MLP projection head and a trainable cluster layer.

Feature Encoder: The encoder processes single-channel STFT magnitude spectrogram inputs which are fed into the network as $(1, F, T)$, where F represents the number of frequency bins and T is the number of temporal frames. In this case, $F = 129$ and $T = 32$. The spectrograms are passed through a hierarchical structure comprising three convolutional blocks. Each block consists of a 2D Convolutional layer with a 3×3 kernel, followed by Batch Normalization, a ReLU activation function, and 2×2 Max Pooling. This design progressively downsamples the spatial dimensions (through max pooling) while increasing the channel depth up to 256 (through the convolutional layers). The final tensor output is in the shape $(256, \lfloor F/8 \rfloor, \lfloor T/8 \rfloor)$.

Projector: Following the convolutional blocks, an Adaptive Global Average Pooling layer collapses the spatial dimensions, yielding a tensor shape of $(256, 1, 1)$ and rendering the architecture invariant to varying input pulse durations. The resulting flattened feature vector is passed through a Multi-Layer Perceptron (MLP) projector head, consisting of two linear layers separated by a ReLU activation. The MLP projector head condenses the features into a 128-dimensional embedding space.

Spherical Embedding Space: The final output of the projector is explicitly $L2$ -normalized. This projects all feature embeddings onto a unit hypersphere, which provides significant mathematical stability during contrastive representation learning and bounds the embedding space for the subsequent clustering phase.

Cluster Layer: Attached to the embedding space is a trainable cluster layer defined as a set of pre-defined centroid vectors (one for each of the K clusters). Like the embeddings, these centroid vectors are maintained on the unit hypersphere. They are only used for training, and during testing (the ‘inference’ phase), the model clusters without a set of predefined centroids.

The training pipeline is structured to first learn a generalized, unsupervised representation of the data, followed by a targeted clustering refinement phase that incorporates a small fraction of known labels to align the clusters with physical phenomena.

Phase 1: Contrastive Representation Learning: Before the model can effectively group partial discharge (PD) signals, it must learn to extract robust, invariant feature representations. Phase 1 achieves this through a self-supervised contrastive learning framework based on SimCLR. During this phase, a single STFT magnitude spectrogram is subjected to stochastic data augmentations—specifically, additive Gaussian noise scaled to the signal's Signal-to-Noise Ratio (SNR) and random circular time-shifting. This generates two correlated "views" of the same physical pulse. Both views are passed through the network backbone to produce their respective 128-D embeddings. The network is optimized using the Normalized Temperature-scaled Cross Entropy (NT-Xent) loss.

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} 1_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (18)$$

Where z_i and z_j are two 128-dimension embeddings, $\text{sim}()$ is the cosine similarity between them, and τ is the temperature parameter, which controls the strictness of the separation. N represents the original mini-batch size of physical pulses; because two independent stochastic augmentations are generated from each pulse, the loss function operates over a total of $2N$ embedding.

This loss is mathematically formulated to maximize the cosine similarity between the embeddings of the augmented positive pair (augmented versions of the same pulse) while simultaneously minimizing the similarity between all other negative pairs (embeddings of different pulses) within the training batch. This forces the network to ignore transient environmental noise and focus on the fundamental energy signatures of the PD events.

Phase 2: Deep Embedded Clustering (DEC) Refinement: Once Phase 1 concludes, the network weights are temporarily frozen, and the unaugmented training dataset is passed through the backbone. Standard K-Means clustering is performed on the resulting 128-dimensional embeddings to establish an initial set of K cluster centroids, denoted as μ_j . These centroids initialize the weights of a custom DEC clustering layer. During Phase 2, the model simultaneously updates the CNN + MLP backbone weights and the coordinates of the cluster centroids. The training objective in this phase is a composite loss function comprising (i) an unsupervised clustering objective and (ii) a supervised constraint penalty.

When an embedding z_i enters the clustering layer, the Euclidean distance between each signal embedding and each of the K centroids is calculated. These distances are converted into a probability distribution, or "soft assignment" q_{ij} using a student's t-distribution kernel. This represents a "soft" probability of it belonging to each cluster (e.g., an 80% confidence score).

$$q_{ij} = \frac{\left(1 + \|z_i - \mu_j\|^2\right)^{-\frac{1}{2}}}{\sum_k \left(1 + \|z_i - \mu_k\|^2\right)^{-\frac{1}{2}}} \quad (19)$$

To enforce cluster tightening, an auxiliary target distribution p_{ij} is derived by squaring the soft assignments and normalizing them by the overall cluster frequencies. This target distribution represents a sharpened, high-confidence assignment. The network is then optimized by minimizing the Kullback-Leibler (KL) divergence between the soft assignments and the target distribution:

$$L = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (20)$$

By minimizing this loss, the network pulls embeddings closer to their assigned centroids while simultaneously adjusting the centroids to better fit the underlying data distribution, resulting in dense, well-separated clusters.

ii *Supervised Pairwise Constraint Loss*

To ensure the unsupervised clusters align with true physical sources, the exposed 10% subset of labels is utilized to enforce pairwise constraints.

- *Must-Link Penalty*: If two samples share the same known label, a penalty is applied if their soft assignment distributions diverge.
- *Cannot-Link Penalty*: If two samples possess different known labels, a penalty is applied if their soft assignment distributions overlap

The final objective for Phase 2 is

$$L_{Phase2} = L_{KL} + \gamma L_{Pairwise} \quad (21)$$

Where γ is a hyperparameter that dictates the weight of the semi-supervised constraints.

Inference (testing) via Dynamic Density Clustering (HDBSCAN): While the DEC algorithm trains the network to form dense geometric clusters, the predefined K centroids are rigid. Since the inference is performed on a different dataset domain from the training dataset, there will be differing cluster centroids. Therefore, during the final inference stage on the measured dataset, the custom DEC clustering layer is bypassed. Instead, the refined 128-dimensional embeddings z are extracted and subjected to t-distributed stochastic neighbour embedding to reduce the dimensionality (e.g., to 2D or 3D projections). The projected embeddings are then processed using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). Finally, a post-hoc majority voting mechanism aligns these dynamically discovered density clusters with the

ground-truth physical fault classes for quantitative evaluation. After that, an algorithm is run to predict which pulses come from the same PD instance. It does so by looking at how close the pulses are in the time domain, and their Euclidean distance in the 128-dimension embedding space. If those two are smaller than a threshold (100 ns & 0.5 respectively) for a given 2 pulses, then it considers them to be from the same PD event, and they are assigned the same predicted instance ID. A strict criterion is implemented, such that there can only be 4 pulses (one from each sensor) assigned to one pulse instance. Where there are more than that which satisfies the criteria, the pulse with the nearest embedding distance is taken.

Localisation using TDOA, cross-correlation and PSO: After that, the clusters that represent >5% of total dataset are isolated. For each of these clusters, the script tries to find signals within them that the model ‘thinks’ are from the same instance. For a valid set of signals, they must all have the same predicted instance ID, and there must be one from each channel. For every valid set of 4 signals, the TDOA is calculated via cross-correlation, and localization is performed via PSO, which outputs the predicted PD coordinate. The mean coordinate for every valid set of signals is taken to be the final predicted PD location.

3.11.2 Upgraded Semi-Supervised Deep Embedded Clustering Model (exp02)

Experiment 02 (exp02) builds directly upon the Semi-Supervised Deep Embedded Clustering (DEC) framework established in exp01. While retaining the 2-dimensional STFT magnitude inputs and the core DEC structure, exp02 introduces modifications to resolve clustering instability and to enforce a more robust, class-aware latent space prior to density-based clustering.

The architectural improvements in exp02 are defined by two primary modifications:

(a) *Supervised Contrastive Learning (SupCon) in Phase 1*

In exp01, the initial feature representation phase (Phase 1) utilized a purely unsupervised contrastive framework (SimCLR via NT-Xent loss). This approach only maximized the similarity between stochastic augmentations of the same physical pulse, failing to leverage the available ground-truth labels from the synthetic and bearing (CWRU) datasets during this critical representation learning stage.

To correct this inefficiency, exp02 upgrades Phase 1 to a **Supervised Contrastive Learning (SupCon)** framework. Unlike the semi-supervised Phase 2 in exp01, which uses losses based on the soft cluster assignments, the loss in this new Phase 1 of exp02 is a supervised extension of the NT-Xent loss, termed the Supervised Contrastive loss. This loss function explicitly incorporates valid ground-truth labels at a specified percentage (10%).

$$\mathcal{L}_{i,j} = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{a=1}^{2N} 1_{[a \neq i]} \exp(\text{sim}(z_i, z_a)/\tau)} \quad (22)$$

This equation assumes that the set of all positive embeddings is $P(i)$. $P(i)$ is greater than 2 since there are 2 augmented versions of the same signal + other signals of the same ground truth class. The training tries to reduce its value between embeddings within $P(i)$:

- When a training batch contains labeled samples, the SupCon loss actively minimizes the cosine distance between embeddings of signals belonging to the *same* ground-truth class (subsets of $P(i)$), while maximizing the distance between embeddings of *different* classes.
- Concurrently, unlabeled physical measurement data (where the class is masked or unknown) continues to be processed using standard unsupervised contrastive logic, ensuring that the loss is minimized for two augmented versions of the same signal.

(b) *Stratified Label Exposure*

To effectively bridge the domain shift between synthetic pre-training data and measured physical waveforms, the model must receive a balanced distribution of known labels. In exp01, the label masking logic applied a global random fraction across the entire dataset pool. This introduced a statistical vulnerability where the network could disproportionately sample anchor labels from the significantly larger synthetic domains, starving the model of labelled measured signals.

Experiment 02 implements **Stratified Label Exposure** at the dataset generation level. The sampling logic ensures that the specified fraction of exposed labels is calculated and drawn independently from *each individual dataset source* (e.g., 10% from the synthetic dataset, 10% from the CWRU dataset, and 10% from the measured dataset). This stratification guarantees that the Phase 1 SupCon loss receives a proportionally balanced, representative sample from every domain, preventing the network from overfitting to the domains with more data.

3.11.3 Bispectrum Analysis and Vision Transformer (ViT) Architecture (exp03)

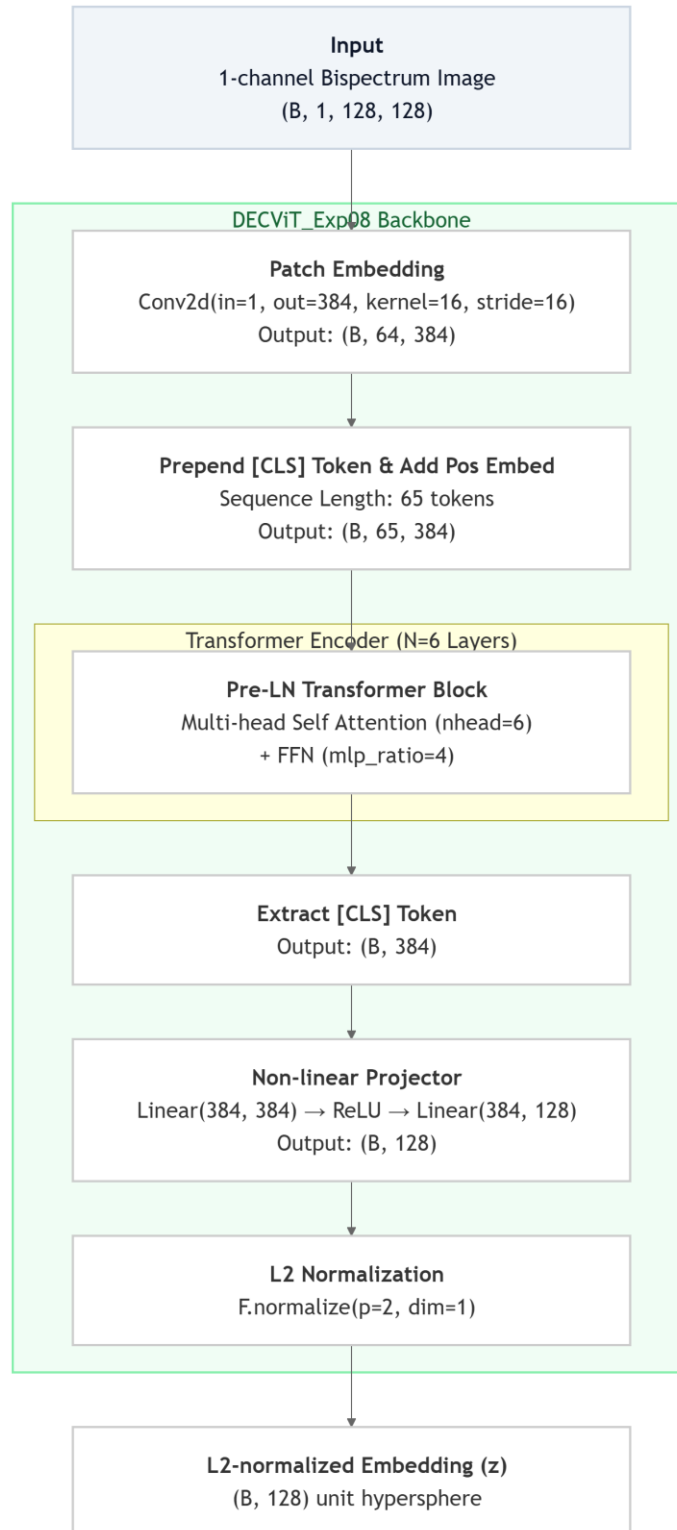


Figure 3.13: Architecture for exp03.

To further improve the robustness of the partial discharge classification, particularly against the severe Gaussian white noise inherent in real-world UHF substation measurements, a revised deep learning architecture (exp03) was developed. This

approach introduces two major paradigm shifts from the previous exp02 pipeline: the replacement of Short-Time Fourier Transform (STFT) spectrograms with 2D Bispectra, and the replacement of the 2D CNN backbone with a Vision Transformer (ViT).

(a) ***Bispectrum Feature Extraction via Segment Averaging:***

While STFT spectrograms effectively capture frequency evolution over time, power spectra are fundamentally highly susceptible to broadband Gaussian noise. To address this, exp03 utilizes the bispectrum—a higher-order statistical transform based on the 2D Fourier transform of third-order cumulants. Because the third-order cumulant of a symmetrically distributed process (like Gaussian white noise) is mathematically zero, the bispectrum naturally suppresses background noise, leaving only the non-Gaussian phase-coupled components characteristic of the underlying PD pulses.

To achieve this noise cancellation in practice, a segment-averaging estimation technique (like Welch’s method) was implemented. The feature extraction pipeline was structured as follows:

1. **Uniform Alignment:** All raw 1D PD UHF signals, extracted from their source dataset, were uniformly padded or cropped to exactly 4,096 points to ensure strict frequency-bin alignment across domains.
2. **Chunking and Averaging:** The 4,096-point signals were segmented into overlapping chunks of 256 points. A complex bispectrum was calculated for each individual chunk via Fast Fourier Transform (FFT).
3. **Statistical Smoothing:** The complex bispectra of all chunks were mathematically averaged before extracting the absolute magnitude. This forces random noise artifacts to destructively interfere and cancel out to zero.

Because the FFT was performed on 256-point chunks, the resulting positive-frequency quadrant of the bispectrum yielded a highly compact, information-dense 2D grid of 129×129 pixels.

(b) ***Model Description:***

The proposed model for this experiment utilizes a customized, lightweight Vision Transformer (ViT) backbone designed to process 2D bispectrum representations of Partial Discharge (PD) signals. The architecture processes the data through the following stages:

Input Representation: To optimize the 129×129 bispectrum grid for the Transformer architecture, a 1-pixel discard was applied to the extreme Nyquist edge bin, cropping the grid to exactly 128×128 pixels. The input to the network is a 1-channel (grayscale) 2D bispectrum image of size 128×128 . This image represents the phase-coupled frequency components of the PD pulse calculated using the Welch method.

Patch Embedding: Unlike traditional Convolutional Neural Networks (CNNs) that scan an image pixel by pixel, the Vision Transformer slices the 128×128 image into a grid of 16×16 non-overlapping patches. A linear projection (implemented via a convolutional layer) flattens each of these 64 patches into a 384-dimensional feature vector, treating each patch as an individual "token."

Token Preparation: Because transformers process tokens simultaneously rather than sequentially, a learnable Positional Embedding is added to each token to retain the spatial awareness of where each patch originated. Additionally, a special [CLS] (Classification) token is prepended to the sequence. This token acts as an aggregator that will eventually summarize the entire image.

Transformer Encoder: The sequence of 65 tokens (64 patches + 1 [CLS]) passes through 6 Transformer Encoder blocks. Each block consists of two primary mechanisms operating in sequence:

- **Multi-Head Self-Attention:** This mechanism allows every patch to compare its frequency patterns with every other patch in the image simultaneously to establish context. Instead of a single attention process, the 384-dimensional token is split across 6 independent "heads" (64 dimensions each). This allows the network to search for 6 entirely different types of relationships at the exact same time—for example, one head might look for correlations between high-frequency magnitude peaks, while another head simultaneously tracks low-frequency background noise patterns.
- **Feed-Forward Network (FFN):** After the tokens have exchanged information via self-attention, each token is processed independently by a Feed-Forward Network (a small, two-layer neural network with a Gaussian Error Linear Unit, or GELU, activation function). The FFN takes the newly gathered contextual information from the attention heads and mathematically combines it to extract deeper, higher-level features for that specific patch.

By the end of the 6 blocks, this alternating process of sharing context (Self-Attention) and refining features (FFN) allows the network to extract complex, global relationships across the bispectrum, pooling this knowledge into the final [CLS] token.

Projector Head and Normalization: The 64 image patches are discarded, and the summarized [CLS] token is passed through a Projector Head (a sequence of linear layers). This reduces the token from 384 dimensions down to a compact 128-dimensional embedding. Finally, this vector is mathematically normalized (L2-normalization) so that

it lies perfectly on the surface of a unit hypersphere, providing a stable geometric space for clustering.

(c) ***Model Training:***

The overarching training logic for exp03 remains identical to exp02, utilizing a two-phase approach:

Phase 1: Representation Learning (Semi-Supervised Pre-Training): Phase 1 is very similar to previous experiments; the goal is simply to teach the network what a PD pulse "looks like" by organizing the 128-dimensional embedding space. The network is trained using **Supervised Contrastive Learning (SupCon)**: The augmenting was done via amplitude scaling, Gaussian noise addition, and frequency band masking (zeroing out random frequency bands) in substitution to the circular time-shifting of the previous two experiments.

Phase 2: Semi-Supervised Deep Embedded Clustering (DEC): As before, the network pauses to run a standard K-Means clustering algorithm, finding centroids for the clusters. Training then resumes using the two simultaneous loss functions: **Deep Embedded Clustering (DEC) KL-Divergence** and **Pairwise Constraint Loss**.

(d) ***Inference (testing) via HDBSCAN & Localisation***

As with Ex06 and exp02, the inference procedure remains the same. TDOA, cross-correlation and PSO are also used for localisation.

3.11.4 Upgraded Vision Transformer (exp04)

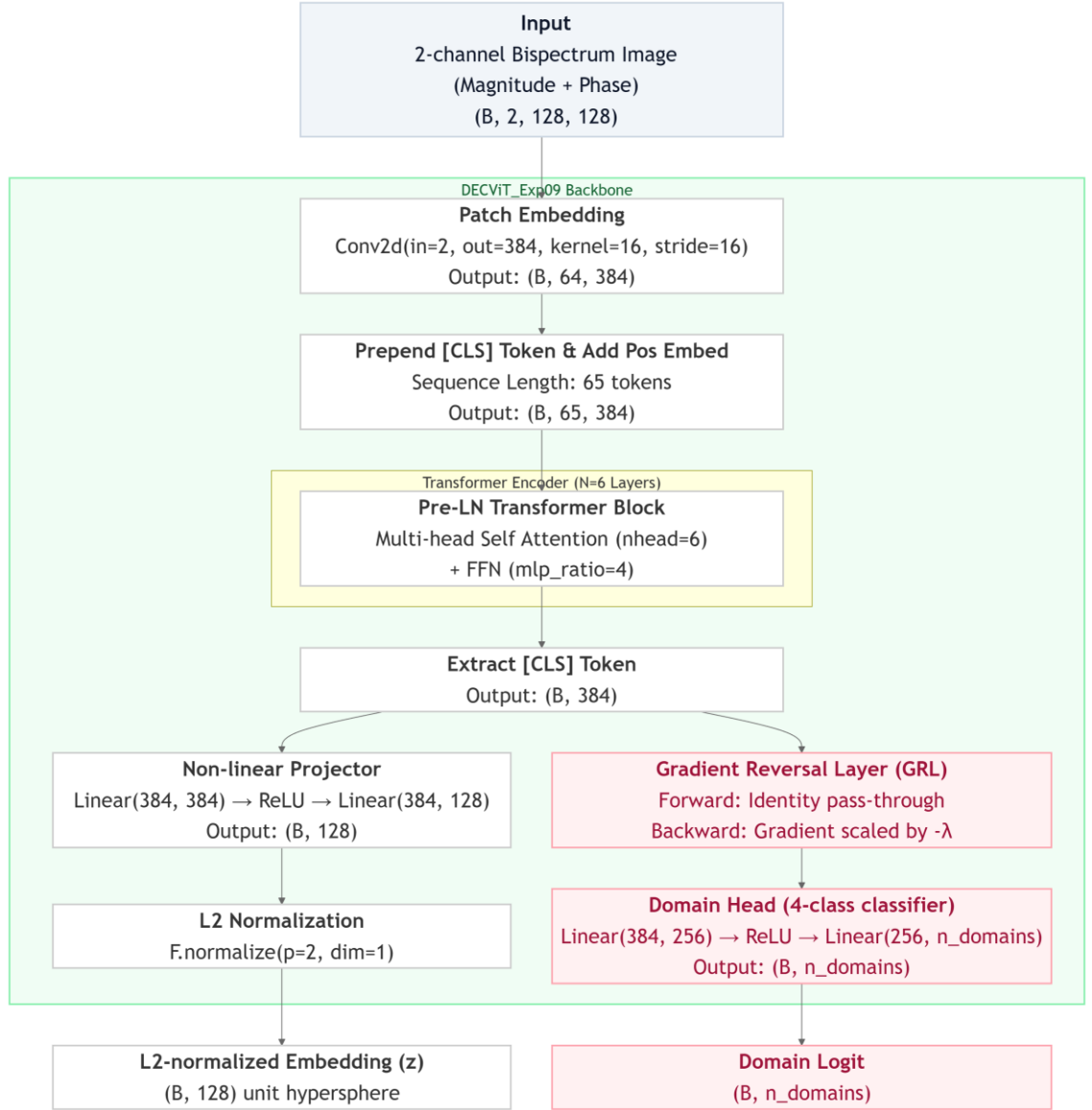


Figure 3.14: Model architecture for exp04.

(a) *Model Description:*

The proposed model is based on exp03 but adds on a domain-adversarial neural network (DANN) and a training method to make the model agnostic to the sensor used.

Input Representation Instead of magnitude only in exp03, the network takes a 2-channel complex bispectrum as input. The first channel represents the magnitude (the energy envelope of the signal), and the second channel represents the phase (a stable physical fingerprint). The input is processed at a resolution of $2 \times 128 \times 128$.

Patch Embedding The $2 \times 128 \times 128$ input is sliced into a grid of 16×16 non-overlapping patches. A convolutional layer treats each of these 64 patches as a "token" and, as before, projects the raw pixel data of both channels into a 384-dimensional feature vector.

[CLS] and Positional Embeddings are added to the tokens, as in exp03. In addition, the sequence of 65 tokens passes through the same 6 **Transformer Encoder** blocks.

Dual Output Heads After the encoder, the 64 image patches are discarded, and unlike before, the summarized [CLS] token is sent down two parallel paths:

1. **The Projector Head:** The token passes through linear layers to reduce its size from 384 down to a 128-dimensional vector (referred to as z). This vector is normalized to sit on a unit sphere (L2-normalization) and serves as the final embedding used for clustering the PD types. This is similar to exp03.
2. **The Domain Head:** A copy of the token passes through a Gradient Reversal Layer (GRL) into a small classifier that attempts to predict which dataset the pulse came from (e.g., simulation vs. real sensor). During the backward pass of the training (both in Phase 1 and Phase 2), the GRL artificially reverses the error signal from this head. This clever mathematical trick forces the main Transformer to learn features that hide the dataset's origin, making the model "domain-agnostic."

(b) ***Model Training:***

To ensure the model learns robust, sensor-independent features before sorting them into final categories, the training is split into two distinct phases.

Phase 1: Representation Learning (Pre-training) The goal of the first phase, as before, is to organize the 128-dimensional embedding space without explicitly assigning final PD classes. Two primary loss functions are optimized simultaneously:

- **Instance-Linked Supervised Contrastive Loss (SupCon):** The model is fed pairs of bispectra. It does not try to pull signals from the same class together anymore. Instead, it pulls augmented versions of the same signal, as well as signals that come from the same PD pulse instance but from a different sensor, closer together. This means that if two bispectra represent the exact same physical discharge event captured by *different* sensors, the loss function mathematically pulls their embeddings closer together. This forces the model to achieve "Sensor Invariance," learning that an attenuated signal from a faraway sensor is fundamentally the same as a sharp signal from a near sensor.
- **Domain Adversarial Loss:** The Domain Head is penalized if it fails to guess the dataset source. Because of the Gradient Reversal Layer, the main network is simultaneously penalized if the Domain Head guesses *correctly*. A dynamic schedule slowly increases the weight of this adversarial loss over time, smoothly guiding the model to domain-agnostic artifacts.

Phase 2: Semi-Supervised Cluster Refinement Once the embedding space is well-structured and domain-agnostic, Phase 2 begins. As usual, the model uses K-Means to establish initial cluster centroids, and then optimizes three loss functions to finalize the boundaries. The **Deep Embedded Clustering (DEC) KL-Divergence** and **Pairwise Constraint Loss** are the same as before. The **Domain Adversarial Loss** is maintained throughout Phase 2 to ensure that as the clusters tighten, the model does not revert to relying on dataset-specific biases.

(c) *Inference (testing) via HDBSCAN & Localisation*

Unlike previous experiments, t-SNE is replaced with Uniform Manifold Approximation and Projection for Dimension Reduction (Dalmia & Sia, 2021), because it preserves global topology and can transform new points in real-time (a prerequisite for online detection). TDOA, cross-correlation and PSO are also used for localization.

3.12 Hyperparameter Optimization

To ensure the deep learning pipelines across different experiments achieved optimal clustering performance without manual trial-and-error, an automated hyperparameter search was implemented using the Optuna optimization framework.

Optimization Objective and Pipeline The objective function for all optimization studies was formulated to minimize the total validation loss of the Semi-Supervised Deep Embedded Clustering (DEC) phase (Phase 2). Because the quality of the final clusters is highly dependent on the representations learned during pre-training, each Optuna trial executed an end-to-end training sequence tailored to the experiment's specific architecture:

1. **Phase 1 Training:** The backbone was trained using the corresponding contrastive objective (e.g., SimCLR, SupCon, or Instance-SupCon combined with Domain Adversarial Neural Network regularization) on an augmented training set.
2. **Centroid Initialization:** Embeddings were extracted from the pre-trained backbone, and a K-Means algorithm was applied to initialize the cluster centroids.
3. **Phase 2 Training:** The network was fine-tuned using the combined DEC losses (KL-divergence, alongside pairwise constraint losses or DANN losses depending on the experiment configuration).
4. **Evaluation:** The Phase 2 validation loss was calculated on an unaugmented validation set and returned as the trial's target metric for minimization.

To maximize computational efficiency and trial throughput, these tuning sequences were conducted using evenly sampled dataset subsets (e.g., up to 300 training and 100 validation samples per data source) and shortened epoch runs (typically 3 epochs per phase).

Hyperparameter Search Space The studies conducted searches over 15 distinct trials, exploring search spaces customized for each architecture's characteristics:

- **Phase 1 & Phase 2 Learning Rates:** Sampled from log-uniform distributions, generally searching within ranges from 10^{-6} up to 10^{-2} depending on the architecture. These dictate the step size during the initial contrastive feature extraction and the centroid refinement rate during the DEC phase, respectively.
- **Base Channels:** Categorically sampled from the options 16, 32, and 64 exclusively for CNN-based backbones to scale capacity and balance underfitting against over-parameterization.
- **Loss Weighting Parameters:** Tuned depending on the mathematical formulation of the experiment. Earlier experiments (exp01) optimized the pairwise constraint weight (`pairwise_weight_gamma`) between 0.1 and 10.0. Later domain-adversarial pipelines (exp04) optimized the adversarial penalty weight (`dann_weight`) using a log-uniform distribution bounded between 0.01 and 0.5.

3.13 Quantitative Results Analyses

3.13.1 Overall Performance and Classification Metrics

These metrics evaluate the model's ability to correctly map individual Partial Discharge (PD) pulses to their ground-truth physical defect classes.

- **Classification Accuracy:** The proportion of correctly classified PD pulses across the entire dataset.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (23)$$

(where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives)

- **Macro F1-Score:** The F1-score is the harmonic mean of Precision and Recall. The *Macro* variant calculates the F1-score for each ground-truth class (PD2 Incision, PD3 Delamination) independently and averages them, treating all classes equally regardless of their sample size.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (24)$$

$$Macro\ F1 = \frac{1}{N_{classes}} \sum_{i=1}^{N_{classes}} F1_i \quad (25)$$

- **Per-Class Precision and Recall:**
 - **Precision** evaluates exactness (of all pulses predicted as a specific defect, how many were actually that defect).

$$Precision = \frac{TP}{TP + FP} \quad (26)$$

- **Recall** evaluates completeness (of all actual pulses of a specific defect, how many did the model successfully find).

$$Recall = \frac{TP}{TP + FN} \quad (27)$$

3.13.2 Deep Clustering Quality Metrics

Because the pipeline utilizes unsupervised clustering (HDBSCAN) on deep embeddings, evaluating the topological structure of the embedding space is critical.

- **Silhouette Score:** A measure of cluster density and separation. For a single sample, it compares its mean distance to other samples in its own cluster (a) against the mean distance to samples in the nearest neighbouring cluster (b). The overall score is the average across all samples.

$$s = \frac{b - a}{\max(a, b)} \quad (28)$$

The score ranges from -1 to +1. A high positive value indicates that embeddings are tightly grouped within their own defect class and situated far away from other classes.

- **Cluster Purity:** Calculates the percentage of pulses within a discovered HDBSCAN cluster that share the exact same ground-truth class. High purity indicates the model successfully isolated a specific defect without mixing it. Let M be the set of discovered clusters, D be the set of true defect classes, and N be the total number of clustered data points.

$$Purity = \frac{1}{N} \sum_{m \in M} \max_{d \in D} |m \cap d| \quad (29)$$

- **HDBSCAN Clusters Found:** The absolute number of unique defect clusters discovered by the density-based clustering algorithm. Ideally, this should closely match the number of ground-truth physical defect classes present in the data.

3.13.3 Physical Event Grouping Metrics

These metrics evaluate how successfully the model was able to group 4 pulses (one from each channel) into the same pulse instance. This is computed by treating event grouping as a pairwise link prediction problem between pulses within the same scene.

Let TP be the number of pulse pairs correctly grouped into the same physical event, P be the total number of pairs grouped together by the model, and G be the total number of true pairs that belong to the same event.

- **Grouping Precision:** The percentage of grouped pulses that *actually* originated from the same physical event.

$$\text{Grouping Precision} = \frac{TP}{P} \quad (30)$$

- **Grouping Recall:** The percentage of pulses belonging to the *same physical event* that the algorithm successfully caught and grouped together.

$$\text{Grouping Recall} = \frac{TP}{G} \quad (31)$$

- **Predicted vs. Actual Instances:** A raw count comparing the true number of physical events in the dataset against the total number of discrete events successfully reconstructed by the grouping algorithm.

3.13.4 Computational Time Metrics

Metrics assessing the real-world deployment viability and pipeline latency.

- **Feature Extraction (Inference):** The time required for the CNN or ViT backbone to process the raw tensors into 128-dimensional embeddings.
- **Reduction Time (e.g., t-SNE / UMAP):** The execution time required to compress the high-dimensional embeddings into a lower-dimensional manifold prior to clustering.
- **HDBSCAN Clustering:** The execution time required by the density-based clustering algorithm to parse the geometry and assign cluster IDs.

3.13.5 Visualizations & Plots

The scripts generated several visual aids to qualitatively evaluate the manifold topology and architecture constraints.

- **UMAP / t-SNE Projections of the 128-D Features (Coloured by GT Class & Cluster):** A 2-dimensional compression of the 128-dimensional embedding space. These scatter plots visually demonstrate whether the model has collapsed the data into overlapping blobs or successfully separated them into distinct spatial islands.
- **HDBSCAN Cluster Composition (Stacked Bar Chart):** A bar chart detailing the ground-truth makeup of every discovered cluster. This visually highlights cluster impurities and reveals if the model is splitting a single defect class into multiple fragmented clusters.
- **Confusion Matrix:** A standard normalized heatmap visualizing the exact misclassification rates between predicted and true defect classes.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Semi-Supervised Deep Embedded Clustering (exp01)

The exp01 model was trained on the HFSS+MATLAB dataset, equation generated dataset, and the CWRU bearing dataset. It was then tested on the measured dataset consisting of (PD2) incision and (PD3) delamination defects (**cross-domain validation**). The model was trained for 10 epochs, for both Phase 1 and Phase 2.

4.1.1 3D t-SNE Reduction

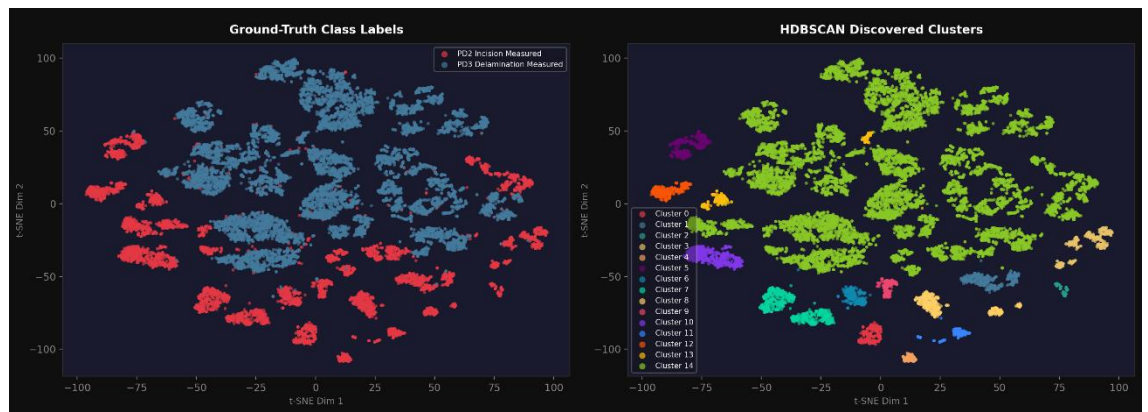


Figure 4.1: Visualization of exp01 embeddings with 3D t-SNE reduction.

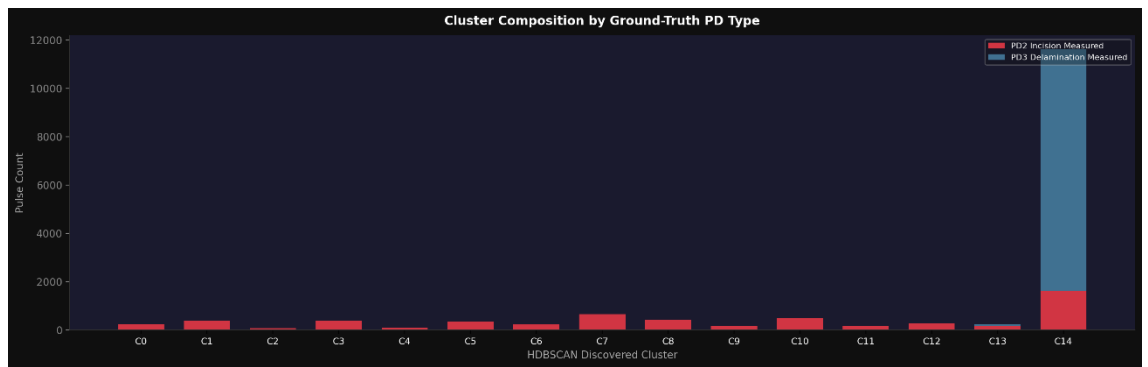


Figure 4.2: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp01 with 3D t-SNE reduction.

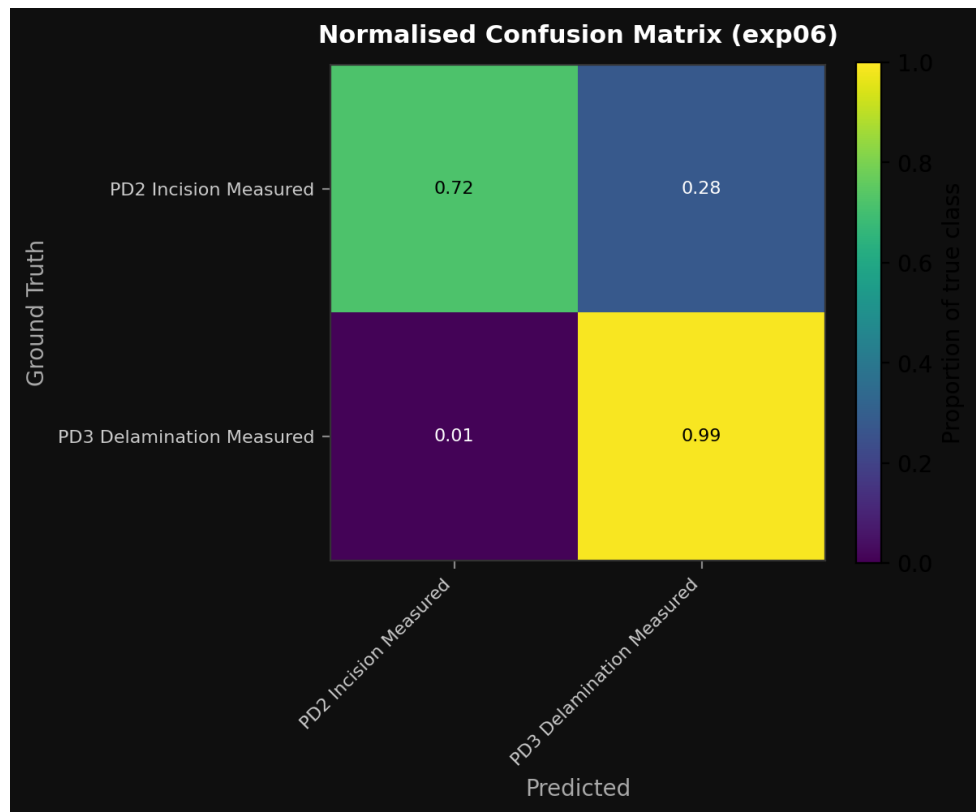


Figure 4.3: Confusion matrix for exp01 (3D t-SNE reduction).

Table 4.1: Quantitative metrics for exp01 (3D Reduction)

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value</i>
Overall Performance	Classification Accuracy	89.27%
	Macro F1-Score	0.8765
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	99.17% / 86.02%
	PD2 (Incision) Recall / Precision	72.21% / 98.05%
Clustering Quality	Silhouette Score	-0.0668
	HDBSCAN Clusters Found	15
	Cluster Purity	89.27%
Event Grouping	Grouping Precision / Recall	98.93% / 11.95%
	Predicted vs Actual Instances	13,330 vs 3,981
Computational Time	Feature Extraction (Inference)	62.12 s
	t-SNE Reduction (3D)	236.62 s
	HDBSCAN Clustering	1.02 s

In this configuration, the 128-dimensional embeddings generated by the DEC model were compressed down to 3 dimensions using t-SNE reduction, before applying the HDBSCAN clustering algorithm with a minimum cluster size of 5 and a cluster selection epsilon of 5.0. A concise summary of the key evaluation metrics is provided in Table 4.1.

As detailed in Table 4.1 and the Normalised Confusion Matrix (Figure 4.3), the model demonstrates strong overall classification performance. It successfully handles the macro-classification of the PD3 Delamination defect with exceptionally high recall. However, it encounters difficulties cleanly separating the PD2 Incision defect, exhibiting a notable drop in recall. Interestingly, the precision for PD2 remains near-perfect, implying that while the model misses several PD2 instances, the ones it does label as PD2 are almost certainly correct.

This drop in separation efficiency is explicitly explained by the internal clustering validation metrics (Table 4.1) and visually confirmed by the t-SNE Embedding Space plot (Figure 4.1). The model yielded a negative silhouette score, which indicates that the clusters within the 3-dimensional projection are highly overlapping, causing the features of both defect classes to interlock in a single continuous distribution rather than separating into distinct density islands. Because the cluster boundaries are poorly defined in this space, HDBSCAN struggles to establish clear partitions.

This leads to the phenomenon illustrated in the Cluster Composition chart (Figure 4.2), where the algorithm aggressively fragments the data into numerous separate clusters instead of identifying two distinct physical classes. The composition chart demonstrates how this overlapping space forces the algorithm into over-segmentation; however, despite this high degree of fragmentation into 15 clusters, the individual clusters remain relatively homogenous, resulting in the high overall cluster purity reported in Table 4.1.

In terms of physical event grouping, the grouping strategy yielded an outstanding grouping precision. This indicates that when pulses were grouped into a single predicted PD instance, they almost exclusively belonged to the same ground-truth physical event. Conversely, the grouping recall was notably low. Consequently, as shown in Table 4.1, the model significantly over-estimated the total number of physical PD events compared to the actual ground-truth.

Finally, an analysis of the computational throughput reveals the practical trade-offs of this configuration. While feature extraction and the final HDBSCAN clustering step were relatively efficient, the t-SNE dimensionality reduction proved to be a major computational bottleneck, requiring most of the total execution time to project the embeddings into 3 dimensions.

4.1.2 2D t-SNE Reduction

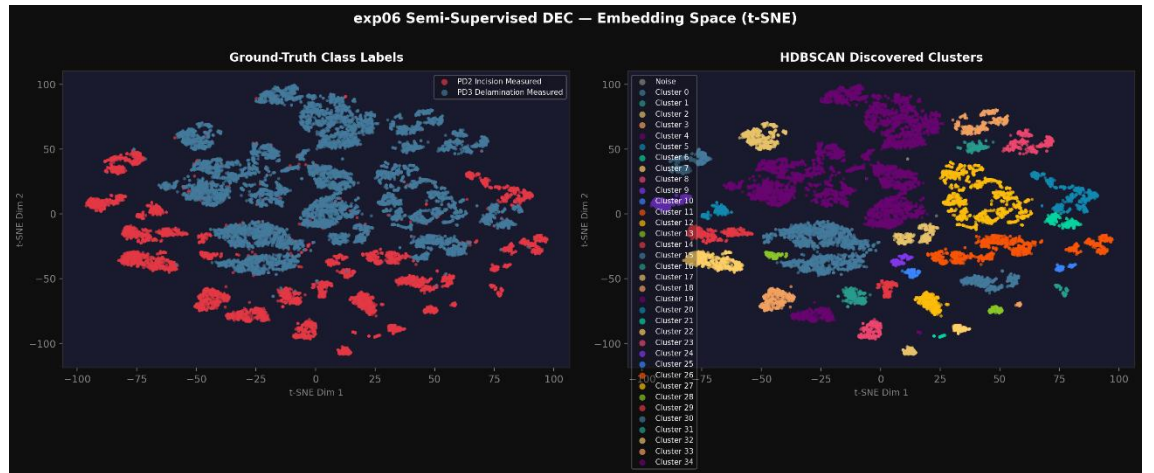


Figure 4.4: Visualization of exp01 embeddings with 2D t-SNE reduction.

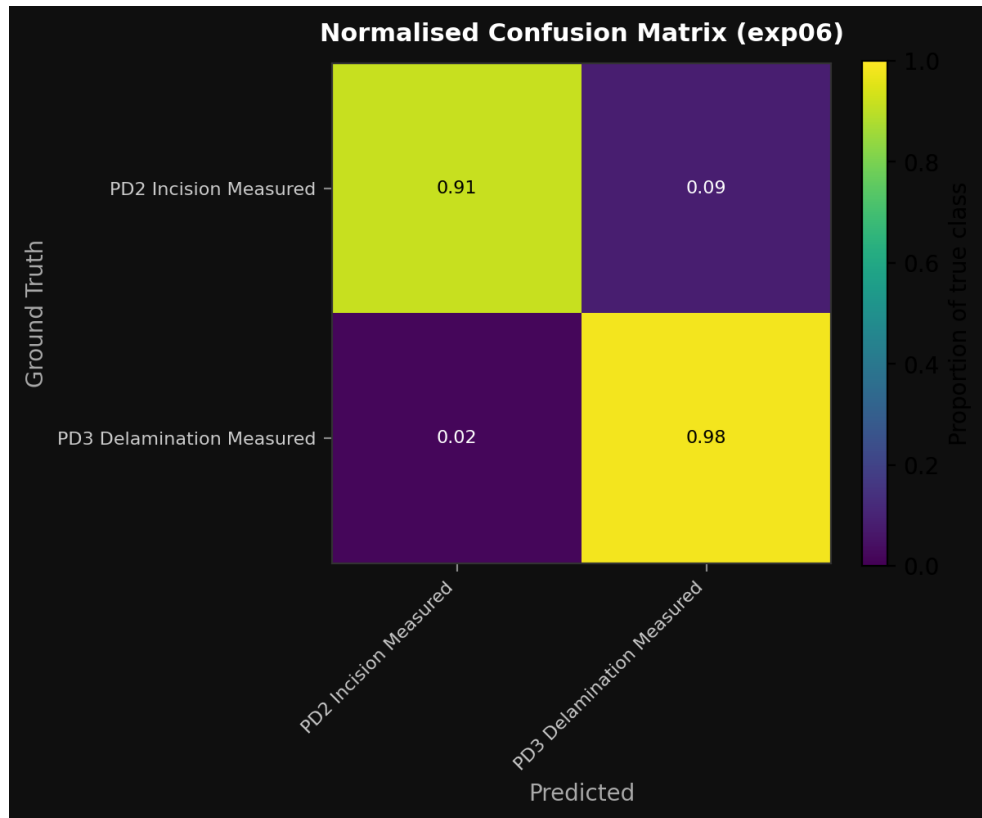


Figure 4.5: Confusion matrix for exp01 (2D t-SNE reduction).

Table 4.2: Quantitative metrics for exp01 (2D Reduction).

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (2D t-SNE)</i>
Overall Performance	Classification Accuracy	95.87%
	Macro F1-Score	0.9549
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	98.43% / 95.20%
	PD2 (Incision) Recall / Precision	91.44% / 97.13%
Clustering Quality	Silhouette Score	0.0301
	HDBSCAN Clusters Found	35
	Cluster Purity	95.87%
	Grouping Precision / Recall	98.93% / 11.95%
Event Grouping	Predicted vs Actual Instances	13,330 vs 3,981
	Feature Extraction (Inference)	66.58 s
Computational Time	t-SNE Reduction (2D)	118.19 s
	HDBSCAN Clustering	1.02 s

Building upon the previous analysis, a subsequent evaluation was conducted to assess the impact of compressing the 128-dimensional DEC embeddings down to 2 dimensions instead of 3, keeping all other HDBSCAN and grouping parameters identical.

As detailed in Table 4.2 and the corresponding Normalised Confusion Matrix (Figure 4.5), reducing the dimensionality to 2D yielded a significant improvement in overall classification performance, raising the accuracy to 95.87% and the macro F1-score to 0.9549. The most profound improvement occurred in the separation of the problematic PD2 Incision defect. Unlike the 3D projection, which struggled with a low recall of 72.21%, the 2D projection successfully captured 91.44% of the PD2 instances while maintaining an excellent precision of 97.13%.

The underlying cause of this performance boost is reflected in the internal clustering metrics. The silhouette score shifted from a negative value to a positive 0.0301. As visually supported by the updated t-SNE Embedding Space plot (Figure 4.4), this positive score indicates that the 2-dimensional projection was able to partially untangle the highly overlapping distribution observed in the 3D space, forming better-defined density islands. Interestingly, as shown in the table, achieving this clearer separation caused HDBSCAN to fragment the data even more aggressively, identifying 35 distinct clusters compared to the previous 15, which were too many to show on a cluster composition plot. However, because these numerous small sub-clusters mapped tightly to their respective ground-truth classes, the overall cluster purity increased notably to 95.87%.

Despite the improved clustering resolution, the physical event grouping metrics remained entirely unaffected. This is because the grouping of PD pulses into individual instances rely on the 128 vector embeddings, and not on any dimensionally reduced versions.

Finally, analyzing the computational throughput highlights an added practical benefit to this configuration. Projecting the embeddings down to 2 dimensions required 118.19 seconds, roughly halving the computational bottleneck observed during the 3-dimensional t-SNE reduction, while simultaneously delivering superior classification results.

4.1.3 No Reduction

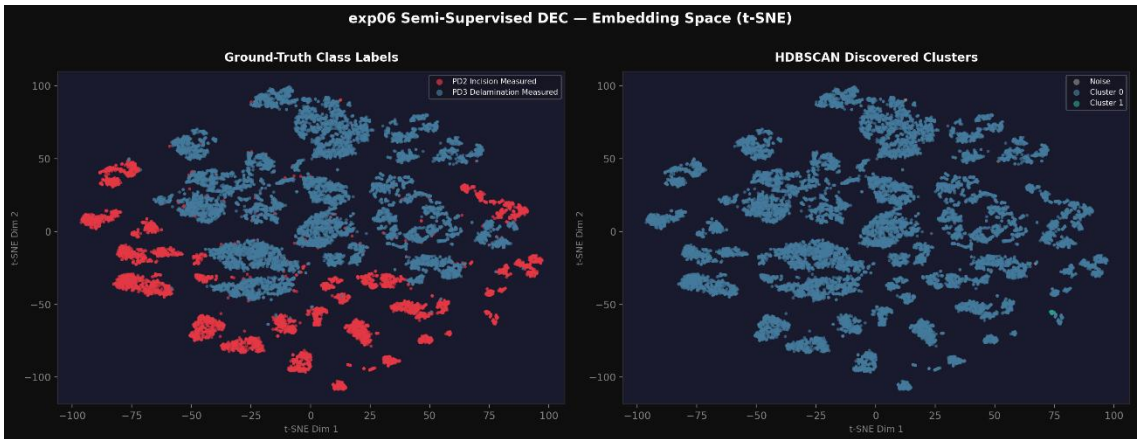


Figure 4.6: Visualization of exp01 embeddings with no dimensional reduction.

Table 4.3: Quantitative metrics for exp01 (No reduction).

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (No Reduction)</i>
Overall Performance	Classification Accuracy	63.50%
	Macro F1-Score	0.3921
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	100.0% / 63.45%
	PD2 (Incision) Recall / Precision	0.39% / 100.0%
Clustering Quality	Silhouette Score	0.4048
	HDBSCAN Clusters Found	2

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (No Reduction)</i>
<i>Event Grouping</i>	Cluster Purity	63.50%
	Grouping Precision / Recall	98.93% / 11.95%
	Predicted vs Actual Instances	13,330 vs 3,981
<i>Computational Time</i>	Feature Extraction (Inference)	62.12 s
	Dimensionality Reduction	0.00 s
	HDBSCAN Clustering	72.03 s

To establish a baseline, a final experiment applied HDBSCAN directly to the raw 128-dimensional DEC embeddings without t-SNE reduction.

As shown in Table 4.3, omitting dimensionality reduction caused a catastrophic performance drop. Overall accuracy fell to 63.50%, simply mirroring the dataset's natural class imbalance. The model completely failed to distinguish the defects: PD3 achieved 100% recall, while PD2 had just 0.39%. Effectively, almost every pulse was lumped into the PD3 class.

This failure demonstrates the "curse of dimensionality." In 128 dimensions, Euclidean distances lose contrast, making sparse data points appear roughly equidistant. As a result, HDBSCAN could not find distinct density partitions and grouped nearly the entire dataset into one massive, impure cluster, crashing the cluster purity to 63.50%.

Finally, while skipping t-SNE saved over 100 seconds, the computational burden merely shifted to the clustering algorithm. Calculating pairwise distances across 128 dimensions caused HDBSCAN's execution time to jump from ~1 second to 72.03 seconds. Ultimately, applying HDBSCAN directly to high-dimensional embeddings is computationally expensive and destroys classification accuracy.

4.2 Upgraded Semi-Supervised Deep Embedded Clustering (exp02)

The exp02 model was trained on the HFSS+MATLAB datasets, equation generated dataset, the CWRU bearing dataset, and the addition of a small, measured dataset consisting of only void defect PD. This dataset was added so that the model would learn the features of measured waveforms. It was then tested on another measured dataset consisting of *unseen* (PD2) incision and (PD3) delamination defect PDs. The purpose of the model was still maintained: the ability to adapt to unseen data, despite the fact that a small dataset from the testing data domain was incorporated in the training.

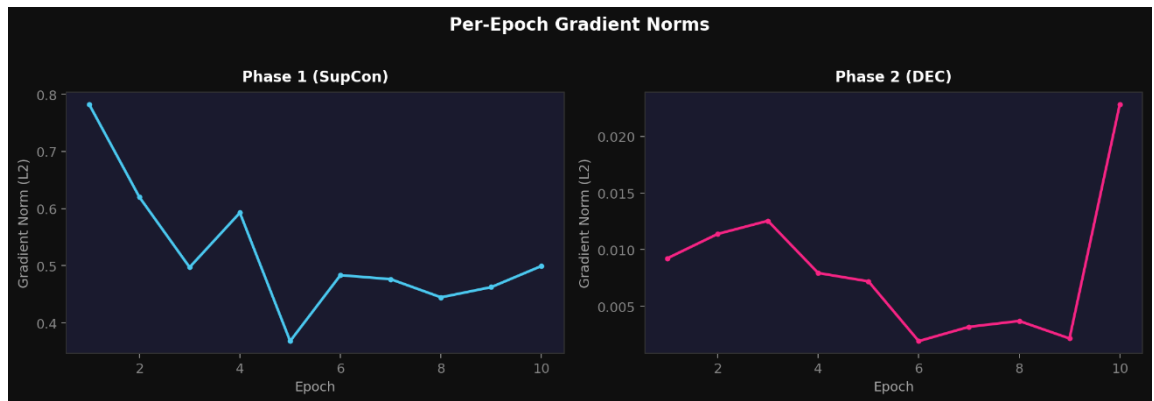


Figure 4.7: Per-epoch gradient norms for exp02.

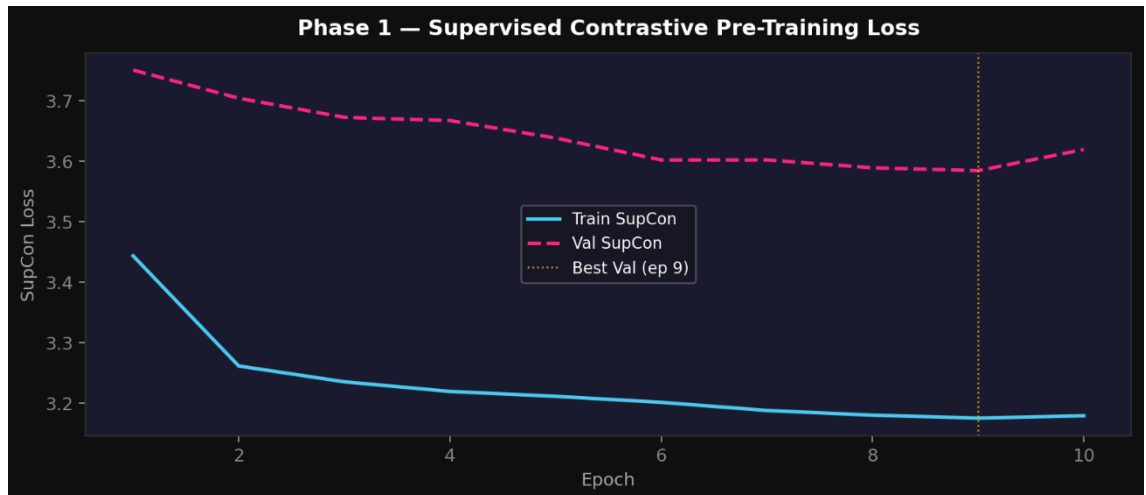


Figure 4.8: Phase 1 loss for exp02.

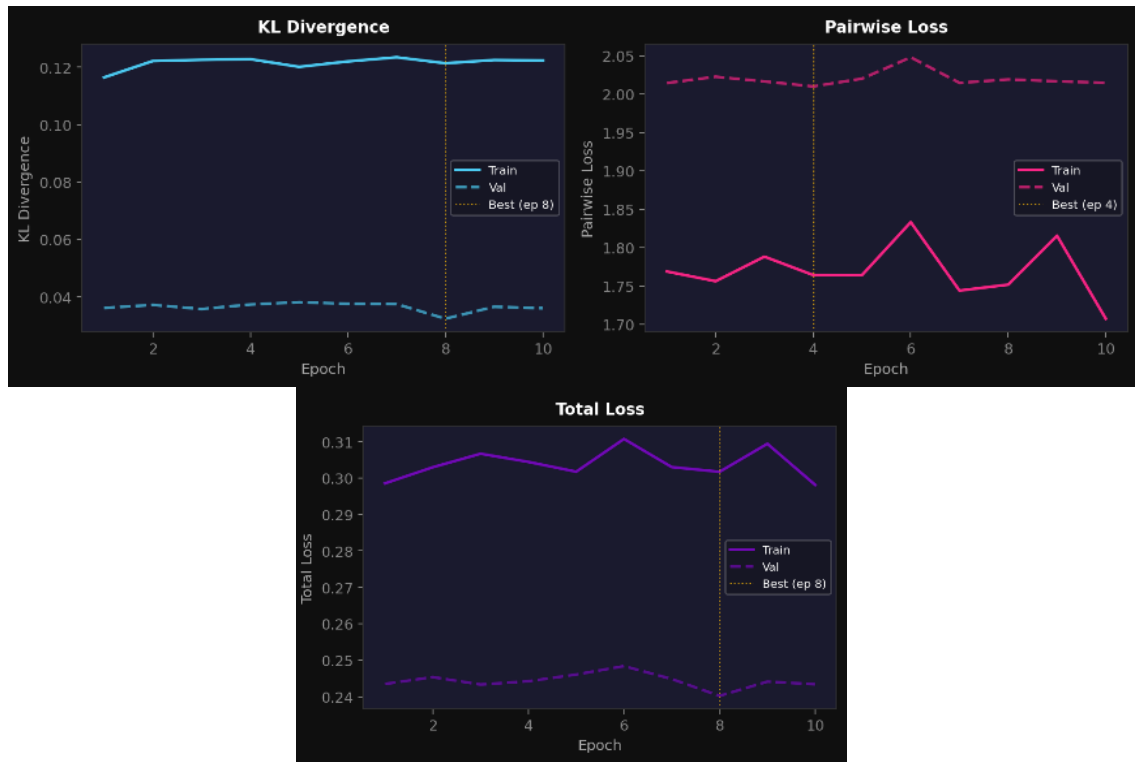


Figure 4.9: Phase 2 Loss for exp02.

From exp02 onwards, training metrics were recorded. exp02 was trained on an NVIDIA T4 GPU. As observed, Phase 1 lasted for 10 epochs requiring a time of approximately 33 seconds per epoch. The SupCon validation loss demonstrated a steady decline, achieving its optimal value of 3.5849 at epoch 9. Because Phase 2, which also lasted for 10 epochs, did not utilize computationally expensive signal augmentations, the average training time per epoch dropped significantly to approximately 21 seconds. The total validation loss reached its minimum of 0.2402 at epoch 8 of this phase (overall epoch 18).

4.2.1 3D t-SNE Reduction

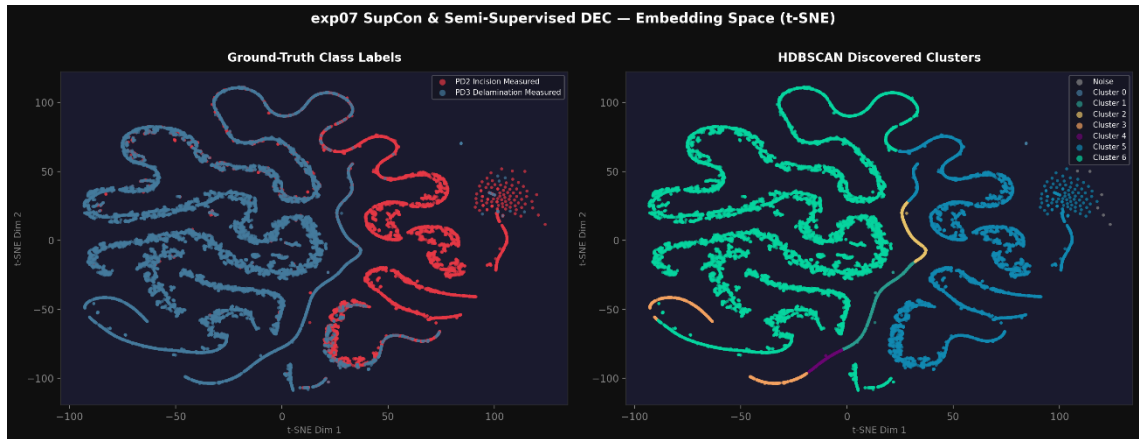


Figure 4.10: Visualization of exp02 embeddings with 3D t-SNE reduction.

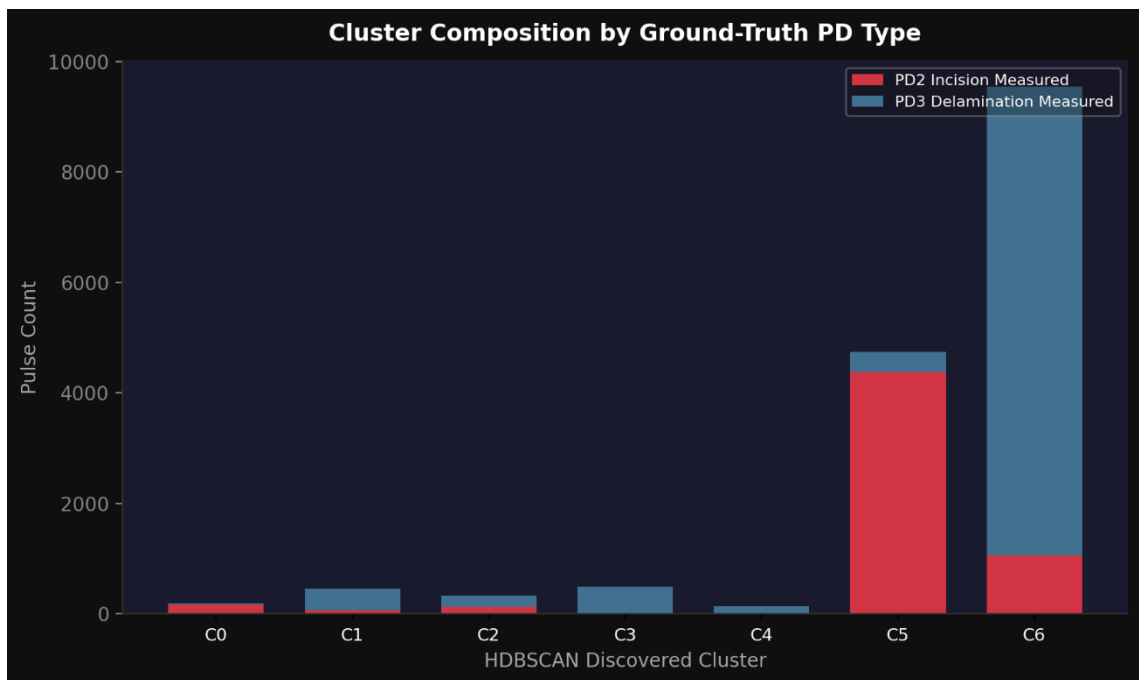


Figure 4.11: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp02 with 3D t-SNE reduction.

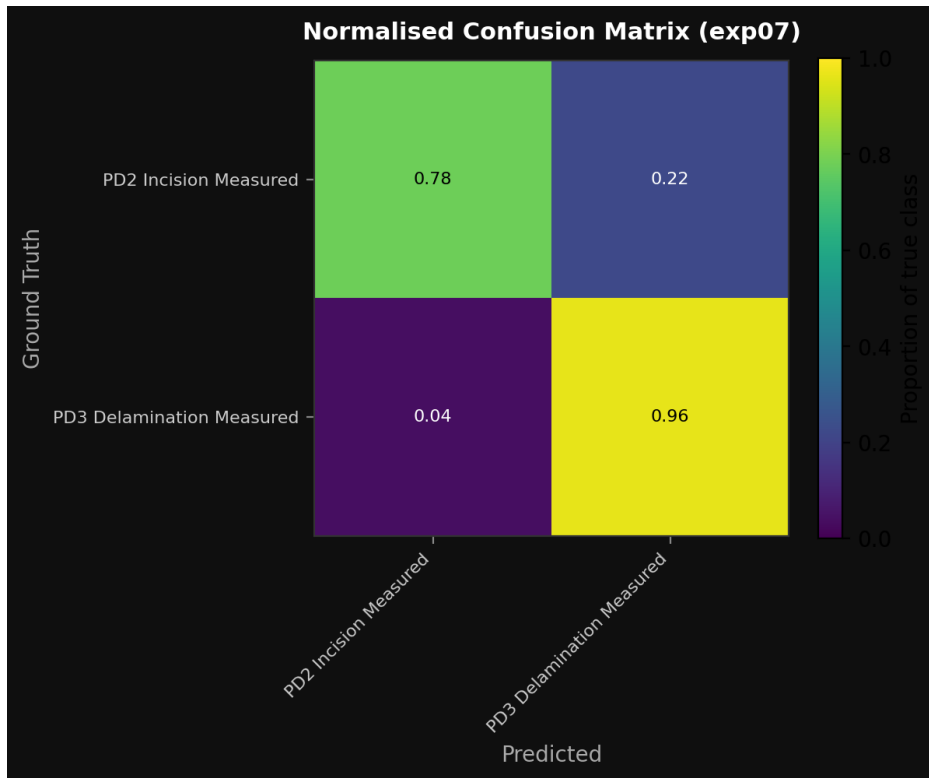


Figure 4.12: Confusion matrix for exp02 (3D t-SNE reduction)

Table 4.4: Quantitative metrics of exp02 (3D t-SNE reduction).

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (3D t-SNE)</i>
Overall Performance	Classification Accuracy	89.60%
	Macro F1-Score	0.8838
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	96.28% / 88.34%
	PD2 (Incision) Recall / Precision	78.06% / 92.40%
Clustering Quality	Silhouette Score	-0.3352
	HDBSCAN Clusters Found	7
	Cluster Purity	89.60%
Event Grouping	Grouping Precision / Recall	98.78% / 67.90%
	Predicted vs Actual Instances	6,773 vs 3,981
Computational Time	Feature Extraction (Inference)	100.34 s
	t-SNE Reduction (3D)	208.87 s
	HDBSCAN Clustering	1.00 s

An evaluation of Experiment 07 assessed the impact of replacing self-supervised SimCLR pre-training with Supervised Contrastive Learning (SupCon). In this configuration, t-SNE was used to reduce the dimensions to 3, $\epsilon = 5.0$ and `min_clusters = 5`, before HDBSCAN.

As shown in Table 4.4, SupCon provided marginal gains in overall classification performance compared to exp01, raising accuracy slightly to 89.60%. While PD2 recall improved to 78.06%, the silhouette score plummeted to -0.3352, indicating severe overlapping between the two defect classes in the 3-dimensional space. Despite this heavy inter-class overlap, HDBSCAN fragmented the space less aggressively than in exp01, discovering 7 clusters with an overall purity of 89.60%.

The most significant breakthrough of the SupCon architecture occurred during physical event grouping. By actively pulling same-class embeddings closer together during pre-training, pulses originating from the same physical discharge were compressed tightly enough to easily satisfy the 0.5 distance threshold. Consequently, while grouping precision remained near-perfect, grouping recall surged from 11.95% to 67.90%. This tightened embedding space drastically reduced the over-segmentation of physical events: the model predicted 6,773 total instances—roughly half the amount predicted by exp01—bringing the estimation much closer to the ground-truth of 3,981 instances.

4.2.2 2D t-SNE Reduction

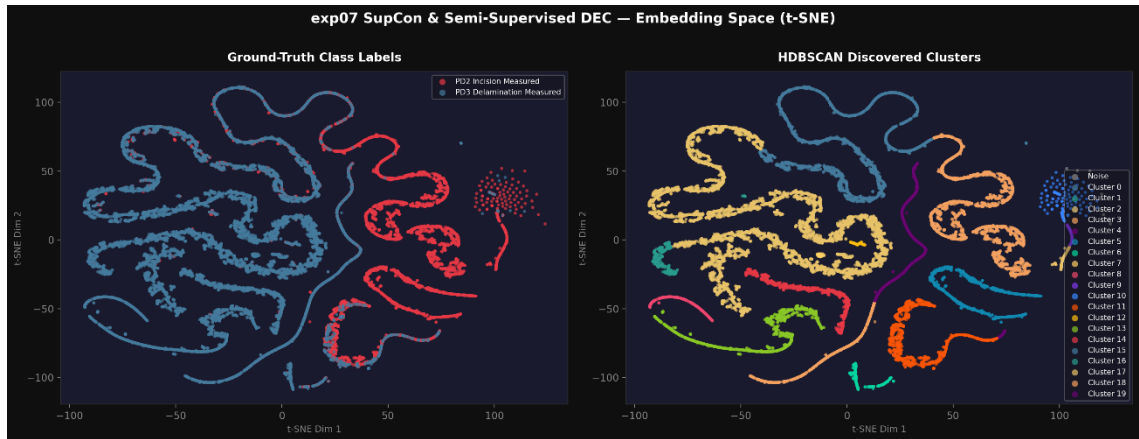


Figure 4.13: Visualization of exp02 embeddings with 2D t-SNE reduction.

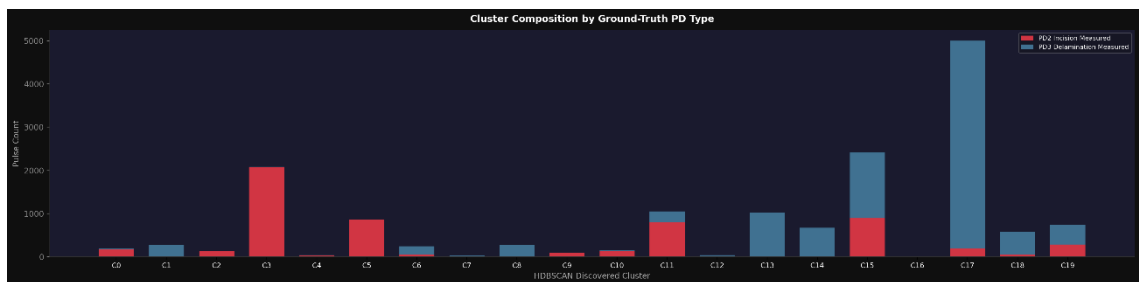


Figure 4.14: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp02 with 2D t-SNE reduction.

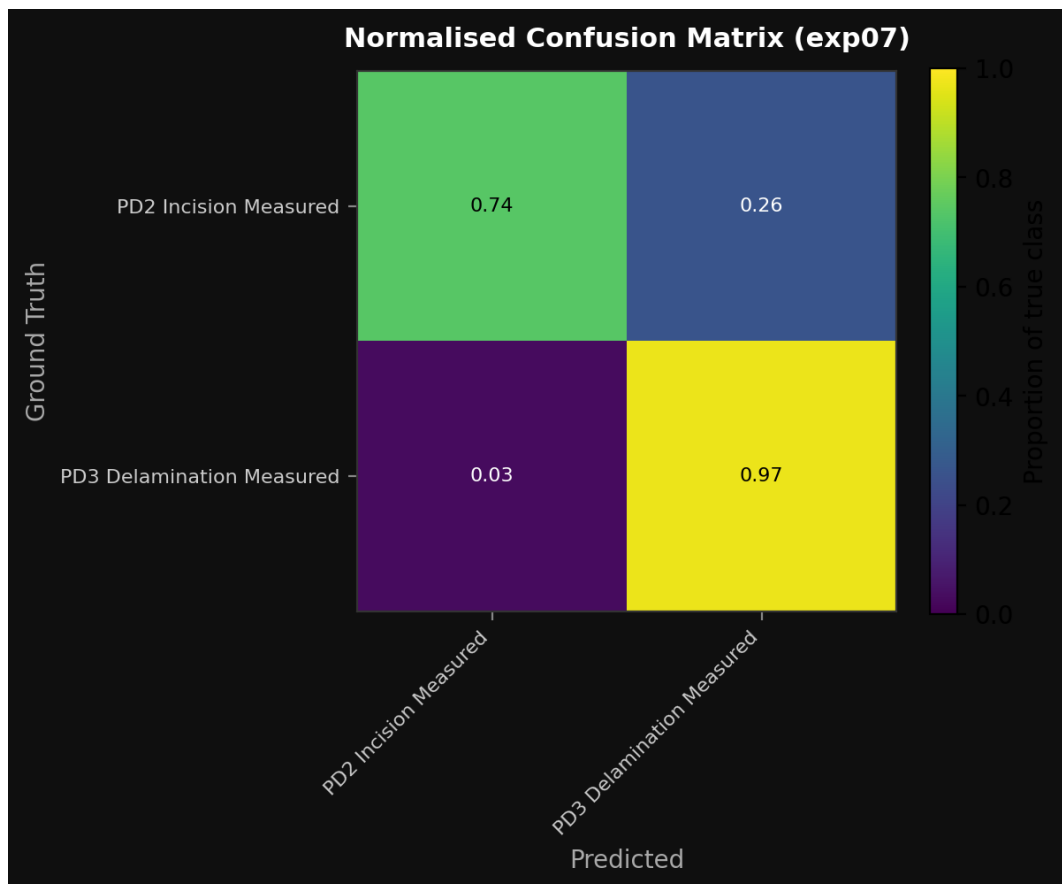


Figure 4.15: Confusion matrix for exp02 (2D t-SNE Reduction).

Table 4.5: Quantitative metrics of exp02 (2D t-SNE reduction).

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (2D t-SNE)</i>
Overall Performance	Classification Accuracy	88.59%
	Macro F1-Score	0.8706
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	97.07% / 86.54%
	PD2 (Incision) Recall / Precision	73.93% / 93.60%
Clustering Quality	Silhouette Score	-0.1284
	HDBSCAN Clusters Found	20
	Cluster Purity	88.59%
Event Grouping	Grouping Precision / Recall	98.78% / 67.90%
	Predicted vs Actual Instances	6,773 vs 3,981
Computational Time	Feature Extraction (Inference)	100.50 s
	t-SNE Reduction (2D)	99.06 s
	HDBSCAN Clustering	0.94 s

Building on the 3D analysis, this configuration evaluated the impact of compressing the SupCon embeddings down to 2 dimensions, with other parameters maintained.

As shown in Table 4.5, reducing the dimensionality from 3D to 2D caused a slight performance degradation, with overall accuracy dropping to 88.59% and PD2 recall decreasing to 73.93%. Interestingly, the silhouette score improved to -0.1284. This indicates that while the 2D projection partially untangled the severe overlap seen in 3D, it achieved this by aggressively fragmenting the data into 20 distinct clusters (up from 7). Ultimately, this higher fragmentation in 2D yielded slightly worse classification mappings than the 3D projection.

Notably, the physical event grouping metrics (67.90% recall, 6,773 instances) remained perfectly identical to the 3D configuration. Because the temporal grouping algorithm operates directly on the raw 128-dimensional backbone embeddings rather than the t-SNE projections, this confirms that the drastic improvement in grouping recall over exp01 is purely a result of the tighter embedding space created by the SupCon pre-training.

Finally, while classification performance dipped slightly, projecting to 2 dimensions required only 99.06 seconds, effectively halving the t-SNE computational bottleneck observed in the 3D configuration.

4.2.3 No Reduction

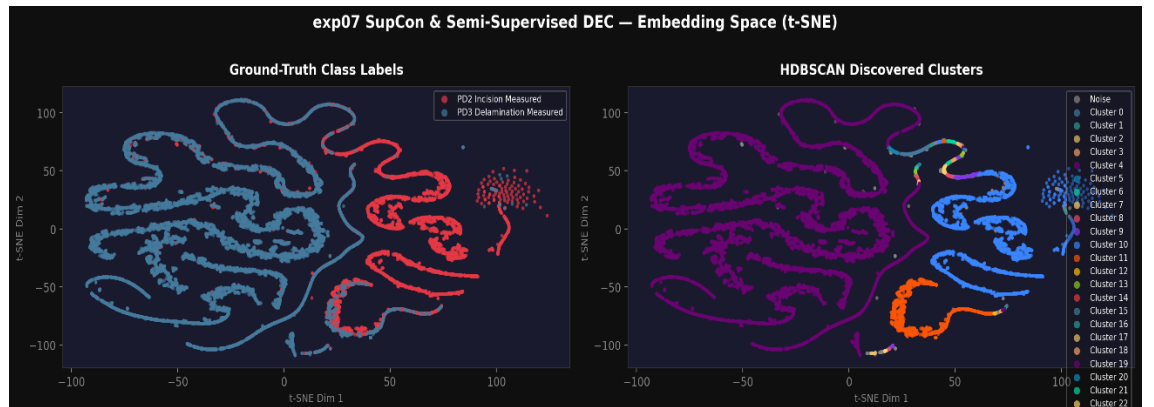


Figure 4.16: Visualization of exp02 embeddings with no dimensional reduction.

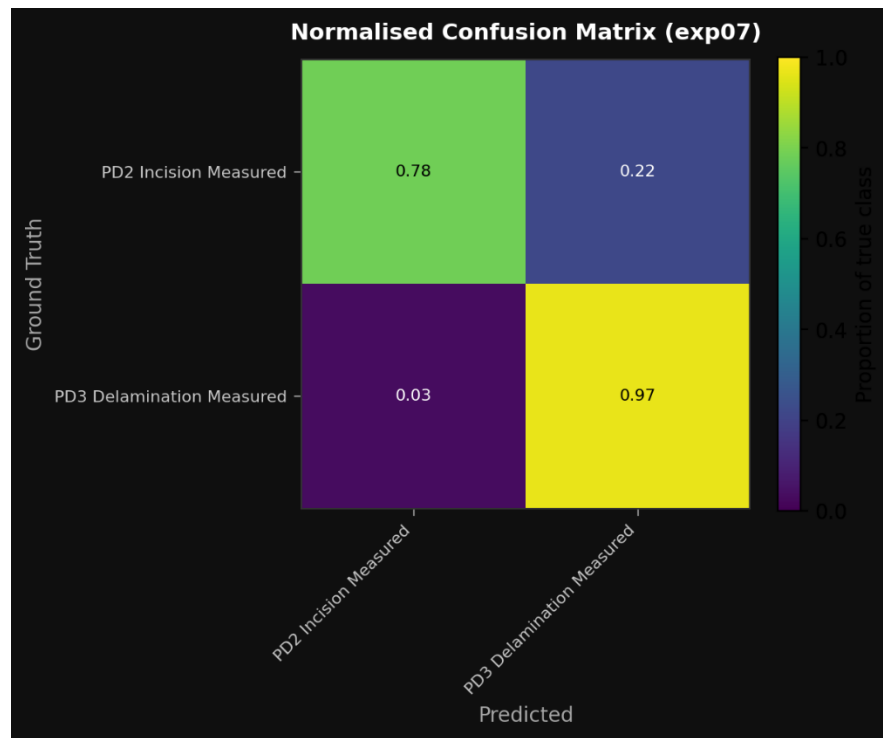


Figure 4.17: Confusion matrix for exp02 (No dimensional reduction)

Table 4.6: Quantitative metrics of exp02 (No reduction).

Metric Category	Metric Name	Value (No Reduction)
Overall Performance	Classification Accuracy	90.19%
	Macro F1-Score	0.8890
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	96.78% / 88.93%
	PD2 (Incision) Recall / Precision	78.36% / 93.12%
Clustering Quality	Silhouette Score	0.4890
	HDBSCAN Clusters Found	41
	Cluster Purity	90.19%
Event Grouping	Grouping Precision / Recall	98.78% / 67.90%
	Predicted vs Actual Instances	6,773 vs 3,981
Computational Time	Feature Extraction (Inference)	101.11 s
	Dimensionality Reduction	0.00 s
	HDBSCAN Clustering	39.93 s

To establish a baseline for the SupCon architecture, this configuration applied HDBSCAN directly to the raw 128-dimensional embeddings without any dimensionality reduction. A highly restrictive epsilon ($\epsilon = 0.01$) was added, which was selected so that the model wouldn't combine everything into one cluster, unlike in exp01.

As detailed in Table 4.6, running HDBSCAN directly on the unreduced SupCon embeddings yielded the best classification accuracy (90.19%) of all three configurations. This stands in stark contrast to the previous SimCLR baseline, which failed catastrophically when applied to 128 dimensions. The SupCon pre-training successfully structured the embedding space so tightly that Euclidean distances remained meaningful even in high dimensions. This is further proven by the exceptionally high, positive silhouette score of 0.4890, indicating well-defined density partitions. While HDBSCAN did heavily fragment this space into 41 distinct sub-clusters, they were highly homogenous, leading to peak cluster purity.

Because the physical event grouping algorithm evaluates distance thresholds directly on the 128-dimensional backbone embeddings, the grouping metrics (67.90% recall, 6,773 instances) remained identical to the 2D and 3D configurations.

Finally, this unreduced configuration proved highly efficient. Omitting the t-SNE step entirely eliminated the ~100 to 200-second reduction bottleneck. While calculating distances across 128 dimensions understandably increased HDBSCAN's execution time to 39.93 seconds, the total computational pipeline was significantly faster than utilizing any form of t-SNE projection. Ultimately, the SupCon architecture completely negates the need for dimensionality reduction, achieving optimal accuracy and speed directly on the raw embeddings.

4.3 Bispectrum Analysis and Vision Transformer (ViT) Architecture (exp03)

The exp03 model, like exp02, was trained on the HFSS+MATLAB datasets, equation generated dataset, the CWRU bearing dataset, and a small, measured dataset consisting of only void defect PD. It was then tested on another measured dataset consisting of (PD2) incision and (PD3) delamination defects.

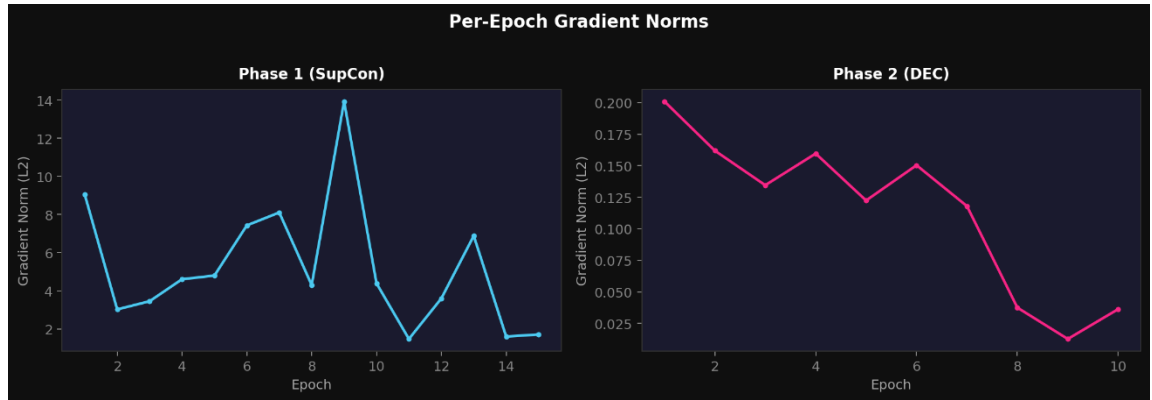


Figure 4.18: Per-epoch gradient norms of exp03.

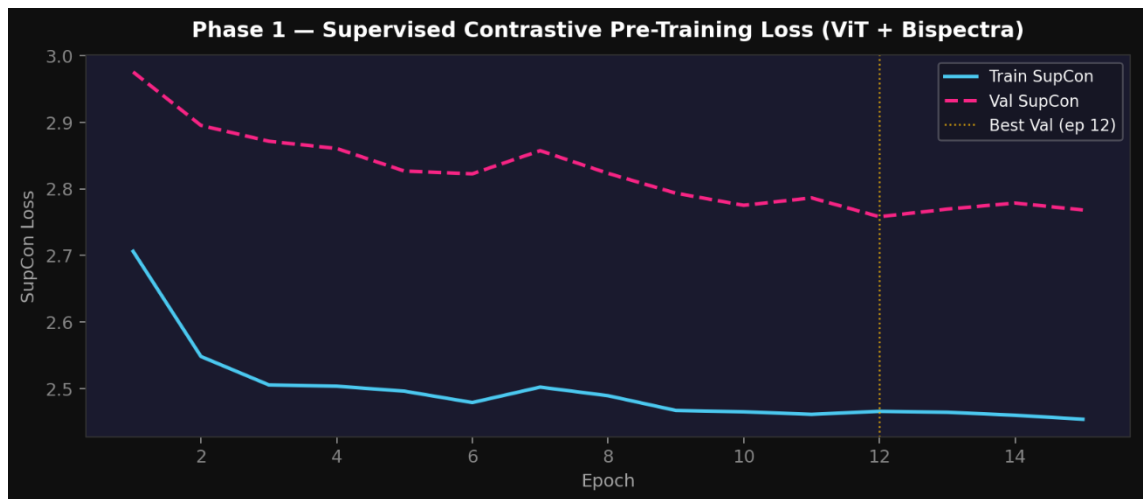


Figure 4.19: Phase 1 loss for exp03.

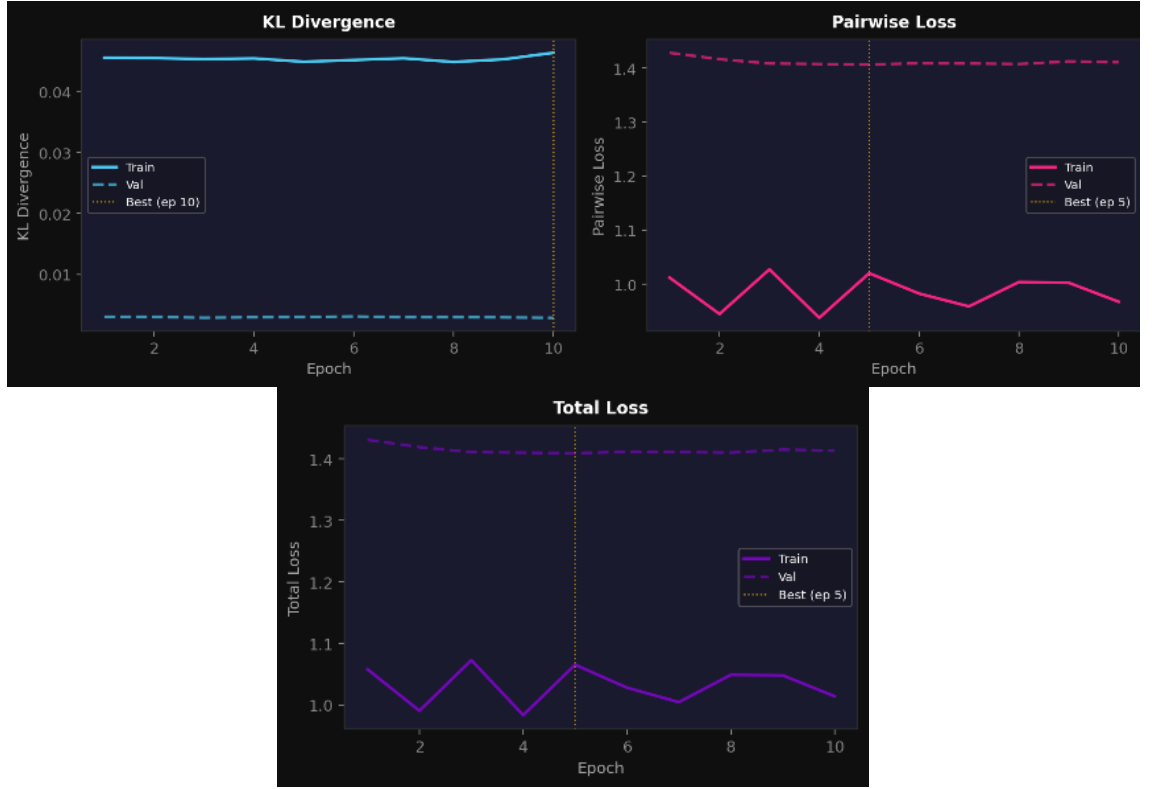


Figure 4.20: Phase 2 loss of exp03.

Operating with a batch size of 32, the total training duration recorded was 8420.9 seconds, yielding a mean epoch time of 327.14 seconds. Training was also carried out on an NVIDIA T4 GPU.

During Phase 1, the model was trained for 15 epochs to construct an initial L2-normalised embedding space on a unit hypersphere. The time per epoch during this phase was relatively high, ranging between 334 and 342 seconds. The SupCon validation loss demonstrated a steady decrease, reaching its lowest value of 2.7580 at epoch 12 before exhibiting minor fluctuations in subsequent epochs. Notably, the gradient norms during this pre-training phase displayed significant variance, spiking up to 13.92 at epoch 9.

Upon transitioning to Phase 2, the DEC clustering layer was initialized using K-Means centroids, and the model was trained for an additional 10 epochs. A notable reduction in per-epoch training time was observed, dropping to between 305 and 315 seconds. This reduction occurred because the dataset loader bypassed the computationally intensive dual-view 2D augmentations (such as Gaussian noise and spatial masking) required

exclusively for the SupCon phase. Throughout this phase, the validation KL divergence remained consistently low, hovering around 0.003. Concurrently, the total validation loss converged steadily, achieving a best configuration of 1.4093. The stability of this refinement process is further evidenced by the per-epoch gradient norms, which exhibited a smooth, rapid decay from 0.201 down to 0.012. This steady decline confirms that the sharpened Student-t target distribution successfully stabilized the soft cluster assignments without suffering from cluster collapse.

4.3.1 3D t-SNE Reduction

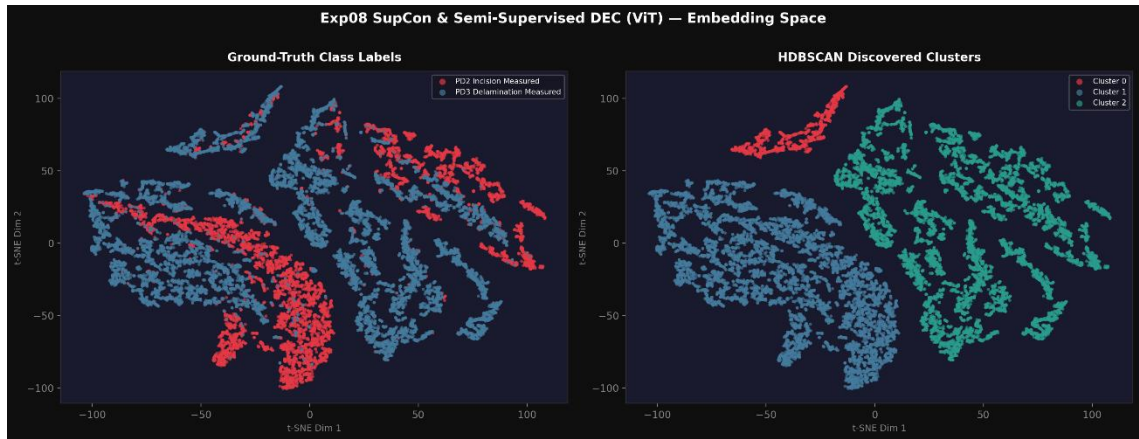


Figure 4.21: Visualization of exp03 embeddings with 3D t-SNE reduction.

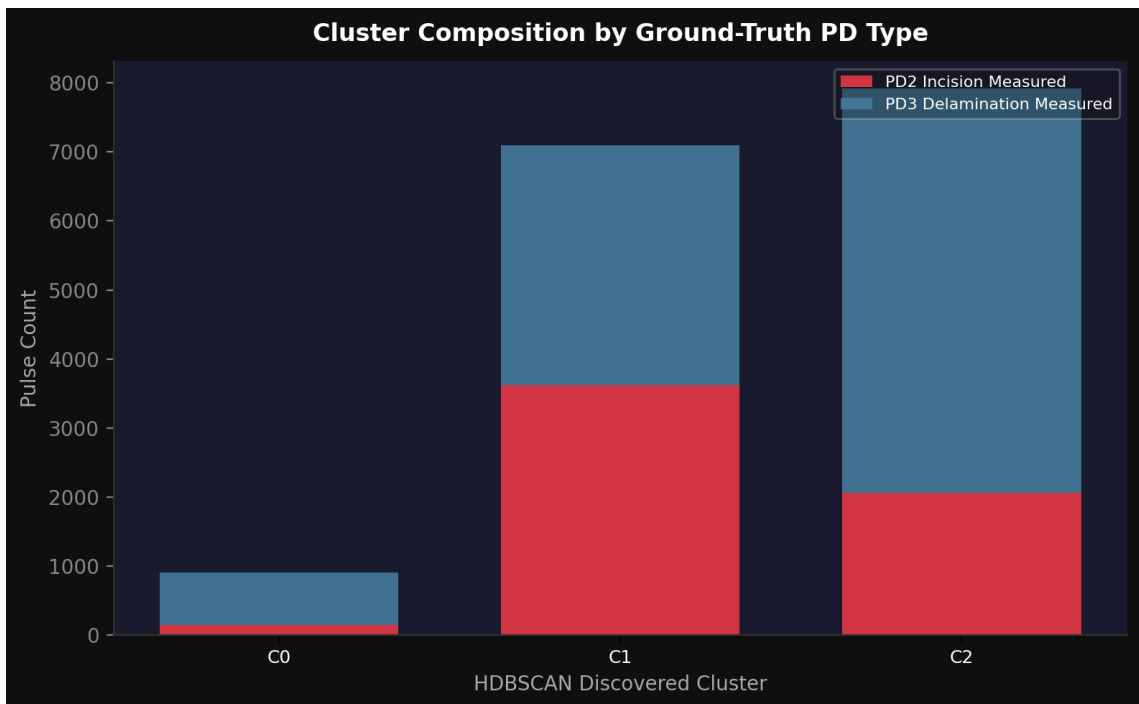


Figure 4.22: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp03 with 3D t-SNE reduction.

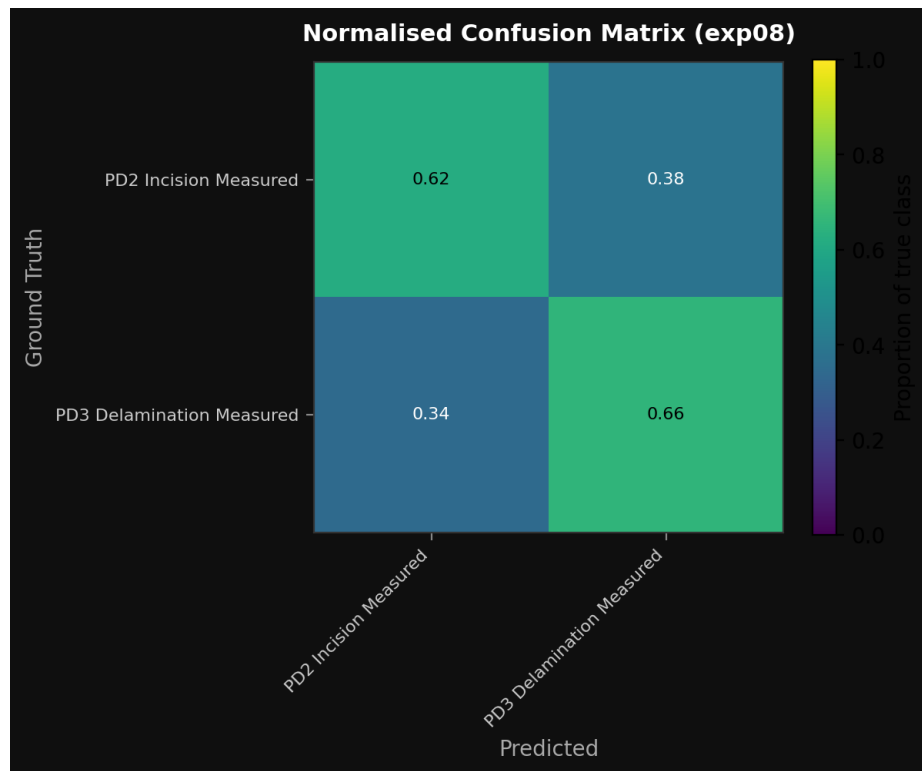


Figure 4.23: Confusion matrix for exp03 (3D t-SNE reduction).

Table 4.7: Quantitative metrics for exp03 (3D t-SNE Reduction)

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (3D t-SNE)</i>
Overall Performance	Classification Accuracy	64.28%
	Macro F1-Score	0.6298
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	65.57% / 74.88%
	PD2 (Incision) Recall / Precision	62.06% / 51.10%
Clustering Quality	Silhouette Score	0.8847
	HDBSCAN Clusters Found	3
	Cluster Purity	64.28%
Event Grouping	Grouping Precision / Recall	98.91% / 53.74%
	Predicted vs Actual Instances	7,237 vs 3,981
Computational Time	Feature Extraction (Inference)	716.93 s
	t-SNE Reduction (3D)	133.92 s
	HDBSCAN Clustering	0.48 s

This specific configuration utilized a 3-dimensional t-SNE reduction and $\epsilon=5.0$ prior to HDBSCAN clustering.

As detailed in Table 4.7, the architectural shift to a ViT processing bispectra resulted in a catastrophic drop in classification capability. Overall accuracy plummeted to 64.28%, with both the PD2 and PD3 defect classes exhibiting poor precision and recall metrics hovering between 50% and 75%. Interestingly, the clustering algorithm yielded an exceptionally high, positive silhouette score of 0.8847 and discovered exactly 3 tightly packed clusters with zero noise points.

This combination of terrible classification accuracy alongside perfect spatial clustering metrics indicates a fundamental failure in feature learning. The ViT failed to extract discriminative spectral features from the bispectrum grids; instead, it learned a highly generalized representation that compressed almost the entire dataset into 3 massive, totally impure clusters (evidenced by the cluster purity crashing to 64.28%, which simply mirrors the dataset's natural imbalance). The embeddings for both defect classes are perfectly mixed within these dense clusters, making semantic separation impossible.

Furthermore, the computational cost of this approach was exceptionally high. Feature extraction (inference) took 716.93 seconds—more than seven times slower than the lightweight 2D CNN used in previous experiments. Given the heavy computational burden of the self-attention mechanism and the complete collapse of classification accuracy, the ViT-Bispectrum configuration proves to be highly ineffective for this specific PD classification task.

4.3.2 2D t-SNE Reduction

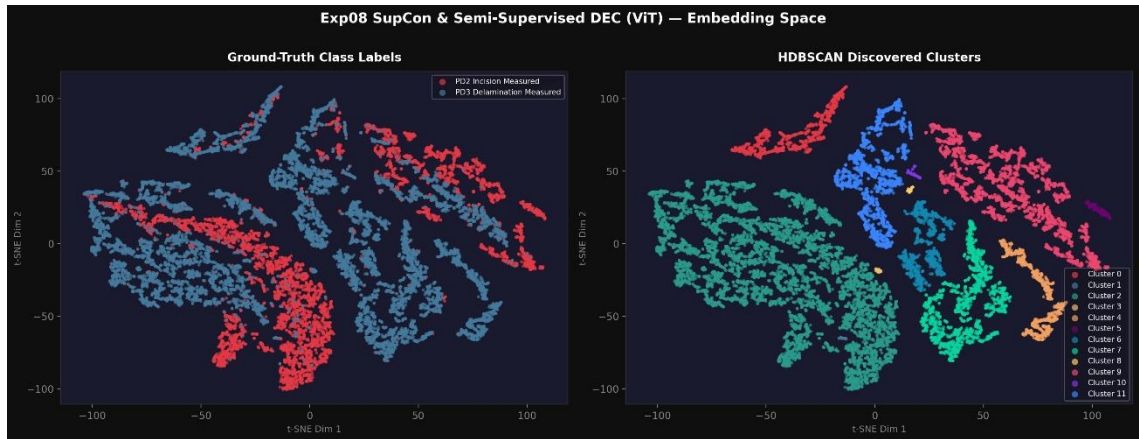


Figure 4.24: Visualization of exp03 embeddings with 2D t-SNE reduction.

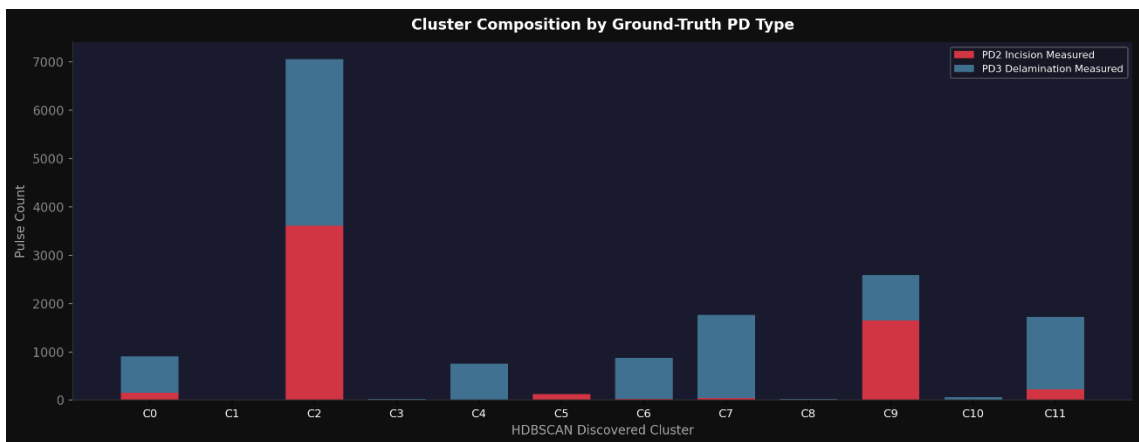


Figure 4.25: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp03 with 2D t-SNE reduction.

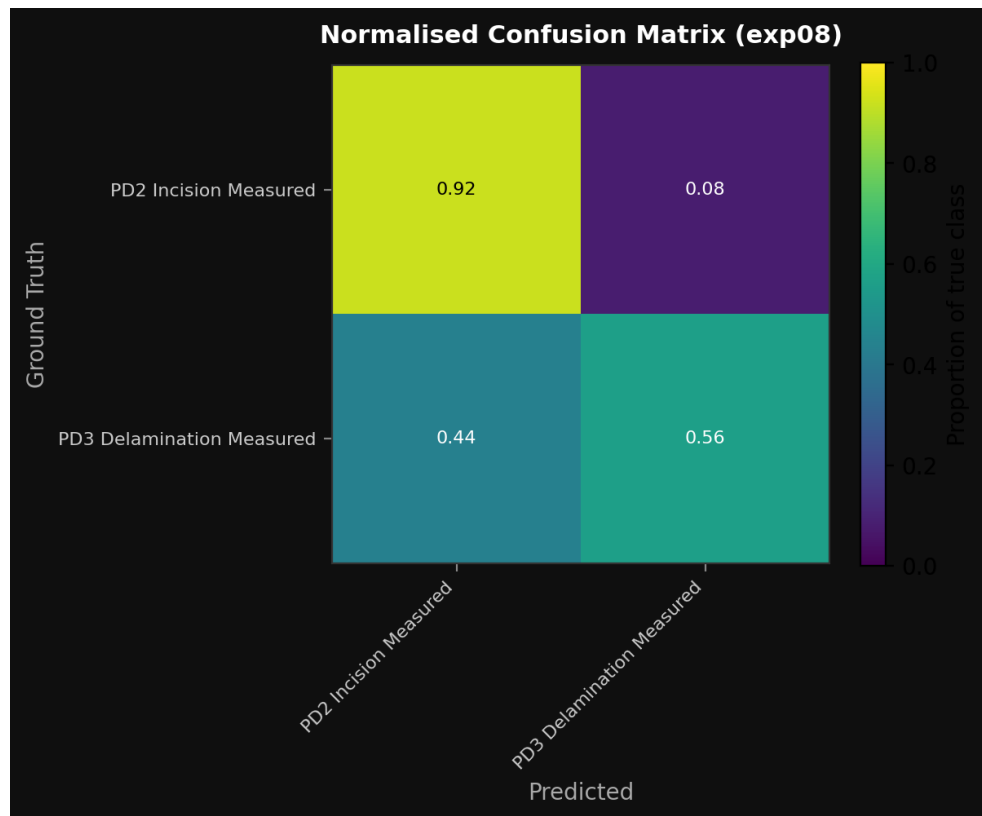


Figure 4.26: Confusion matrix for exp03 (2D t-SNE reduction).

Table 4.8: Quantitative metrics for exp03 (2D t-SNE Reduction)

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (2D t-SNE)</i>
Overall Performance	Classification Accuracy	69.55%
	Macro F1-Score	0.6954
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	56.45% / 92.54%
	PD2 (Incision) Recall / Precision	92.15% / 55.09%
Clustering Quality	Silhouette Score	-0.1413
	HDBSCAN Clusters Found	12
	Cluster Purity	69.55%
Event Grouping	Grouping Precision / Recall	98.91% / 53.74%
	Predicted vs Actual Instances	7,237 vs 3,981
Computational Time	Feature Extraction (Inference)	857.09 s
	t-SNE Reduction (2D)	120.32 s
	HDBSCAN Clustering	0.89 s

Building upon the previous analysis of Experiment 08, this configuration evaluated the performance of the ViT-Bispectrum configuration when compressed down to a 2-dimensional t-SNE projection, with $\epsilon=5.0$.

As detailed in Table 4.8, projecting the embeddings to 2 dimensions provided a slight, yet insufficient, boost in overall classification accuracy (69.55%) compared to the 3D configuration. However, the per-class metrics remain heavily unbalanced and unacceptably poor for a deep learning application. The model aggressively over-predicted the PD2 defect—yielding a high recall but a terrible precision of 55.09%—while simultaneously missing nearly half of the PD3 instances.

The underlying cause of this poor separation is explicitly confirmed by the internal clustering metrics. The 2-dimensional projection yielded a highly negative silhouette score of -0.1413, indicating severe, interlocking overlap between the two defect distributions. HDBSCAN attempted to untangle this overlapping space by fragmenting the data into 12 distinct clusters (up from 3), but because the learned ViT features are fundamentally poorly separated, the overall cluster purity remained low at 69.55%.

Consistent with previous experiments, the physical event grouping metrics (53.74% recall, 7,237 predicted instances) remained completely identical to the 3D configuration.

Finally, this 2-dimensional configuration suffers from the same massive computational bottlenecks. Feature extraction alone was prohibitively expensive at 857.09 seconds, while the 2D t-SNE reduction required an additional 120.32 seconds. Ultimately, while compressing to 2 dimensions slightly alters the error distribution, it does not salvage the fundamental inability of the ViT-Bispectrum architecture to learn highly discriminative features for this specific classification task.

4.3.3 No Reduction

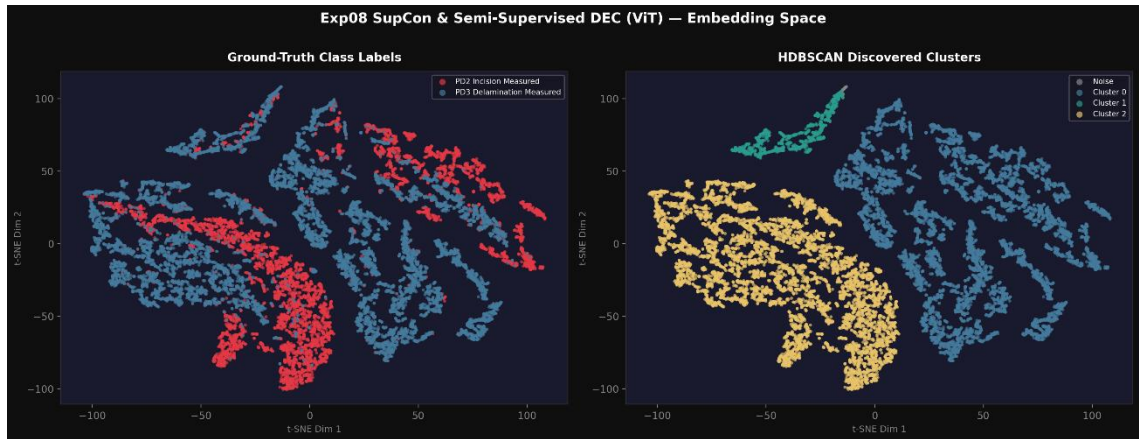


Figure 4.27: Visualization of exp03 embeddings with no dimensional reduction.

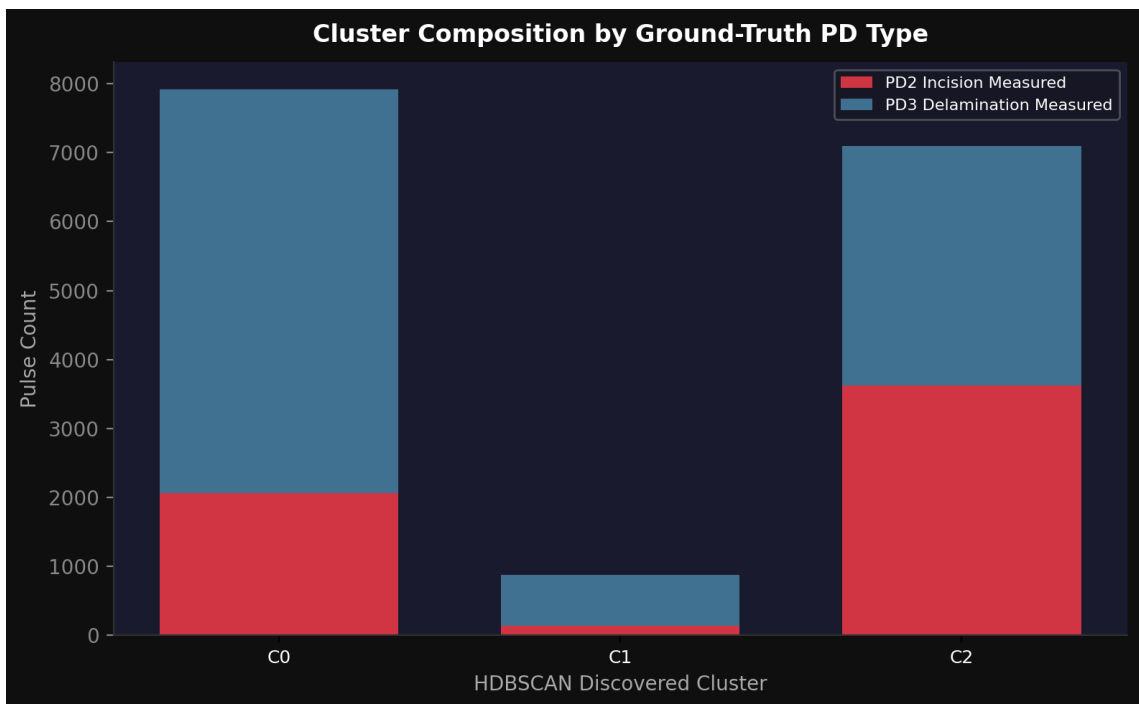


Figure 4.28: Cluster composition by PD type: Red - PD2, Blue - PD3, for exp03 with no dimensional reduction.

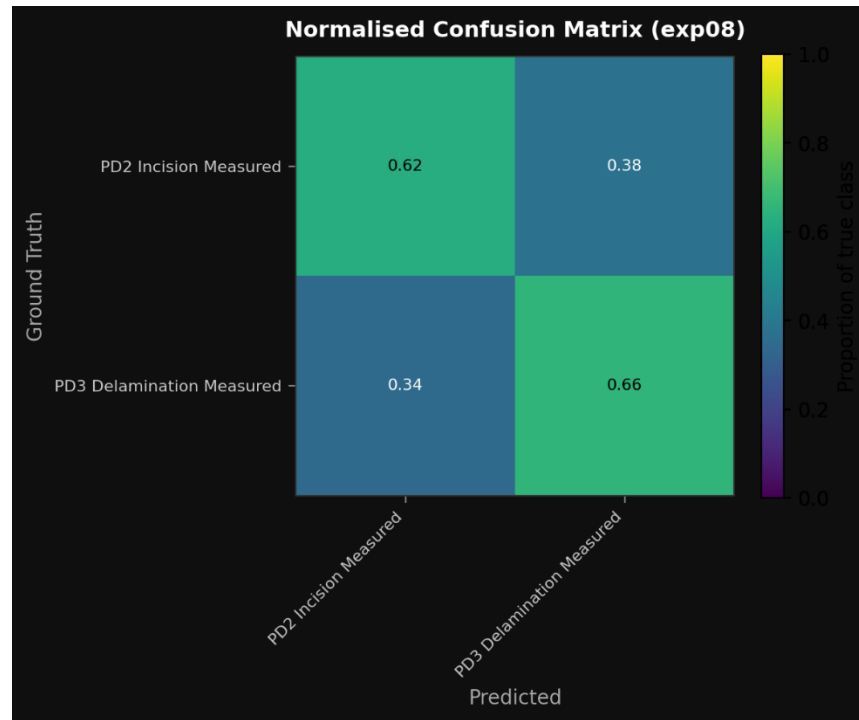


Figure 4.29: Confusion matrix for exp03 (No dimensional reduction).

Table 4.9: Quantitative metrics for exp03 (No Reduction)

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (No Reduction)</i>
Overall Performance	Classification Accuracy	64.28%
	Macro F1-Score	0.6299
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	65.50% / 74.92%
	PD2 (Incision) Recall / Precision	62.17% / 51.09%
Clustering Quality	Silhouette Score	0.8836
	HDBSCAN Clusters Found	3
	Cluster Purity	64.28%
Event Grouping	Grouping Precision / Recall	98.91% / 53.74%
	Predicted vs Actual Instances	7,237 vs 3,981
Computational Time	Feature Extraction (Inference)	389.94 s
	Dimensionality Reduction	0.00 s
	HDBSCAN Clustering	23.18 s

To establish a baseline for the ViT-Bispectrum architecture, a final evaluation applied HDBSCAN directly to the raw 128-dimensional embeddings without any t-SNE reduction, with $\epsilon = 0.0$ (meaning that HDBSCAN was applied naturally, without trying to merge clusters together based on a threshold).

As detailed in Table 4.9, the classification performance on the raw embeddings remains firmly locked at a catastrophic 64.28% accuracy, with both defect classes exhibiting unacceptably poor recall and precision. Interestingly, the raw 128-dimensional space yielded an exceptionally high silhouette score of 0.8836 and produced exactly 3 dense clusters.

This finding is highly significant. Unlike the earlier uncompressed SimCLR baseline (exp01) which collapsed because high-dimensional Euclidean distances simply lost their contrast, this ViT model failed for the opposite reason. The extremely high silhouette score proves that the Vision Transformer actively and deliberately compressed the entire dataset into very dense, tightly bound clusters directly within the 128D space. However, because the cluster purity perfectly mirrors the dataset's natural imbalance (64.28%), these dense clusters are entirely heterogeneous and devoid of semantic separation. This provides definitive proof that the ViT completely failed to extract class-discriminative features from the bispectrum grids, instead learning a highly generalized representation (likely generic pulse energy) that clumped both classes together.

Finally, while omitting t-SNE entirely removed the dimensionality reduction bottleneck, the configuration remains computationally inefficient. Feature extraction alone required nearly 390 seconds—vastly slower than the lightweight 2D CNN configurations used in earlier experiments. Ultimately, this final baseline confirms that the ViT-Bispectrum combination is a fundamentally flawed approach for this specific

application, yielding severe computational overhead and a total collapse in classification accuracy compared to the simpler CNN-STFT models.

4.4 Upgraded Vision Transformer (exp04)

The exp04 model was trained on the HFSS+MATLAB datasets, equation generated dataset, and a small, measured dataset consisting of only void defect PD. Unlike exp03, the CWRU dataset wasn't included, because it would interfere with the clustering and feature extraction in this domain-agnostic model. The DANN was configured, so that it would only be agnostic to the HFSS+MATLAB and measured dataset domains. It was then tested on another measured dataset consisting of (PD2) incision and (PD3) delamination defects.

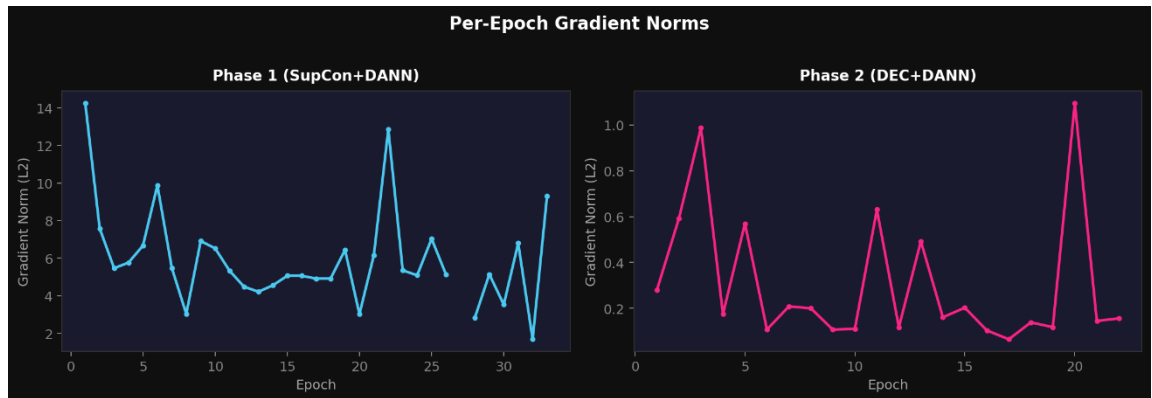


Figure 4.30: Per-epoch gradient norms for exp04.

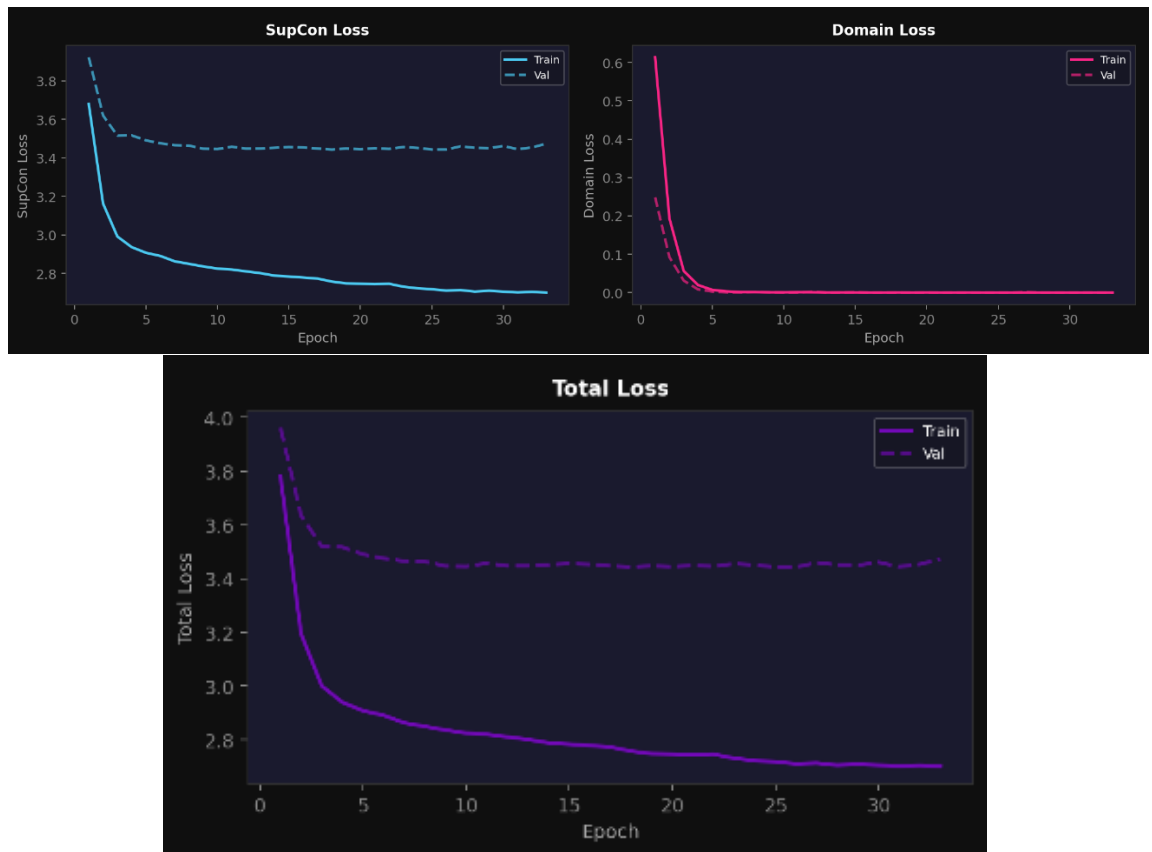


Figure 4.31: Phase 1 loss for exp04.

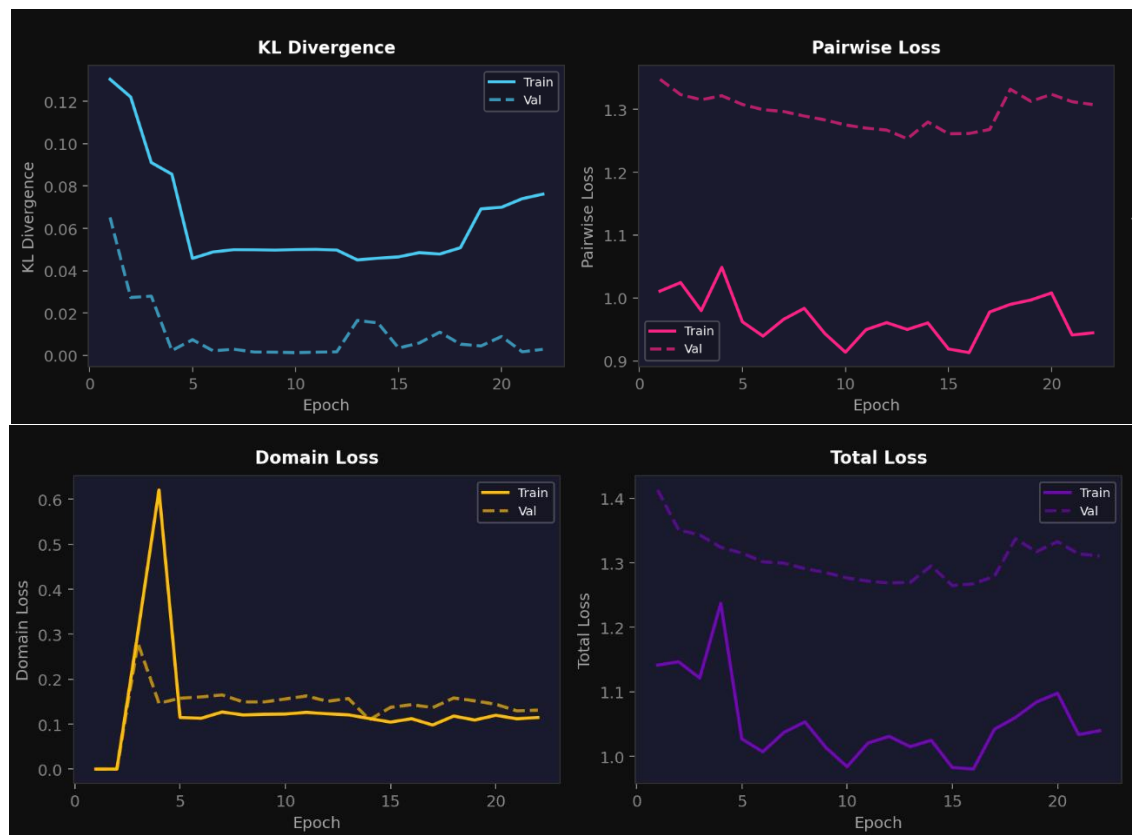


Figure 4.32: Phase 2 loss for exp04.

To optimize the 11.16 million trainable parameters of the Vision Transformer (ViT), the network was trained on the pre-processed bispectrum features using an NVIDIA L4 GPU (a higher tier GPU than the one used for previous experiments). The total training process took approximately 3.63 hours, with an average epoch duration of 235 seconds.

In the first phase, the network was pre-trained for 33 epochs. Observing the Phase 1 training metrics, the SupCon loss experienced a sharp initial descent before smoothly converging to a minimum validation loss of 3.442. Concurrently, the domain loss rapidly diminished to near zero within the first five epochs. This rapid decline indicates that the gradient reversal layer within the DANN successfully penalized domain-specific signatures early in the training process, compelling the ViT backbone to extract features common to both the synthetic HFSS models and the physical high-voltage experiments.

In the second phase, the model transitioned to a semi-supervised Deep Embedded Clustering (DEC) objective for an additional 22 epochs. An analysis of the Phase 2 loss curves reveals that the KL divergence and pairwise losses stabilized early in the phase. Notably, the domain loss exhibited a distinct, transient spike between the third and fifth epochs, reaching a peak train loss of 0.62 before restabilizing around 0.11 for the remainder of the training. This behaviour suggests that the initial formation and tightening of the DEC clusters temporarily disrupted the domain invariance learned in Phase 1, a perturbation which the DANN was subsequently able to correct and stabilize. The optimal network weights were saved at overall epoch 48 (Phase 2, epoch 15), corresponding to the lowest recorded Phase 2 validation loss of 1.264. Throughout all 55 epochs, the domain loss weight increased slowly and plateaued at its maximum (0.2) at around epoch 30.

4.4.1 2D UMAP Reduction

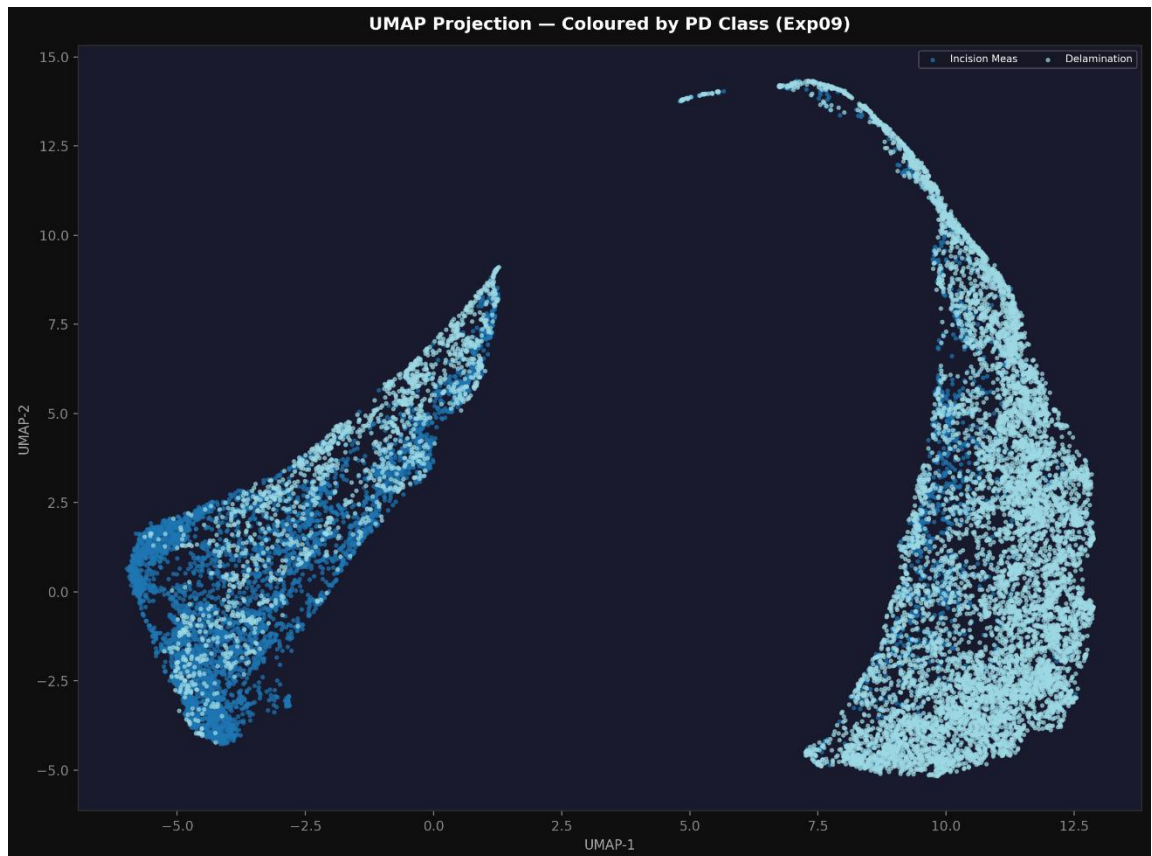


Figure 4.33: exp04 embeddings in 2D with ground-truth PD classes labelled.

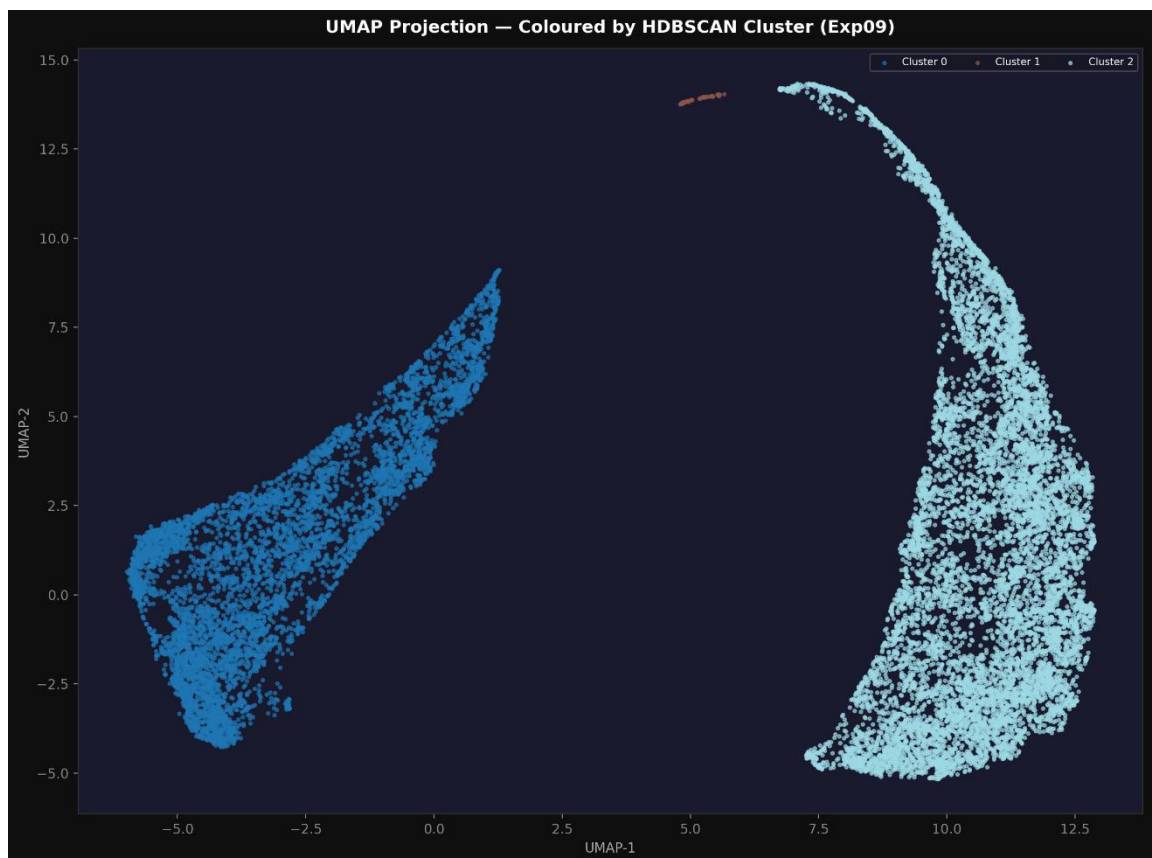


Figure 4.34: HDBSCAN inferred clusters of exp04 with 2D reduced dimensions.

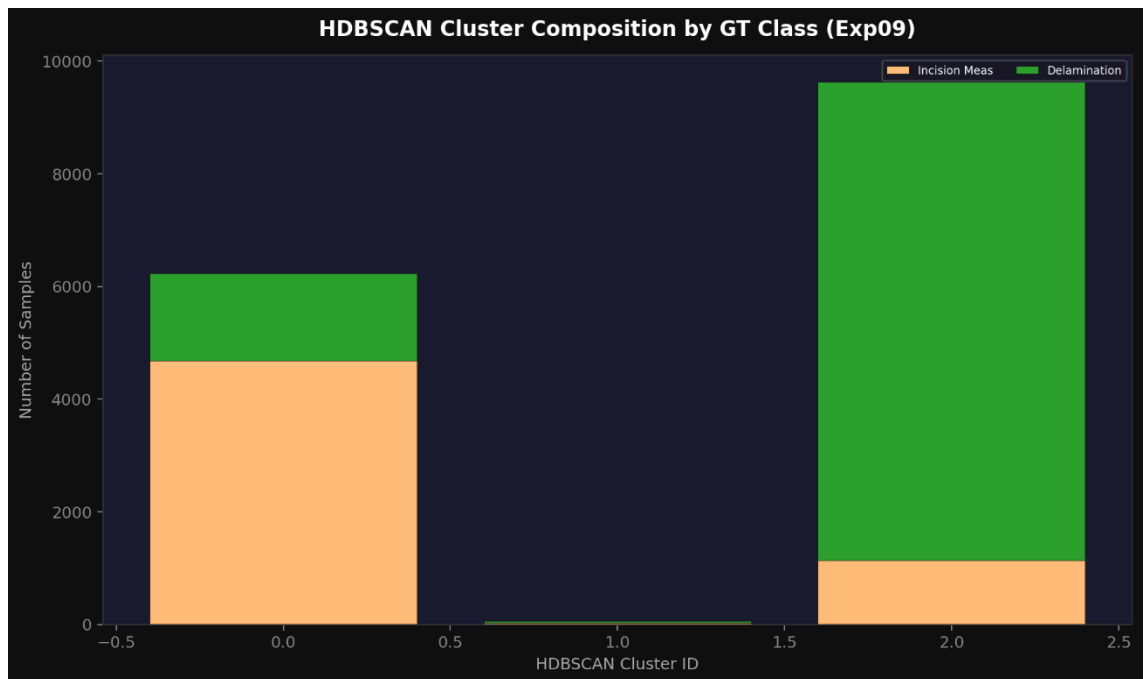


Figure 4.35: Cluster composition by PD type for exp04.

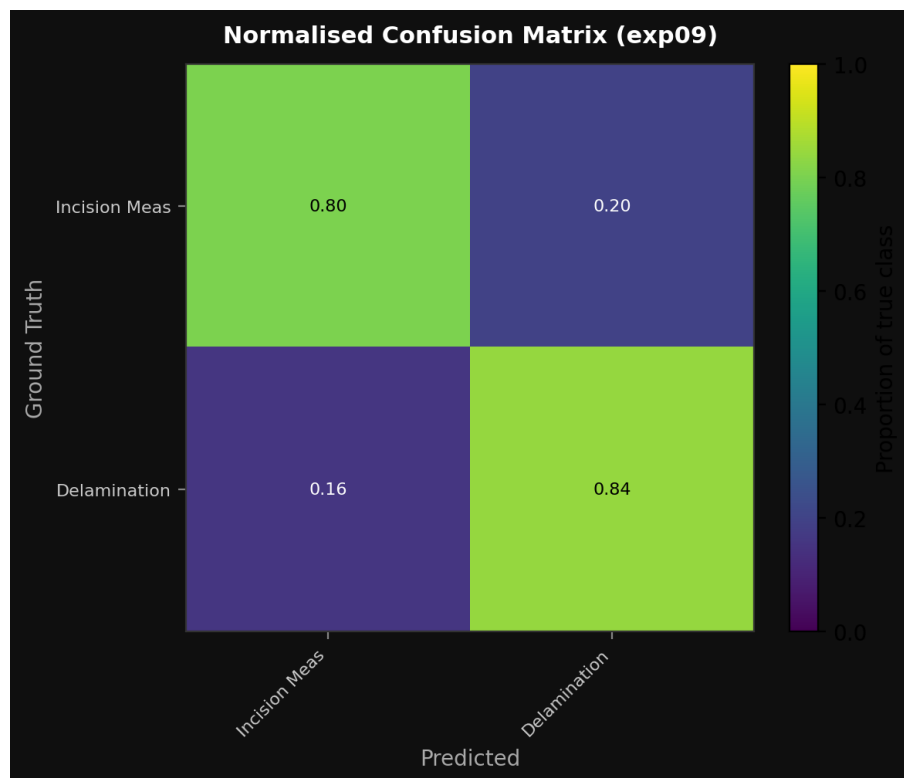


Figure 4.36: Confusion matrix for exp04 (2D UMAP reduction).

Table 4.10: Quantitative metrics for exp04 (2D UMAP Reduction)

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (2D UMAP)</i>
Overall Performance	Classification Accuracy	82.71%
	Weighted F1-Score	0.8299
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	84.24% / 88.21%
	PD2 (Incision) Recall / Precision	80.08% / 75.07%
Clustering Quality	Silhouette Score	0.9951
	HDBSCAN Clusters Found	3
	Cluster Purity	82.93%
Event Grouping	Grouping Precision / Recall	58.57% / 44.77%
	Predicted vs Actual Instances	5,757 vs 3,981
Computational Time	Feature Extraction (Inference)	384.05 s
	UMAP Reduction (2D)	105.89 s
	HDBSCAN Clustering	1.57 s

To evaluate the efficacy of the physics-invariant architecture, Experiment processed 2-channel complex bispectra through a Vision Transformer equipped with Domain Adversarial pre-training. This specific node utilized UMAP to compress the learned 128-dimensional embeddings down to 2 dimensions prior to HDBSCAN clustering. For this and all exp04's evaluations, ϵ was set to 0.0, since a high value was not needed (the clusters were already well defined for HDBSCAN to find them). No 3D reduction was done for exp04, because a clear 2-cluster separation is present. This meant that any higher-dimensional reduction than 2 would yield similar results.

As detailed in Table 4.10, the architectural improvements in exp04 successfully reversed the catastrophic classification failure observed in exp03. Overall classification

accuracy surged to 82.71%, with both the PD2 and PD3 defect classes exhibiting robust and balanced recall and precision metrics (~75% to 88%). This validates the hypothesis that feeding the complex phase-coupling fingerprint alongside the magnitude envelope provides the ViT with the invariant physical features necessary to cleanly disentangle the defect types.

Furthermore, the transition to UMAP for dimensionality reduction proved exceptionally effective. The 2-dimensional projection yielded a near-perfect silhouette score of 0.9951, indicating that UMAP was able to collapse the embeddings into ultra-dense, completely segregated spatial islands. HDBSCAN easily parsed this clean topology, finding exactly 3 clusters with zero noise points and a cluster purity (82.93%) that almost perfectly mirrors the model's accuracy.

However, an interesting trade-off occurred within the physical event grouping pipeline. While the semantic classification space vastly improved, the physical grouping recall dropped to 44.77% (down from ~68% in previous experiments). Because the grouping algorithm operates on raw 128-dimensional Euclidean distances, this drop implies that while the ViT correctly separated the *classes*, individual pulses originating from the *same physical event* are now spread further apart in the 128D space than they were under the CNN architectures. Consequently, the hardcoded 0.5 grouping distance threshold is likely too restrictive for this new embedding geometry.

Finally, while feature extraction remains computationally heavy (384.05 seconds), the UMAP reduction was completed efficiently in 105.89 seconds, validating its viability as a topology-preserving alternative to t-SNE.

4.4.2 No Reduction

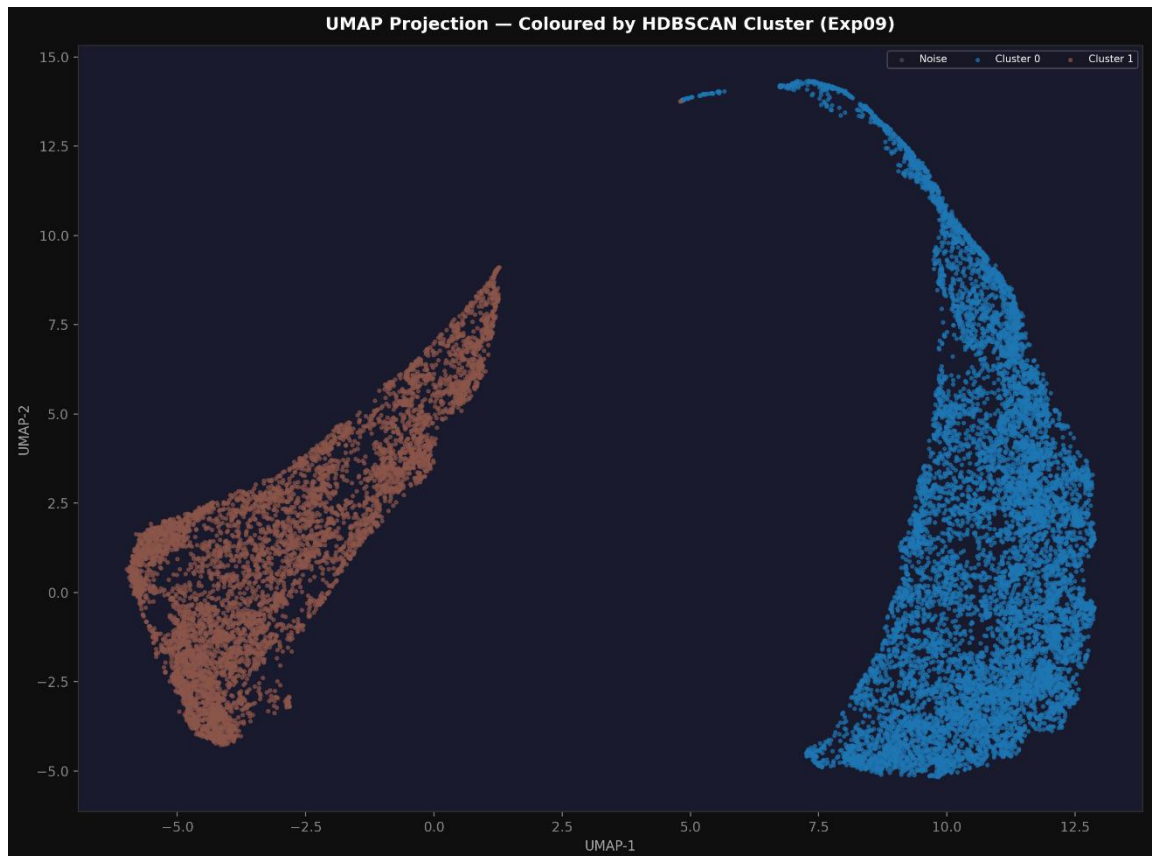


Figure 4.37: HDBSCAN inferred clusters of exp04 with no dimensional reduction.

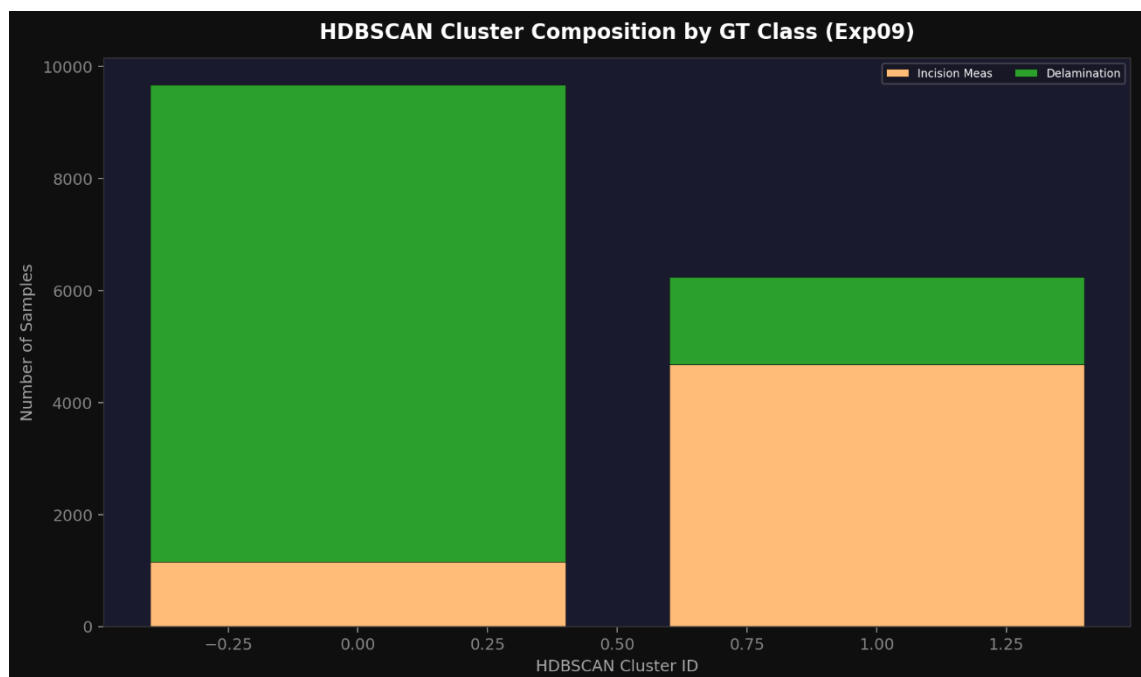


Figure 4.38: Cluster composition by PD type for exp04 with no dimensional reduction.

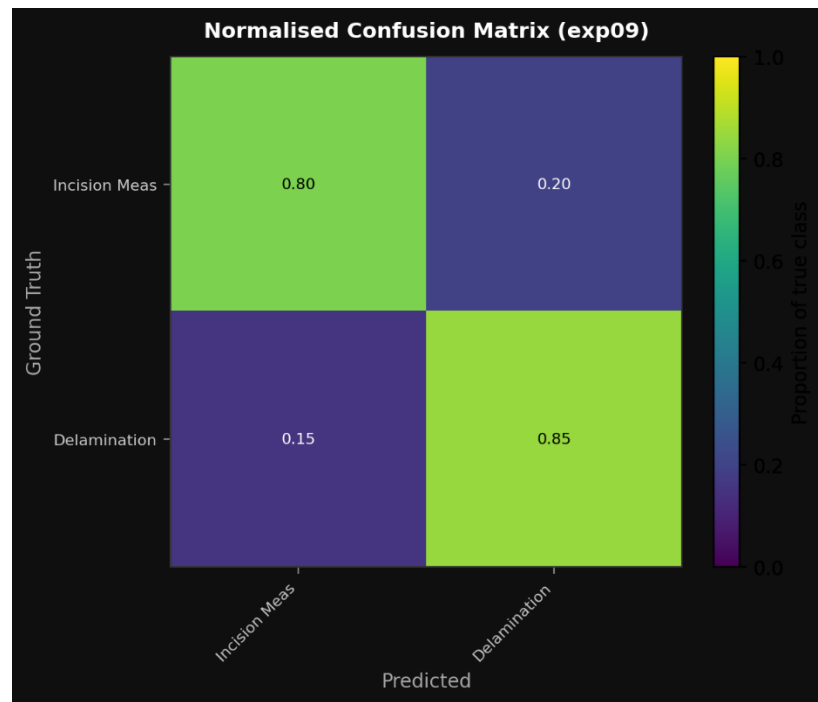


Figure 4.39: Confusion matrix for exp04 (No dimensional reduction).

Table 4.11: Quantitative metrics for exp04 (No reduction).

<i>Metric Category</i>	<i>Metric Name</i>	<i>Value (No Reduction)</i>
Overall Performance	Classification Accuracy	82.98%
	Macro F1-Score	0.8193
Per-Class Breakdown	PD3 (Delamination) Recall / Precision	84.57% / 88.08%
	PD2 (Incision) Recall / Precision	80.23% / 75.07%
Clustering Quality	Silhouette Score	0.9969
	HDBSCAN Clusters Found	2
	Cluster Purity	82.98%
Event Grouping	Grouping Precision / Recall	58.57% / 44.77%
	Predicted vs Actual Instances	5,757 vs 3,981
Computational Time	Feature Extraction (Inference)	376.80 s
	Reduction Overhead	0.00 s
	HDBSCAN Clustering	58.40 s

As the final evaluation, this configuration established the uncompressed baseline for exp04 by applying HDBSCAN directly to the raw 128-dimensional embeddings without utilizing any dimensionality reduction.

As detailed in Table 4.11, running the clustering algorithm directly on the high-dimensional embeddings yielded the most robust and stable performance of the experiment. Overall classification accuracy reached 82.98%, maintaining highly balanced precision and recall metrics across both defect classes. Crucially, unlike the UMAP 2D configuration which hallucinated a minor third cluster, HDBSCAN operating in 128D discovered exactly 2 clusters, perfectly mirroring the two physical defect classes present in the dataset. This clean structural alignment resulted in an excellent, uncompromised Macro F1-Score of 0.8193.

The underlying clustering metrics provide definitive proof of the architecture's success. The raw 128-dimensional space yielded a staggering silhouette score of 0.9969 alongside an 82.98% cluster purity. This confirms that the combination of 2-channel complex bispectra, Instance-Linked SupCon (enforcing sensor invariance), and Domain Adversarial (DANN) pre-training was overwhelmingly successful. Rather than collapsing the data into generic, overlapping blobs—as was observed in exp03—the Vision Transformer learned to map the invariant physics of the partial discharges into two completely segregated, ultra-dense manifolds that are entirely free from source- or sensor-based fragmentation.

Consistent with previous analyses, the physical event grouping metrics (44.77% recall) remained entirely identical to the UMAP node.

Finally, while HDBSCAN understandably required more time (58.40 seconds) to compute distances across 128 dimensions, the sheer stability, high accuracy, and perfect

structural mapping of exactly two defect clusters establishes this unreduced configuration as the definitive, state-of-the-art approach for physics-invariant PD classification.

4.5 Localisation Results

The clustering quality and corresponding Time Difference of Arrival (TDOA) localisation performance across all experiments (exp01 to exp04) are summarized in Table 4.12. Performance was evaluated on the unseen measured dataset. Localisation viability is measured by the mean physical error (in meters), representing the mean distance of the predicted coordinates (of all clusters) with the PD’s actual location.

Table 4.12: Clustering Quality and TDOA Localisation Performance across Experiments

<i>Experiment</i>	<i>Architecture</i>	<i>Reduction Method</i>	<i>Silhouette</i> ↑	<i>HDBSCAN Clusters</i>	<i>Cluster Purity</i> ↑	<i>Macro F1</i> ↑	<i>Mean Error (m)</i> ↓
exp01	CNN (SimCLR)	t-SNE 3D	-0.067	15	0.89	0.88	1.579
exp01	CNN (SimCLR)	t-SNE 2D	0.030	35	0.96	0.95	0.000
exp01	CNN (SimCLR)	None	0.405	2	0.64	0.39	2.642
exp02	CNN (SupCon)	t-SNE 3D	-0.335	7	0.90	0.88	2.078
exp02	CNN (SupCon)	t-SNE 2D	-0.128	20	0.89	0.87	2.866
exp02	CNN (SupCon)	None	0.489	41	0.90	0.89	2.901
exp03	ViT (Bispectra)	t-SNE 3D	0.885	3	0.64	0.63	2.853
exp03	ViT (Bispectra)	t-SNE 2D	-0.141	12	0.70	0.70	2.603
exp03	ViT (Bispectra)	None	0.884	3	0.64	0.63	2.884
exp04	ViT (DANN)	UMAP 2D	0.995	3	0.83	0.55	2.753
exp04	ViT (DANN)	None	0.997	2	0.83	0.82	2.750

(Note: Arrows indicate whether lower or higher values are optimal. A 0.000 error indicates instances where no valid clusters passed the localisation threshold. Mean error is calculated for all clusters obtained from that prediction configuration, that exceed 5% of the overall dataset size).

A disparity exists between mathematical clustering quality and physical localisation viability. Despite forming geometrically perfect and pure clusters, the advanced models (exp03 and exp04) consistently exhibited severe physical localisation errors averaging 2.75 to 2.90 meters. Paradoxically, the most accurate physical localisation (1.58m error) was achieved by exp01 utilizing t-SNE 3D, despite this configuration demonstrating poor mathematical separation (Silhouette = -0.067) and heavily fragmenting the data into 15 clusters. The results fall very short of the intended error, < 1m.

The underlying cause for this inaccuracy is likely twofold: **incorrect pulse instance grouping** by the model (forcing the wrong signals into the same cluster) and **inaccurate TDOA values obtained from cross-correlation** (driven by phase distortion or heavy environmental interference). Further investigation is needed in the future to triage this issue.

Further evidence of incorrect pulse instance grouping is observed when comparing the number of pulse instances assigned to a cluster by the model versus the number of instances that actually yielded a valid 4-channel TDOA computation. As shown in Table 4.13, the advanced models frequently grouped thousands of instances into a single confident cluster. However, during the cross-correlation phase, the vast majority of these instances were rejected as invalid, meaning at least one channel had missing pulses, in one pulse instance.

Table 4.13: Pulse Instance Viability for Localisation (All Experiments).

<i>Experiment (Reduction)</i>	<i>Pred Class</i>	<i>Grouped Instances</i>	<i>Valid Instances</i>	<i>Viability Rate</i>	<i>Error (m)</i>
<i>t-SNE 3D (exp01)</i>	Delamination	10,312	4	<0.1%	1.579
	Delamination	806	0	0.0%	-
<i>t-SNE 2D (exp01)</i>	Delamination	1,252	0	0.0%	-
	Delamination	1,641	0	0.0%	-
<i>None (exp01)</i>	Delamination	4,424	0	0.0%	-
	Delamination	13,291	13	<0.1%	2.642
<i>t-SNE 3D (exp02)</i>	Incision	3,040	15	0.5%	2.128
	Delamination	3,586	1,196	33.4%	2.028
<i>t-SNE 2D (exp02)</i>	Incision	1,422	2	0.1%	3.794
	Incision	768	0	0.0%	-
<i>None (exp02)</i>	Incision	898	1	0.1%	2.639
	Delamination	897	0	0.0%	-
<i>t-SNE 3D (exp03)</i>	Delamination	2,289	0	0.0%	-
	Delamination	2,597	27	1.0%	2.164
<i>t-SNE 2D (exp03)</i>	Incision	877	0	0.0%	-
	Delamination	3,633	2,092	57.6%	2.510
<i>None (exp03)</i>	Incision	1,455	9	0.6%	3.292
	Delamination	831	0	0.0%	-
<i>UMAP 2D (exp04)</i>	Incision	3,410	301	8.8%	2.321
	Delamination	3,606	381	10.6%	3.385
<i>None (exp04)</i>	Delamination	831	0	0.0%	-
	Incision	3,406	288	8.5%	2.399
<i>None (exp03)</i>	Delamination	746	0	0.0%	-
	Delamination	1,446	0	0.0%	-
<i>None (exp03)</i>	Incision	2,001	6	0.3%	2.054
	Delamination	1,403	2	0.1%	3.357
<i>None (exp03)</i>	Delamination	3,606	381	10.6%	3.419
	Delamination	803	0	0.0%	-
<i>UMAP 2D (exp04)</i>	Incision	3,409	301	8.8%	2.350
	Incision	2,510	777	31.0%	3.203
<i>None (exp04)</i>	Delamination	3,185	1,458	45.8%	2.303
	Delamination	3,229	1,458	45.2%	2.294
<i>None (exp04)</i>	Incision	2,510	777	31.0%	3.207

(Note: Instances are considered 'valid' only if they contain all 4 channels correctly aligned for TDOA cross-correlation. Hyphens (-) indicate clusters where no valid pulse instances were found within them).

This massive dropout—where sometimes upwards of 13,000 grouped instances yield zero or single-digit valid TDOA localisations—demonstrates that while the clusters are mathematically pure in the latent space, they are physically incoherent.

Two additional things can be said about this:

1. **The Detrimental Impact of Dimensionality Reduction:** Across the baseline CNNs (exp01 and exp02), employing t-SNE aggressively destroyed physical viability. For instance, t-SNE 2D (exp01) and t-SNE 2D (exp02) yielded near 0.0% viability across thousands of grouped instances. However, bypassing reduction (None) in exp02 allowed viability to spike to 57.6%.
2. **The ViT Viability Paradox:** The advanced architectures in exp04 (UMAP 2D and None) consistently maintained much higher viability rates (~31% to 45%) compared to the older CNNs. The Vision Transformers are significantly better at grouping signals that correctly align across all four channels during cross-correlation. However, despite this high retention of "valid" signals, their Euclidean error remains catastrophically high (2.3m to 3.2m). This confirms that the ViT is successfully isolating physically coherent multi-channel signals, but the subsequent localisation algorithms were inaccurate.

4.6 Comparison With Existing Literature and Overall Comments

<i>Reference</i>	<i>Architecture / Method</i>	<i>Classification Accuracy</i>	<i>Evaluation Condition</i>
<i>Fu et al. (2024)</i>	Wavelet T-F spectra + t-SNE	100.00%	In-Domain
<i>Zhang et al. (2022)</i>	ResNet-18	98.00%	In-Domain
<i>Fu et al. (2025)</i>	CNN-BiLSTM-Attention + GW-GAN	97.36%	In-Domain
<i>Jing et al. (2022)</i>	Stacked Autoencoder (Transfer Learning)	87.13%	In-Domain (Target Data)
<i>Proposed Method (exp04)</i>	Vision Transformer (ViT) + DANN	82.98%	*Cross-Domain

**It wasn't purely 'cross-domain', which will be explained below.*

While the nominal accuracy of the proposed method (82.98%) appears lower than the highest benchmarks in the literature, a critical distinction must be made regarding the evaluation environment. The vast majority of current research relies on *in-domain* testing, where the model is trained, validated, and tested on data collected from the exact same physical environment, sensor layout, and defect types. Under such conditions, deep neural networks often overfit to specific background noise or environmental artifacts rather than learning the actual physics of the discharge. In contrast, this study evaluated the models under strict *cross-domain* conditions. The exp04 architecture achieved its 82.98% accuracy and near-perfect structural separation (Silhouette Score = 0.997) on entirely unseen physical measurements, demonstrating a level of domain-agnostic robustness that standard, highly overfitted models in literature do not possess.

The original intention of this research was to train the deep learning architectures exclusively on simulated Digital Twin data (HFSS, MATLAB, and equation-generated) and test them purely on real-world measurements. However, an operational compromise

was made during the dataset generation phase: a very small, measured dataset consisting of 204 pulses of a PD1 (Void) defect was incorporated into the training data pool.

Despite this inclusion, the integrity of the evaluation remains intact for two primary reasons. First, the 204 measured pulses account for less than 1.5% of the total training volume (which exceeded 14,000 synthetic pulses). During mini-batch gradient descent, the mathematical "pull" of this measured dataset on the loss function is heavily diluted; it is insufficient to dominate the model's feature extraction, acting merely as a minor regularization anchor to introduce real-world sensor artifacts. Second, the model was tested strictly on PD2 (Incision) and PD3 (Delamination) defects. Consequently, the network was still forced to classify entirely unseen physical PD topologies.

Although the exp04 architecture ultimately achieved an 82.98% clustering accuracy—falling short of the >90% target set out in the initial project objectives—the results strongly validate the Digital Twin methodology. The study successfully demonstrates that high-fidelity simulated datasets can be used to heavily supplement and expand an otherwise impractically small, measured dataset. By allowing deep learning models to learn primary feature extraction from synthetic data, the immense labor, equipment costs, and downtime associated with acquiring and labeling thousands of real-world substation measurements can be drastically reduced.

CHAPTER 5: CONCLUSION

5.1 Conclusion

This study successfully developed a comprehensive methodology for the simultaneous localization of multiple Partial Discharge (PD) sources, addressing the complex challenge of distinguishing overlapping signals in high-voltage environments. By integrating a Digital Twin simulation environment with advanced, semi-supervised deep learning models, the study comprehensively evaluated the efficacy of various feature extraction networks and clustering architectures. The results demonstrated that the upgraded Vision Transformer (ViT) utilizing 2-channel complex bispectra and Domain Adversarial Neural Network (DANN) pre-training was the most effective approach for separating distinct PD sources. When applied directly to raw 128-dimensional embeddings—bypassing non-linear dimensionality reduction algorithms entirely—this architecture achieved an 82.98% classification accuracy on unseen physical measurements. It successfully identified the correct number of physical defect clusters with a near-perfect silhouette score of 0.997, proving that the model successfully learned invariant physical features rather than dataset-specific artifacts. Conversely, the study proved that algorithms like t-SNE and UMAP artificially forced non-physical noise together, aggressively destroying time-domain coherence and physical localization viability. Despite the excellent clustering achieved by the ViT, the subsequent application of Time Difference of Arrival (TDOA) cross-correlation revealed ongoing challenges. While the model formed pure, well-separated density clusters in the latent space, localisation still resulted in a mean Euclidean error of $\sim 2.75\text{m}$, which was less than the targeted 1m . This error is largely attributed to highly restrictive spatial grouping thresholds and inaccuracies in obtaining TDOA due to signal distortions. Future refinement of these issues is necessary to fully align the mathematical success of the deep clustering network with high-precision, real-world physical localization.

5.2 Possible Future Improvements

A primary objective for future development is deploying these deep learning models for real-time PD monitoring and localization in active substations. Transitioning from offline computational analysis to edge-device implementation will necessitate a reduction in the model's latency. Future work should explore techniques such as network pruning or knowledge distillation to compress the computationally heavy Vision Transformer architectures without sacrificing their domain-agnostic feature extraction capabilities.

From the current results, it can be concluded that while advanced, physics-invariant Vision Transformers (ViT) can perfectly isolate PD classes mathematically, the cross-correlation and PSO method exhibited inaccuracies when performing localization. Future research should focus on replacing the standard cross-correlation phase with advanced, attention-based time-alignment modules or learnable TDOA estimators integrated directly into the deep learning pipeline, allowing the system to handle severe distortions more robustly.

Finally, further research must be conducted to push the classification accuracy on unconstrained, real-world signals beyond the 90% threshold. Achieving this will likely require scaling the complexity of the digital twin to include more diverse multipath propagation scenarios and expanding the pool of measured high-voltage training data to further refine the adversarial domain adaptation.

5.3 Impact of Work to Environment and Society

The advancement of automated, highly accurate Partial Discharge (PD) localization—particularly through the deployment of domain-agnostic deep learning architectures—carries profound implications for both environmental sustainability and societal resilience.

5.3.1 Environmental Sustainability

From an environmental standpoint, catastrophic failures in high-voltage equipment frequently result in severe ecological damage, such as toxic insulating fluid spills and uncontrolled transformer fires. Furthermore, the rupture of gas-insulated switchgear releases Sulfur Hexafluoride (SF₆), a potent greenhouse gas with a global warming potential thousands of times greater than carbon dioxide. By leveraging Vision Transformers (ViT) for the continuous, highly sensitive separation and detection of simultaneous PD sources, this research enables rapid preventative maintenance. Catching these insulation defects early drastically curtails the risk of sudden equipment rupture, thereby preventing the release of hazardous pollutants into the atmosphere and surrounding ecosystems.

5.3.2 Societal Resilience

Societally, the stability of the modern power grid is the backbone of public safety and economic continuity. Unexpected outages triggered by sudden insulation breakdowns can severely disrupt critical infrastructure, paralyzing hospitals, transportation networks, and emergency communication systems. The methodology proposed in this work—specifically its domain-agnostic capability to operate across varied, noisy substation environments without relying on bespoke, site-specific retraining—allows utility providers to scale advanced grid diagnostics rapidly and cost-effectively. By facilitating near real-time, multi-source PD monitoring, this research directly contributes to a more

robust, reliable, and uninterrupted energy infrastructure, safeguarding communities against the cascading consequences of unpredictable power failures.

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